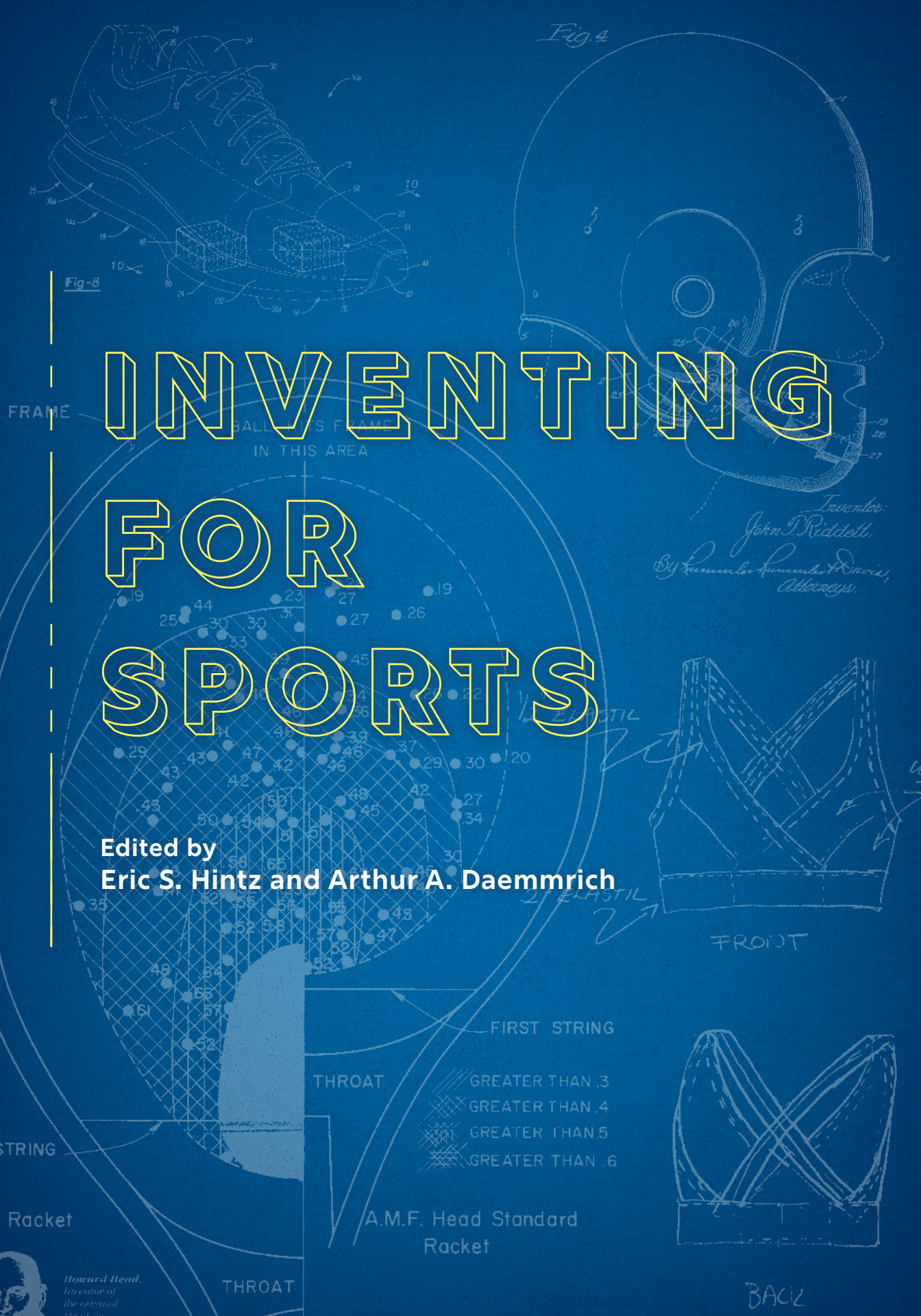

INVENTING FOR SPORTS

Edited by
Eric S. Hintz and Arthur A. Daemmrich

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Contents

Preface vii
Eric S. Hintz

Chapter 1: Introduction 1
Eric S. Hintz and Arthur A. Daemmrich

Chapter 2: Competitive Edge 17
Meg Maher

Chapter 3: Health and Safety 37
Jeffrey L. Brodie

Chapter 4: Fairness and Accuracy 59
Eric S. Hintz

Chapter 5: Fun for Everyone 79
Monica M. Smith

Chapter 6: Fair or Foul? The Governance of Sports Technology 101
Arthur A. Daemmrich

Chapter 7: The Past, Present, and Future of Sports Innovation 121
Arthur A. Daemmrich, Eric S. Hintz, and Jane Rogers

Acknowledgments 141

Bibliography 143

About the Contributors 163

Index 165

Preface

Eric S. Hintz

This book emerged from a multi-year initiative on sports invention and technology at the Smithsonian Institution's Lemelson Center for the Study of Invention and Innovation. Based at the National Museum of American History (NMAH) in Washington, D.C., the Lemelson Center develops exhibitions, public programming, educational materials, and publications that advance the historical study and public understanding of invention and innovation. By sharing new perspectives on inventors past and present, the Lemelson Center aims to empower all people to apply their own inventive creativity to real-world problems.

Our interest in sports innovation began with collections development. During the 1990s and 2000s, the Lemelson Center partnered with our NMAH curatorial colleagues to collect the artifacts and personal papers of several sports inventors, including Howard Head's metal laminate snow skis and oversized tennis rackets, windsurfer Newman Darby's sailboards, Marilyn Hamilton's "Quickie" sports wheelchair, Van Phillips's "Flex-Foot Cheetah" running prostheses, Arthur Ehrat's "Break-A-Way" basketball rim, and Sherman Poppen's "Snurfer," the precursor of modern snowboards.

The Lemelson Center and collaborators across NMAH then featured sports inventors and their innovations in a series of exhibitions. *Sporting Invention* opened in 1999 and traced the development of sports inventions through drawings and prototypes. Darby's sailboard and invention process were prominently featured in the Lemelson Center's first major exhibition, *Invention at Play*, which traveled to dozens of museums between 2003 and 2011.[1] In 2017, the Center's *Inventive Minds* gallery included a "Sporting Invention" rotation that featured Marilyn Hamilton, Howard Head, and other sports inventors.

Our museum colleague and sports curator Jane Rogers also curated four notable sports-themed exhibitions. *Snowboarding* (2012–2013) featured Poppen's Snurfer and traced the evolution of the sport. *Everyone Plays!* (2016–2017) highlighted the history of adaptive sports innovations and the Paralympic movement. *Special Olympics at 50* (2018–2021) introduced visitors to an athletic and social movement that organizes sports training and competitions for people with intellectual disabilities. *We Belong Here* (opened 2022) marked the fiftieth anniversary of the Title IX legislation that created more opportunities for women as well as for nonbinary and transgender athletes to play sports.

We also explored sports innovations through several public programs. In 2013, the Lemelson Center launched its "Innoskate" initiative, which highlights invention and innovation in the skateboarding community. In June 2013, the National Museum of American History installed a skate ramp on its outdoor terrace and treated a crowd of 30,000 visitors to skating demonstrations, panel discussions, and artifact donations from skateboarding legends such as Tony Hawk, Rodney Mullen, Patti McGee, and Cindy Whitehead. Skaters and equipment manufacturers described how technical innovations in wood laminate decks, polyurethane wheels, and skate park design enabled skaters to invent

new tricks and elevate their craft. Since 2013, the Lemelson Center has partnered with local communities to host additional Innoskate festivals in Florida, Washington, Texas, South Carolina, Massachusetts, South Dakota, and London, England, which collectively attracted hundreds of thousands of visitors. The Innoskate initiative underscores the Lemelson Center's core belief that everyone is inherently inventive and creative, and that innovation often flourishes among communities where—at first glance—it might seem unexpected. Innoskate also proved that large, diverse crowds of young people and adults could be attracted to learn about invention and innovation in the context of sports programming.

In 2017, the Lemelson Center hosted "Invention, Sport, Technology, and Society," a two-day workshop that featured presentations by historians, sports inventors, and museum professionals. The group also engaged in structured discussions to gauge the feasibility of pursuing a multi-year research, exhibition, and educational initiative on the topic of sports innovation. Workshop participants enthusiastically affirmed our belief in the topic and agreed to serve as advisers on the project.

In 2018, we developed an intellectual framework for the emerging *Change Your Game / Cambia tu juego* exhibition project and secured a seed grant from the Lemelson Foundation. This financial support enabled us to hire an exhibition design firm, Roto, that helped us produce an early conceptual design. During this formative period, we crystallized our belief that the exhibition should strive to inspire visitors to see themselves as inventive. We then partnered with educational psychologists Dr. Joanna Garner (Old Dominion University) and Dr. Avi Kaplan (Temple University); as we developed the exhibition, we applied their "dynamic systems model of role identity" (DSMRI), which suggests that both classroom-based and informal educational experiences such as museum visits can inspire students to engage in role-playing and adopt new identities.[2]

Given the entrenched lack of diversity in the innovation ecosystem, we identified three primary audiences to engage through the initiative: African American boys aged 10 to 17; girls aged 10 to 17; and people with disabilities of all ages.[3] To meet and talk with a diverse spectrum of students, we developed local community partnerships with the YMCA of Metropolitan Washington, Girl Scouts of the Nation's Capital, Boys & Girls Clubs of Greater Washington, and the Smithsonian's Anacostia Community Museum. We also partnered with Access Smithsonian, which provides accessibility services for thousands of Smithsonian visitors while advocating for best accessibility practices across the entire museum field. Finally, we retained an evaluation firm (Kera Collective, formerly RK&A) to interview focus groups from our primary audiences; these discussions helped us test the educational effectiveness of Roto's early concepts and refine the design.

In 2020, the National Science Foundation (NSF) awarded the Lemelson Center a major grant from the Advancing Informal STEM Learning (AISL) program;[4] we subsequently secured additional grants from the Lemelson Foundation, the United States Patent and Trademark Office, Nike, the Patrick J. McGovern Foundation, the Shō Foundation, ConocoPhillips, and the Hopper-Dean Family Fund. This funding supported ongoing exhibition development as we sought to empirically answer the following question: Can an informal STEM (science, technology, engineering, and mathematics) learning setting activate a visitor's "inventive identity"? In other words, could a young person walk into the *Change Your Game / Cambia tu juego* exhibition, become inspired by the stories of sports innovation, engage with the hands-on educational activities, and leave the exhibition feeling empowered to invent? Encouraging exhibition visitors to adopt an inventive identity became a guiding principle for our story selections and other design decisions.[5]

FIGURE 0.1. The entrance to the *Change Your Game / Cambia tu juego* exhibition. Credit: Jaclyn Nash, National Museum of American History, Smithsonian Institution.

Meanwhile, the Lemelson Center continued to produce public programming and educational content on sports innovation. In partnership with SportTechie and the Qatar Foundation, the Lemelson Center traveled across the United States and around the world visiting various stadiums and arenas. The collaboration produced two educational videos—"12 Game-Changing Stadiums in History" and "Grand Stands: Stadium Technology and the New Fan Experience"—that showcase historical and contemporary innovations in stadium design. When the COVID-19 pandemic forced the museum to close Spark!Lab, the Lemelson Center's invention learning lab, the team developed online challenges, including one in which visitors use the Tinkercad web app to design a virtual stadium.[6]

As exhibition development proceeded apace, the Lemelson Center's *Innovative Lives* speaker series featured several inventors whose sports technologies have enabled increased participation. Notably, we hosted a public conversation with Jogbra inventors Lisa Lindahl, Hinda Schreiber Miller, and Polly Palmer Smith; they described how their innovation enabled more women to participate in running and other sports comfortably and confidently. Subsequent programs featured athletes whose disabilities inspired them to develop adaptive sports technologies. For example, an Innoskate-themed program featured adaptive skateboarders Oscar Loreto, Jr., and Dan Mancina alongside wheelchair motocross pioneer Aaron "Wheelz" Fotheringham. Marilyn Hamilton similarly recalled how she and two colleagues invented the ultra-light Quickie sports wheelchair after a hang-gliding accident left her partially paralyzed. In a conversation with Paralympic alpine skier Sarah Will, competitive snowboarder and motocross racer Mike Schultz described the snowmobile accident that took his lower left leg and inspired his second act as a para-athlete and inventor. Schultz explained how his Moto Knee and Versa Foot prostheses have enabled him and other para-athletes to compete in a variety of adaptive sports. During these inventors' visits to the museum, we collected examples of their sports technologies and recorded detailed oral histories, which would eventually be featured in the exhibition and this book.[7]

In March 2024, the bilingual *Change Your Game / Cambia tu juego* exhibition opened in the Jerome and Dorothy Lemelson Hall of Invention at the Smithsonian's National Museum of American History (Figure 0.1). The 3,500-square-foot gallery features a rich

collection of inventors' stories, historic and contemporary artifacts, and multimedia presentations that help visitors understand the motivations and inventive practices underlying various sports innovations. The exhibition also features several interactive activities that invite visitors to engage directly in the inventive process while considering the broader impacts of those innovations, both inside and outside sports. Collectively, these experiences aim to activate each visitor's "inventive identity" so that they see themselves as game changers. Visitors will leave the exhibition with the confidence to develop creative solutions in their own lives—in sports, in school, at home, and in any other context.

This book is a standalone companion to—not a catalog for—the *Change Your Game / Cambia tu juego* exhibition. The book explores the underlying ideas that visitors encounter in the exhibition while drawing on many of the same case studies they can find in the gallery. However, the book allows for deeper theoretical explorations into sports inventors, their motivations and inspirations, the impacts of their inventions both within and beyond sports, and the dynamic processes underlying the regulation and governance of sports technologies. In addition, the book looks ahead to consider where sports technology might be headed in the future. By sharing multiple engaging stories about a diverse set of sports inventors, we hope the volume will provide an accessible starting point for historians, scientists, and other scholars who wish to learn more about the role of invention and technology in sports. More importantly, we hope these stories will inspire readers to see themselves as inventive problem solvers who can be game changers in their everyday lives.

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Chapter 1

Introduction

Eric S. Hintz and Arthur A. Daemmrich

Every sport and every athlete relies on technology. This is obvious in certain sports, such as Formula 1 auto racing or America's Cup sailing. In these high-tech contests, the chassis setup or hull design can determine who wins, potentially surpassing the relative skill and training of the drivers, the sailors, and their crews.

Yet, increasingly, even a simple, "low-tech" sport like running involves multiple technologies. A runner's shorts, tank top, and sports bra are now constructed of advanced synthetic fabrics designed to wick away perspiration. Their shoes are made with highly engineered materials that store and release energy with each step. During training, a recreational jogger might record her time and heart rate with a sophisticated smart watch. At a 10K road race or marathon, a radio-frequency identification (RFID) microchip built into the race number will automatically record her adherence to the route and finish time. An Olympic sprint race uses even more sophisticated timing equipment, including an automated false start detection system built into the starting blocks, and a digital photo-finish camera that captures up to 20,000 frames per second. After the race, a sprinter might recover by consuming a specially formulated Gatorade sports drink or high-calorie "energy gel." Officials may require the sprinter to provide blood or urine samples so that lab technicians can run chemical assays to test for performance-enhancing drugs.

We can easily observe that both amateur and Olympic-caliber athletes use multiple technologies. But we are probably unaware of *who invented* those technologies and *why*. Who invented the sports bra, the Gatorade sports drink, or the high-speed photo-finish camera? What motivated the inventors to create these new technologies? Was it just to make money, or were there other drivers for change? Who determines whether those new technologies are permitted or banned? Furthermore, in what ways have these innovations changed the games?

* * *

This book invites readers to view sports as an activity inherently and intimately intertwined with advances in science, technology, and medicine. While athletes will always remain central to recreational and competitive sports, we argue that inventors and their innovations are just as crucial and inseparable. By exploring a mix of historical and contemporary case studies, we show that every sport, every competition, and even recreational play are mediated by numerous inventions, including highly engineered sticks, bats, balls, rackets, skates, and skis; sport-specific athletic apparel, shoes, gloves, prostheses, and wheelchairs; helmets, pads, mouth guards, and protective gear; high-tech arenas, playing surfaces, scoreboards, and broadcast technologies; training aids, nutritional supplements,

medicines, and rehabilitation techniques; plus stopwatches, high-speed photo-finish cameras, and other officiating devices.

Athletes and coaches also engage in non-technological innovation when they introduce new bodily techniques, training regimens, and in-game strategies. Some of these sports techniques are invented in response to technology change, while others are co-developed alongside new gear and become integral to the use of new technologies.[1] For amateur and professional athletes alike, these advanced technologies and innovative techniques make the difference between victory and defeat, safety and injury, and participation and exclusion.

Fans and spectators likewise serve multiple roles in the sports innovation ecosystem. Occasionally, "weekend warriors" will invent new sports technologies to improve their amateur performance levels. To keep their fans engaged, sports leagues introduce in-game innovations such as basketball's three-point line or baseball's pitch clock to make games more exciting. Fans also deepen their engagement with teams and athletes through technology-based interfaces, including interactive stadium experiences, diverse social media, online fantasy leagues, and increasingly, sports-betting apps. As consumers of sports equipment and apparel, fans also act as a key market force that drives brands to design products with different features at various price points. While we do not focus specifically on fan engagement technology in this book, numerous case studies identify fans as influential in both stimulating and inhibiting technological change. For example, sports fans have both encouraged the adoption of instant replay to enhance television broadcasts and discouraged its use in officiating because it slows down the games.

There is often disagreement about the individual risks and community impacts of an invention; in fact, some inventions are harmful to athletes or destructive to public perception of a sport. For example, performance-enhancing drugs (PEDs) facilitate cheating, threaten athletes' long-term health, and are generally condemned for undermining the integrity of sports. Nevertheless, individual athletes and teams continue to embrace pharmaceutical innovations including synthetic amphetamines, anabolic steroids, human growth hormone, and erythropoietin to gain a competive edge. As athletes and rogue physicians develop ingenious new doping methods, sports federations employ scientists to develop improved methods to test athletes' bodies to ensure fair play. What emerges from this arms race is a pathological pattern of innovation, in which (1) scientists develop new pharmacological inventions to treat legitimate medical conditions; (2) athletes experiment with non-therapeutic uses for those drugs to gain a competitive advantage; (3) sports federations fund the invention of biomedical tests to detect these new modes of doping; and (4) athletes, coaches, and physicians develop clever new ways of evading anti-doping measures. Although this book does not engage in a deep exploration of PEDs or other negative examples of innovation, several case studies observe that there is rarely unanimous consensus about the value of a particular technology. Indeed, athletes, inventors, and officials regularly engage in vigorous debate about the potential benefits and harmful effects of new sports inventions.

Overall, *Inventing for Sports* shifts our perspective from athletes and athletic contests to the high-tech equipment, apparel, protective gear, and officiating technologies that infuse all modern sports. The book explores the different motivations that inspire athletes and inventors to develop game-changing technologies, their invention processes, and the often-surprising sources of their ideas. It examines the social and cultural reception of sports technologies, the passionate debates they spur, and the myriad ways they change the

games, for better and worse. Finally, it explores different approaches to the assessment, regulation, and governance of sports technologies and examines why game-changing innovations are ultimately adopted or rejected.

Extending Scholarship: Innovation and Sports

This book builds upon and extends a new and growing body of literature on the role of science, technology, medicine, and innovation in sports.[2] Popular accounts include Michael Lewis's bestselling book *Moneyball*, which describes how Oakland Athletics general manager Billy Beane deployed mathematical techniques in an unorthodox manner to find high-performing but undervalued baseball players so his low-payroll team could compete successfully against wealthier clubs. Other popular books, including *The Sports Gene* and *Range* by David Epstein and *Peak* by Marc Bubbs, describe how athletes have improved their performance through high-tech training regimens and evidence-based strategies for optimizing nutrition and sleep.[3] Similarly, celebrity athletes and techno-enthusiasts have written dozens of practical guides recommending a variety of technologies, techniques, and "hacks" to help athletes improve their performance.[4] Meanwhile, much of the relevant academic literature on sports technology has emerged from several overlapping subgenres, including critical sports studies, science and technology studies (STS), and works on regulatory governance.

Critical Sports Studies

Critical sports scholars have developed a sophisticated literature on the role of technology in sports.[5] Much of this literature focuses on technoscientific interventions to protect, repair, enhance, or surveil athletes' bodies.[6] As noted, several studies have traced the use of performance-enhancing drugs in various sports and the development of scientific testing regimes to detect and deter their use.[7] Scholars have also explored the development of high-tech prostheses and wheelchairs in adaptive sports, the contested criteria used to segregate para-athletes by their disabilities, and the genetic enhancement of "trans-human" athletes.[8] Most recently, scholars have examined the controversies surrounding gender verification testing, in which sports federations test for specific chromosomes and hormone levels to determine an athlete's eligibility to compete in sex-segregated events.[9] Collectively, this literature focuses primarily on athletes, their bodies, and the messy negotiations that determine who is eligible to compete. While these studies recognize that technoscientific and medical interventions are implicated in rule-making, they often overlook the motivations and interests of inventors.

Sports historians have also produced a growing literature that examines sports in the context of race, gender, disability, and other aspects of identity. For example, several scholars have criticized an entrenched racial power dynamic in professional sports, wherein Black athletes are often the top competitors, yet the owners, managers, and league officials remain predominately White.[10] Scholars have also documented women's historic exclusion and underrepresentation in various sports and the lower earnings of professional women athletes, while lauding the considerable progress achieved since the passage of Title IX legislation in 1972.[11] Identity can be a double-edged sword in the realm of sports innovation. The spectrum of human differences can motivate sports inventions—like the Jogbra and high-tech prostheses—designed for different body types. Yet inventors,

designers, and entrepreneurs who are women, African American, or disabled often find that pernicious structural barriers extend from the playing field to the engineering laboratory and boardroom.

Kinesiologists and sports engineers have also examined the material culture of various sports to trace the impact of new technologies on athletes and game play. These studies typically describe empirically measurable performance gains achieved through the application of advanced materials and superior product design in sports equipment, such as titanium golf clubs, aerodynamic running shoes, and carbon fiber bike frames.[12] Some scholars go further in recognizing the cultural impacts of new sports technologies. Douglas Booth and Tolga Ozyurtcu, for example, have explored how fiberglass surfboards, neoprene wetsuits, and GoPro cameras have changed the ethos of modern surfing, which has evolved considerably from its indigenous Hawai'ian roots.[13] These nuanced studies trace how athletes, equipment manufacturers, governing federations, and fans engage in passionate debates to determine if new technologies are good for a particular sport or undermine its norms and traditions. Overall, studies of equipment generally focus on the reception of new technologies and how they might impact athletic performances or the integrity of the games. However, like other work in critical sports studies, they often say little about how the technologies themselves emerged in the first place.

Science and Technology Studies

Science and Technology Studies (STS) scholars have paid greater attention to the invention and technical development of sports technologies than researchers in other fields.[14] Most famously, Trevor Pinch and Wiebe Bijker depicted the evolution of high-wheel and "safety" bicycles while developing the framework and analytical approach to the social construction of technology (SCOT).[15] Changes to bicycle technology have been a frequent topic of interest ever since, ranging from economic studies of the late-nineteenth-century bicycle manufacturing bubble to sociological work on user-shaped innovation.[16] Like critical sports scholars, historians of technology have studied the introduction, reception, and impact of new technologies in various sports. They have explored, for example, how the introduction of snow-making equipment created new possibilities for alpine skiing; how the invention of synthetic fibers changed the apparel, gear, and behaviors of hikers and campers; and how high-speed ball-tracking cameras have both aided and complicated the efforts of referees to officiate soccer games.[17]

Among STS scholars, Rayvon Fouché has presented the most sophisticated analysis of what he terms the "the technoscientific revolution in sports." In *Game Changer*, Fouché suggests that there is no such thing as a "pure athletic performance." Rather, he argues that we must also consider technoscientific interventions when evaluating the winners and losers in modern sports competitions. Fouché pays close attention to athletes' bodies in his analyses of performance enhancing drugs, prostheses, and gender verification testing, while simultaneously accounting for the impact of high-tech swimsuits, shoes, bicycle frames, and other sports equipment. Drawing on the SCOT framework and actor-network theory, Fouché also explores the social negotiations that determine whether various innovations are integrated into their sports or rejected. Those negotiations, Fouché observes, occur among four sets of stakeholders: athletes; sports commissioners and governing federations; various "publics," including fans and the media; and "technoscientific actors."[18] *Inventing for Sports* extends the work of Fouché and other STS scholars by focusing specifically on those

"technoscientific actors," namely the inventors, designers, and engineers who develop and introduce innovations in sports. Moreover, the book situates stakeholder debates about sports technologies within a broader context of regulatory governance.

Regulatory Governance

Traditional studies of regulation have focused on the role of federal agencies that oversee financial institutions, workplaces, transportation, food and pharmaceuticals, consumer products, and the environment. Federal regulators exercise authority by determining market approvals, inspecting products and manufacturing processes, imposing fines for failing to meet standards, recalling products, or specifying the claims manufacturers can make.[19] Nevertheless, an agency's regulatory authority remains dependent on public confidence in its legitimacy. As Sheila Jasanoff, Daniel Carpenter, and other scholars have documented, federal regulators work closely with scientific, technical, and medical experts from academia and industry to establish credibility and maintain an aura of apolitical objectivity.[20] Yet, unlike pharmaceuticals, chemicals, or financial products, sports technologies are only rarely subject to national regulations.

Instead, cutting-edge sports technologies—as with early automobiles or contemporary artificial intelligence—often outpace the efforts of traditional government regulators to institute timely safeguards. Breakthrough technologies also cross regulatory boundaries, raising complex questions regarding the locus of oversight. For example, new synthetic wearables, often termed "smart clothing," use embedded sensors to record health data. Regulation of the chemicals in the synthetic fabrics falls under the purview of the Environmental Protection Agency. However, protection of personal health data falls under the aegis of the Department of Health and Human Services. Meanwhile, claims about the benefits of wearing smart clothing would fall under the Federal Trade Commission's Bureau of Consumer Protection. In reponse to uncertainty about which regulatory agency has oversight and what amount of testing would be required, the manufacturer may argue that users are assuming risks in a voluntary way, and that regulation is unnecessary.[21]

In the absence of "command and control" governmental oversight—and in part to preempt it—industry groups and trade associations develop their own quality and safety standards for emerging technologies.[22] Sports leagues and associations historically have focused on promoting their sports to increase participation and revenues. But with the emergence of disruptive sports technologies, leagues and international associations have found it necessary to act as regulators. This voluntary or associational model of self-regulation has been analyzed in a variety of contexts by political scientists, economists, and sociologists who have labeled it "neo-corporatist" for its web of relationships among industry, professions, and associations.[23] The approach offers efficiency and lower costs compared to traditional government-centered regulation, but also runs the risks of inconclusive decisions and a lack of transparency or clear enforcement power.

Sports associations work with multiple stakeholders to develop and police rules concerning the technologies that are allowed or prohibited in competition. Similarly, they establish certification schemes that manufacturers then fund when they pay the associations to review their products. Like government agencies, sports federations have aligned themselves with scientific and medical experts to establish their credibility and visibly demonstrate objectivity as they inspect sports equipment and determine which athletes are eligible to compete.[24]

When viewed holistically, we find that the governance of sports technologies is negotiated by a dynamic social process that is typically nimble and adaptive, especially when compared with traditional command-and-control governmental models. However, absent the transparency of government regulatory processes and legal recourse for manufacturers or consumers, the approach followed in sports runs the risk of prolonging uncertainty for athletes and fans. The case studies in this book explore and contextualize the often opaque and confusing regulatory processes encountered by sports technology inventors.

Inventing for Sports: Motivation, Inspiration, Impact, and Governance

Inventing for Sports focuses primarily on sports inventors, their motivations, and the often-surprising sources of their ideas. Moreover, the book documents what happens when an inventor introduces a new sports technology. It explores why game-changing innovations are adopted or rejected and traces their impacts both inside and outside the realm of sports.

Throughout this book, we refer to both *invention* and *innovation* to explain the emergence of new sports technologies and techniques. How do we define and distinguish between these two commonly used terms? To invent is to create a unique material, device, process, or method; to innovate is to meet an important customer or societal need. An invention is a way of doing something differently, while an innovation is a way of scaling up or commercializing one or more inventions in a way that creates value.

Who invents for sports, and why? Not surprisingly, many sports inventors are current or former athletes who envision new technologies for their own benefit. As such, these athlete-inventors engage in what economist Eric von Hippel has called "user innovation." In *The Sources of Innovation*, von Hippel argued that manufacturers get too much credit in the traditional linear view of innovation management and product development. In that framework, a firm observes an unfilled need in the marketplace and invests in research, development, and marketing of a new product to fill the need. However, when studying various sectors such as scientific instruments, tractors, and plastics, von Hippel found that experts operating on the cutting edge of their disciplines often became frustrated with the limitations of existing tools and products. To compensate, these sophisticated "lead users" hacked and adapted their equipment to improvise solutions that achieved better results. By demanding better performance from the products in their fields, lead users signaled emerging market needs to manufacturers. The manufacturers, in turn, often partnered with their lead users to convert improvised prototypes into next-generation commercial products.[25]

The sports industry has long been a fertile sector for user innovation because amateur and elite athletes demand superior performance from their gear. For example, in 1977, Joe Breeze and his friends invented some of the first mountain bikes when they modified vintage cruisers (nicknamed "clunkers") with sturdier brakes, wheels, and other components to withstand the bumpy mountain trails in Marin County, California. As the niche sport became more popular, the community of mountain bikers eventually grew into a recognizable market. In the early 1980s, Breeze's "Breezer 1" design became the prototype for the first mass-produced mountain bikes offered by vendors such as Gary Fisher, Specialized, and Univega.[26] User innovators attract significant attention and have

particular credibility in sports but, as the book's case studies document, not all sports inventors are athletes. Throughout the book, we highlight the multitude of pathways that people pursue to becoming a sports inventor.

Many of the inventors featured in this book have sought to commercialize their inventions as part of the $416 billion global sports industry. While financial gain is definitely an important motivation for sports inventors, we find that it is of secondary importance.[27] Beyond financial incentives then, what ultimately motivates athletes and engineers to invent for sports? What do they hope to achieve by developing new sports equipment and innovative techniques? As the prior examples suggest, frustration is a key motivator for innovators in general and athletes in particular. Athletes, we find, become frustrated—and invent solutions—when they lose a contest, sustain an injury, or endure a bad call. Inventors with disabilities and different body types also grow frustrated when they are unable to participate fully in their chosen sport, and become motivated to develop sport-specific apparel, prostheses, and wheelchairs. Having fun and trying something new are also universal but important motivators for many sports inventors. Accordingly, we argue that athletes and inventors are driven to develop sports technologies because of one of four primary motivations: (1) to improve performance and gain a competitive advantage, (2) to prevent injuries and enhance safety, (3) to aid officiating, increase measurement precision, and prevent cheating, or (4) to make sports more fun and inclusive for everyone.

While fun and frustration motivate sports inventors, their inspirations and technical solutions often emerge from contexts outside sports. For example, Chris Yu and Mark Cote drew on their backgrounds in aerospace engineering to develop the "Win Tunnel," used to test Specialized's bicycles (Chapter 2). Former NFL cornerback Shawn Springs founded the start-up Windpact to develop safer football helmets after airbags and children's car seats helped his family survive a harrowing automobile accident (Chapter 3). The computer scientist and cricket player Paul Hawkins was inspired to invent his Hawk-Eye ball-tracking system while calculating trajectories for a missile-defense system (Chapter 4). And Frank Nasworthy identified urethane wheels as a solution to challenges facing skateboarding (Chapter 5) not through a systematic study of materials, but through a serendipitous visit to a plastics factory. Andrew Hargadon has described this phenomenon of "recombinant innovation" in his book *How Breakthroughs Happen*. The most successful innovators, Hargadon suggests, are "technology brokers" who apply existing technologies to analogous use cases in new domains.[28]

What happens when an inventor introduces a new sports technology? How does the invention change the game and vice versa? Throughout the book, we find that the relationship between invention and sport is bi-directional and iterative. In many cases, the introduction of a new technology—such as blade prostheses in the Paralympics, for example—has a clear impact within the sport. Nearly every competitor adopts the technology, forcing league officials to either ban it or write new rules to accommodate it. In other instances, intensified competition drives technological change. For example, our human inability to discern time increments of less than one-tenth of a second motivated the invention of high-speed photo-finish cameras and fully automated timing.[29] In most of the cases explored here, there is a mutual, parallel, and iterative process of technological and social change, in which new rules, training regimens, and fan experiences are co-constructed alongside emerging sports inventions.[30]

Nevertheless, an innovation can be disruptive to a sport's norms and traditions, and it may stir passionate debates. Athletes, fans, and members of the media may ask, "Is the

new invention fair or foul? Should it be adopted or banned?" In tennis, for example, critics initially argued against the introduction of Howard Head's metal-frame rackets; they believed the oversized "sweet spot" made the game too easy when compared with traditional wooden rackets. However, with endorsements from influential professional players like Arthur Ashe and Pam Shriver, metal rackets were widely adopted in the 1970s before subsequently being replaced by rackets made of carbon fiber and composite graphite materials. In contrast, the swimming community rejected Speedo's high-tech LZR Racer swimsuit in 2009 after athletes who wore the suit broke an unprecedented number of records. The undue competitive advantage the suit conferred elicited charges of "technological doping" and threatened the sport's integrity. Other innovations are only partially adopted. For example, aluminum baseball bats and certain golf ball dimple designs are prohibited from professional use because they would necessitate constructing larger fields and courses, yet these innovations are seen as acceptable for recreational or youth league play. By examining the competing interests of players, sports federations, and fans, *Inventing for Sports* traces how concerns over safety, ethics, tradition, and economics influence why we accept or reject certain sports innovations.

Chapter Previews

At its core, this book highlights the stories of sports inventors and the motivations that inspire them to develop game-changing technologies (see Box 1). Athletes want to win, of course, and they are always looking for an advantage that will propel them to victory. In Chapter 2, "Competitive Edge," Meg Maher explores how breakthrough technologies and innovative techniques help athletes excel, and the diversity of backgrounds and ambitions that lead people to become inventors of sports gear. The chapter's case studies include Mike Schultz, whose Versa Foot and Moto Knee prostheses help para-athletes compete in a variety of sports; Jake Burton Carpenter, whose snowboard designs surpassed the earlier Snurfer in materials and performance; Arielle Rausin, who learned 3D-printing techniques to create a new wheelchair-racing glove; and Chris Yu and Mark Cote, who built a novel "Win Tunnel" to test and improve the aerodynamics of bicycle designs. Maher finds that athlete-inventors rely on and partner with experts in STEM-related fields. However, her chapter also demonstrates that not everyone who invents for competitive edge in sports has a background in STEM or industry. Common to the case studies is each inventor's ability to accurately identify an aspect of athletic performance that could be improved, to apply first-hand knowledge to develop a solution, and to iterate and collaborate as needed.

An athlete's body is her most important piece of sports "equipment." Playing sports and pushing the physical limits of our bodies carries the potential for injury, pain, and sometimes even death. In Chapter 3, "Health and Safety," Jeffrey L. Brodie examines how sports technologies help athletes enhance their strength and fitness, train more effectively, protect themselves from injuries, and recover more quickly from intense play and competition. The chapter's case studies include Shawn Springs, an ex-NFL player whose Crash Cloud technology formed the basis for a start-up company that is developing safer helmets; Robert Hubbard, whose family connections in auto racing inspired him to invent the HANS (head and neck support) device to reduce serious injuries and deaths from crashes; Arthur Ehrat, a farmer who tinkered with a variety of springs to invent the breakaway basketball rim; Dr. Robert Cade, whose team of scientists invented Gatorade to prevent dehydration among football players at the University of Florida; and Nathan Ramasarma,

BOX 1. *Inventing for Sports* case studies.

Chapter 2: Competitive Edge

Mike Schultz (Versa Foot and Moto Knee)
Jake Burton Carpenter (snowboard)
Arielle Rausin and Adam Bleakney (3D-printed wheelchair gloves)
Chris Yu and Mark Cote (Specialized Win Tunnel)

Chapter 3: Health and Safety

Shawn Springs (Windpact Crash Cloud helmet pads)
Robert Hubbard (HANS device)
Robert Cade (Gatorade)
Arthur Ehrat (Break-A-Way basketball rim)
Nathan Ramasarma (formsense smart apparel)

Chapter 4: Fairness and Accuracy

Bill Parkinson (swim-timing touchpad)
Doug DeAngelis (FinishLynx photo-finish system)
Tony Verna (instant replay)
Paul Hawkins (Hawk-Eye ball-tracking system)

Chapter 5: Fun for Everyone

Lisa Lindahl, Hinda Miller, and Polly Palmer Smith (Jogbra)
Howard Head (Head skis and tennis rackets)
Marilyn Hamilton (Quickie wheelchair)
Frank Nasworthy (urethane skateboard wheels)
Tahira Reid (double-Dutch jump-rope machine)

Chapter 6: Fair or Foul? The Governance of Sports Technology

Regulating full-body swimsuits
Regulating football helmets
Regulating prostheses in track and field
Regulating e-bike trail access

whose wearable formsense "smart apparel" helps people maintain better biometric form to prevent injuries. Brodie finds that inventors of safety equipment were strongly motivated by personal experiences, asked a wide diversity of questions that helped identify critical problems, and crossed disciplinary boundaries. They examined problems from different perspectives, transferred knowledge and expertise from one area to another, and worked in collaboration.

No one cheers for the referees, but most sports fans agree that athletic contests are only meaningful if they are officiated honestly, equitably, and accurately. In Chapter 4, "Fairness and Accuracy," Eric S. Hintz examines how numerous inventions help officials and leagues keep time, keep score, and review close calls to ensure fair play. The chapter's case studies include Bill Parkinson, whose underwater touchpad helped eliminate human errors in timing swim races; Doug DeAngelis, whose FinishLynx digital photo-finish

camera improved on film-based systems to offer fast, reliable race results; television producer Tony Verna, whose instant replay video technique revolutionized the way we watch and officiate sports; and Paul Hawkins, whose Hawk-Eye ball- and player-tracking system has aided officials in tennis, soccer, and a dozen other sports. Hintz observes that these officiating technologies supplement human perceptions, yet still require a significant degree of human interpretation to help officials make the right calls. Not surprisingly, league rules and officiating technologies often provoke spirited debates among athletes, coaches, and fans. Indeed, some critics argue against technological officiating; they believe that human imprecision, missed calls, and the occasional controversy make games more interesting and entertaining.

More than anything, both competitive and recreational athletes want to move, play, and have fun. Yet inventors often have to overcome physical and societal limitations to increase access and participation. In Chapter 5, "Fun for Everyone," Monica M. Smith explores the universal impulse to invent new sports, games, and gear. The chapter's case studies include Lisa Lindahl, Hinda Schreiber Miller, and Polly Palmer Smith, who invented the Jogbra, which enabled millions of women to participate in sports; Howard Head, who developed aluminum skis and tennis rackets; Marilyn Hamilton, who created the Quickie wheelchair, which enabled thousands of athletes with disabilities to play tennis, basketball, and other sports; Frank Nasworthy, who brought polyurethane wheels to skateboarding and revolutionized the sport; and Tahira Reid, who envisioned a machine to turn double-Dutch jump ropes in third grade and later built and patented the device as an undergraduate at Rensselaer Polytechnic Institute. Smith finds that these inventors offer insight into the playful creativity that is innate to all humans. While some inventors achieved success in the marketplace, others elected not to commercialize their inventions. These inventors, Smith concludes, were not motivated by profits, but rather the desire to make sports more fun and inclusive.

The introduction of many sports technologies, however, has also spurred vigorous debates and contested rulings by leagues, associations, and governments. In Chapter 6, "Fair or Foul? The Governance of Sports Technology," Arthur A. Daemmrich explores why some sports technologies are accepted as beneficial and fair, while others are rejected as unsafe or unfair. Daemmrich finds that calls for regulation grow when fans and officials perceive technology to overshadow athletes' "pure" human performance, when innovations challenge the heritage and traditions of a sport, or when injuries necessitate safety mandates. The chapter's case studies include regulatory disputes concerning the Speedo LZR Racer and other full-body swimsuits; football helmets and controversies regarding concussion protection; para-athletes who use the C-shaped lower-leg "blade" prosthesis and seek to compete against able-bodied athletes; and disputes between e-bike enthusiasts and traditional mountain bikers over off-road trail access. Daemmrich concludes that sports technology governance by leagues and international associations generally succeeds at balancing the interests of athletes, teams, fans, and inventors, but absent clearer procedural frameworks, regulatory decisions lack transparency and predictability that are important to the work of inventors.

Collectively, the book's case studies document the long history and contemporary relevance of innovation in sports. As inventors continue to push the technological envelope and a new class of athlete-investors funds a wave of sports-tech start-ups, the pace of sports innovation may well be accelerating. In Chapter 7, "The Past, Present, and Future of Sports Innovation," Arthur A. Daemmrich, Eric S. Hintz, and Jane Rogers survey current

trends and explore future motivations for sports inventors. They observe that the global COVID-19 pandemic forced the cancellation of many athletic contests, but also accelerated innovations in certain sports technologies. Going forward, they see opportunities for inventions in the realms of big data, unified and all-ability sports, and climate change adaptations. As always, athletes, fans, and the media will continue to debate whether new leading-edge sports technologies are safe, fair, and "good for the sport."

* * *

As you reflect on the stories in the pages that follow, we hope you, the reader, will appreciate the central role of invention and technology in sports. We believe you will gain a better understanding of the motivations, inspirations, and creativity that propel sports inventors to change their games. And by critically evaluating the stakeholders, norms, and traditions of various sports, you should develop a deeper understanding of the dynamic social and regulatory processes that determine whether new sports technologies are ultimately adopted or rejected.

Then, as you learn from a diverse array of sports inventors and come to understand their motivations, we hope you will begin to see your own potential to be a game changer. As the case studies throughout this book illustrate, every individual has the capacity to be inventive and to develop creative solutions—in sports and in other areas of everyday life. Everyone is inventive, and that includes you. As you read ahead, we hope you will be inspired to change *your* game, whatever that may be.

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Chapter 2

Competitive Edge

Meg Maher

Who doesn't want to win? This universal desire to be the best is a powerful motivation, especially in sports, where those on top of their game can reap significant rewards. In a realm where the slightest edge can mean the difference between the thrill of victory and the agony of defeat, athletes have increasingly drawn on the expertise of fellow athletes, coaches, trainers, doctors, scientists, and engineers to gain a competitive advantage. As a result, inventive thinking has helped bring about significant improvements in methods, strategies, and technologies designed to boost performance and outshine rivals.

This chapter explores the role of inventors in helping athletes enhance their performance, gain an edge on the competition, and push the boundaries of what is physically possible.

Four case studies are presented: Mike Schultz and his development of prostheses to compete in extreme sports; Jake Burton Carpenter and his transformation of the recreational Snurfer into a competition snowboard; Arielle Rausin and her application of 3D printing to make performance-enhancing wheelchair marathon race gloves; and Chris Yu and Mark Cote of Specialized Bicycle Components, who created a slower-speed wind tunnel to test and improve the aerodynamic performance of cyclists and their equipment. Collectively, these examples illustrate a range of individuals motivated by different circumstances who used a variety of inventive strategies to help themselves or other athletes gain a competitive advantage in sports.

Mike Schultz: Extreme Sports Prostheses

Competitive athletes possess an innate drive to improve and be their best, if not *the* best. They are invested in winning and eager to try new technologies and techniques to reach their highest potential. Mike Schultz is one such athlete and inventor. Born in 1981, Schultz grew up on a farm in Minnesota. As a young teen, he began racing in BMX competitions and soon switched to motocross racing. To improve his skills so he could compete, Schultz and his brother designed a motocross track on the farm, clearing trees, constructing the course, and even installing lighting so they could ride after dark.[1]

Schultz inherited his father's high motivation, work ethic, and propensity for tinkering. During these formative years, he learned how to weld, cut metal, and repair motors. Schultz recalled, "I grew up hanging out in the shop with my dad, trying to fix things, taking apart lawn mowers and putting them back together. Sometimes they worked, sometimes they didn't. He just let me figure things out on my own and kind of stood

back."[2] As a teen, he applied what he was learning in drafting class—understanding how machine parts fit together—to a job working in a metal fabrication shop.[3]

Schultz's early tinkering experiences fixing farm and construction equipment helped him maintain the motorcycles and snowmobiles he raced. In 2003, while still a teenager, he received sponsorships and began racing snowmobiles professionally. Schultz became known for his tenacity and aggressive riding style and embraced the nickname "Monster Mike." Going into the 2008–2009 season, Schultz was ranked fourth in the national championship series. At an AMSOIL Championship Snocross race held in December 2008, he was thrown from his snowmobile during a qualifying round. Schultz suffered a compound fracture in his left leg, severing the femoral artery and cutting off the blood supply to his lower leg. The injury was not repairable and was deemed life-threatening. With no other options, he agreed to an above-the-knee amputation.[4]

Schultz was determined to continue racing, but soon learned that the available prostheses did not have the flexion, tension, or durability needed for extreme sports. Rather than give up competing, he became inspired to invent. At an appointment with a prosthetist, he recalled thinking that he had "ridden mountain bikes as cross-training for years" and was "familiar with the suspension components." He began to imagine how he might deploy bicycle suspensions in a prosthesis to cushion the impacts his legs previously absorbed as he went over bumps on his snowmobile. "[I had] spent nearly 10 years tuning my own equipment," Schultz recalled, "learning how it works so I can better communicate with my mechanics and suspension techs to get the needed performance." He left the appointment "wondering if, by using a mountain bike shock, a leg could be fabricated that would permit more than 90 degrees of flex."[5]

After mulling over designs in his head, Schultz holed himself up in a spare bedroom of his house and began to sketch. "For hours at a time," he remembered, "I [sat] at my desk, drawing obsessively, calculating, erasing, and redrawing, to get all the angles right. I [had] 15 vertical inches to work with—the length of my leg from the knee to the floor."[6] Schultz considered what could provide the range of motion of a normal knee and would be small, lightweight, and adjustable. In his design, he incorporated a 7.5-inch bicycle shock set at 250 psi. (By way of comparison, a 180-pound mountain biker typically sets their front shocks in the range of 85 to 120 psi, depending on the manufacturer and their riding intensity.) With 2 inches of travel, the shock provided the correct amount of suspension. However, he had difficulty creating sufficient angular motion, which he calculated to be 130 degrees. He then designed the prosthesis with an open construction to reduce its weight but was concerned "whether debris [would] cause it to lock up, or if there [would] be too much free play between the rollers and track surface."[7]

Schultz next built models out of cardboard with thumbtacks for hinges to rotate the moving parts. After various iterations, he found that a simple track-rolling system worked best. After five weeks of sketching, and with insights gained through preliminary modeling, Schultz settled on the design, dimensions, parts, and method of assembly for his prosthesis. He realized it was "time to start cutting."[8]

It was at this stage of the process—creating a prototype—that Schultz needed to acquire new skills and expertise. Despite his mechanical aptitude and earlier job in a machine shop, he was not skilled at fabricating parts. Schultz connected with friends who, conveniently, worked for shocks manufacturer Fox Shox. One friend gave Schultz the mountain bike shock he needed for his prosthesis and access to a mill and lathe that the company used for research and development. Another friend taught Schultz how to

operate the equipment. After a week of fabricating parts, Schultz assembled his first prototype. The result was not pretty. Still, Schultz put the prosthetic knee on, stood up and flexed it, then took a test ride on his dirt bike. He was amazed at how well his "Moto Knee" performed. The shocks needed better calibration, Schultz recalled, but "the mechanics [felt] more natural than I could have imagined." [9]

Only five months after his life-threatening injury, Schultz began racing in adaptive motocross. However, at a qualifying event for the highly competitive X Games, he was racing "on this insane super cross track . . . with 90-foot jumps . . . pushing myself to the max." At one point, Schultz "fell out of the sky" and landed hard after a big jump. The standard walking foot that he used for everyday activities, which was attached to the Moto Knee, nearly snapped in two and "lost all of its strength."[10] Schultz hung on to place second, but realized if he wanted to compete at the X Games two months later, he would need a superior prosthetic foot to pair with his prototype knee.

Motivated by the success of his Moto Knee and his impressive return to extreme sports, Schultz embarked on developing a foot prosthesis for racing. He applied his drafting skills and mechanical know-how once again, this time adapting a smaller Fox mountain bike shock absorber to simulate the toe pressure and ankle tension needed for competition. Schultz had a prototype foot prosthesis like the Moto Knee, but with fewer parts and a simpler design, ready by early 2010. "[I] designed the foot to endure the rigors of racing," he said, and to be "customizable depending on which sport I'm doing. I can adjust the shock absorption, swap out sole plates, and configure ankle positions." Since the versatile prosthesis could be tuned to a variety of settings for use on dirt bikes and snowmobiles, Schultz called it the "Versa Foot." "Paired with my Moto Knee," he recalled, "I had a complete leg that [was] customizable to any race conditions" (Figure 2.1).[11]

Being able to adjust the resistance and size of the prosthetic foot helped Schultz improve his biomechanics and stability. He found that "having a smaller foot length [was] beneficial for finding my stance on the snowmobile's running board." This enabled him to "rotate my leg as I lean[ed] . . . in the corners to help counterbalance the machine." As Schultz raced his snowmobile over rough terrain and big jumps, the prosthesis compressed "while keeping the sole of the foot flat on the running board."[12] This allowed the center of Schultz's knee to shift forward over the foot so he could maintain his balance.

Working in tandem, the prostheses provided Schultz with the flexion, tension, extension, and rotation necessary for extreme sports. After officials approved the Moto Knee and Versa Foot for use in competitions, Schultz found that the inventions enhanced his performance. Whereas Schultz's competitors used conventional prostheses designed for walking, his equipment allowed him "to stand on my running board or the pegs of my dirt bike, giving me an advantage."[13] Consequently, at the 2010 X Games, Schultz placed first in adaptive snocross. At the following summer's X Games, he won gold in adaptive motocross.

With these victories, Schultz realized his inventions could help others. In 2010, he started the company BioDapt to make the Moto Knee and Versa Foot commercially available. Schultz also toured war zones in the Middle East in 2011. As he worked with veterans to help them return to rigorous physical activity, Schultz was introduced to snowboarding. He quickly picked up the sport. With minimal training, he placed fifth in adaptive snowboardcross at the 2012 X Games.[14]

A month after winning a gold medal at the 2014 Paralympics, Evan Strong wore the Versa Foot to victory at the national snowboarding championship. As a result, interest in

FIGURE 2.1. The Moto Knee and Versa Foot (shown at right beside Mike Schultz's prosthetic socket) each feature a mountain bike shock absorber, air spring, and patented linkage system. These components control the speed and resistance of the artificial joints' bending and extending during extreme sports. Credit: Hugh Talman, National Museum of American History, Smithsonian Institution.

BioDapt equipment began to pick up.[15] Strong and other athletes encouraged Schultz to train more seriously, and he became a member of the U.S. Paralympic snowboarding team in December 2014. Starting in 2015, Schultz suffered a string of injuries that limited his ability to train and compete. He returned to racing at the 2017 national snowboarding competition. There, he bested his rivals—some of whom also were using BioDapt prostheses—to win both boardercross and banked slalom events.[16]

FIGURE 2.2. Mike Schultz on his way to a gold medal at the 2018 Paralympics. Credit: Joe Kusumoto.

At the 2018 Paralympics, Schultz and eight teammates competed using BioDapt equipment (Figure 2.2). Schultz earned a gold medal in snowboardcross and a silver in banked slalom. By the 2022 Games, 26 Paralympians from 11 countries, fully one-third of all snowboarders, were using Schultz's products.[17] "It's extremely rewarding," Schultz told the *Wall Street Journal*. "I [had] a unique situation where I [was] an athlete, but I [was] also [the founder of] a company that [helped] other athletes." The 40-year-old Schultz's performance at the 2022 Paralympics—finishing in second place in snowboardcross and fifth in banked slalom behind rivals using prostheses he had made—was bittersweet. "It [could] be a little hard to swallow as an athlete, helping out my competition," Schultz reflected. "But for me, it [was] so much more about the bigger picture and what I [was] doing to help the sport overall. Winning medals [was] really cool. But when you [could] really create something that [helped] other people achieve their dreams, that [was] even deeper."[18]

Jake Burton Carpenter: The Snowboard

Not every sports innovator demonstrates the hallmarks of inventive thinking as clearly as "Monster Mike." Sometimes, ingenious ideas emerge from seemingly unlikely sources. Jake Burton Carpenter is a prime example. Born in 1954, Carpenter spent his childhood just outside New York City, where at an early age he showed little curiosity, problem-solving aptitude, or entrepreneurial initiative. His father recalled, "he wasn't the type of kid who set up lemonade stands."[19] Enrolled at the same boarding school his father and brother had attended, Carpenter developed a reputation as a slacker and troublemaker. "[I] got expelled [from boarding school] after two years," he recalled. "I was the proverbial

'underachiever' and wise ass. I did LSD with my friends and stayed up all night roughly twice a week."[20]

Carpenter's life would change course in 1968 when he first tried a "Snurfer," a wood laminate board ridden on snow. It had a rope tether attached to the front, which the rider used to steer and keep balance while standing sideways, as if on a surfboard.[21] The Snurfer, a portmanteau of "snow" and "surfer," was developed in the mid-1960s by Sherman Poppen, who had enlisted the bowling and billiards manufacturer Brunswick Corporation and later the JEM Corporation to handle production and marketing. To Poppen's chagrin, Brunswick decided to market the Snurfer as a child's toy rather than a piece of sports equipment.[22] Carpenter could not have agreed more with Poppen. "It just had so much soul right from the beginning," he reminisced years later. "We'd get high and go out on the local sledding hill and ride Snurfers straight down. It was a rodeo-type thing. Just fourteen or fifteen years old, bombing down and taking horrendous wipes."[23] While his thrill-seeking friends rode their Snurfers purely for fun, Carpenter saw the potential of snow surfing as a competitive sport. And looking at it from that perspective, he became increasingly frustrated by the Snurfer's poor performance. Meant to be ridden by a child, it was small, narrow, and had no bindings of any kind. These factors made it difficult for riders to go fast and stay balanced.

Life circumstances prevented Carpenter from immediately trying to improve the Snurfer. At the time, he had been sent to a "second chance" boarding school and was grieving his older brother, a Marine who had died serving in Vietnam, and his mother, who had succumbed to cancer. Their deaths were sobering, but ultimately constructive. Carpenter later reflected, "The losses made for two things: real independence and an ability to persevere."[24] He applied himself academically, graduated as high school valedictorian, and partnered with a friend to run a landscaping business over the summer. Carpenter then attended the University of Colorado with the intent of skiing for the school's NCAA championship team. When he failed to make the team, Carpenter transferred to New York University, became captain of the swim team and graduated with a degree in economics.

While working at a Wall Street investment bank in his first job out of college, Carpenter realized he had what it took to start a business. In helping entrepreneurs raise capital, he found, "the people I was dealing with didn't bowl me over with their capabilities. They seemed like normal people, and I was like, 'Shit, I could've done that.'"[25] Remembering his love of snow surfing and aspirations for it as a sport, Carpenter began to turn his inventive and business ideas into a reality. In December 1977, he moved from New York to Vermont. Using a small inheritance, Carpenter embarked on his self-described "get-rich-quick scheme" and started Burton Snowboards.[26]

Carpenter initially spent his days making prototypes and his evenings bartending for extra cash. Since snow surfing was barred from ski resorts at the time, he tested his designs on the back hills of southern Vermont. Carpenter experimented with a variety of shapes, woods, laminates, and bindings. He also tried different manufacturing processes, explaining, "I made prototypes from a furniture making angle (steam bent solid ash) to boat construction (fiberglass chop) to surfboard construction."[27] His father recalled his son's newly found focus, passion, determination, and perseverance: "Once he had the idea for this board in his head, he put every bit of his energy into it."[28]

Two years and one hundred prototypes later, Carpenter had created a snow surfer that through testing and tweaking addressed the deficiencies of the Snurfer. He called his invention the "Backhill" (Figure 2.3). The wood laminate board was six inches longer and an inch or so wider than a Snurfer to better distribute a rider's weight and increase stability.

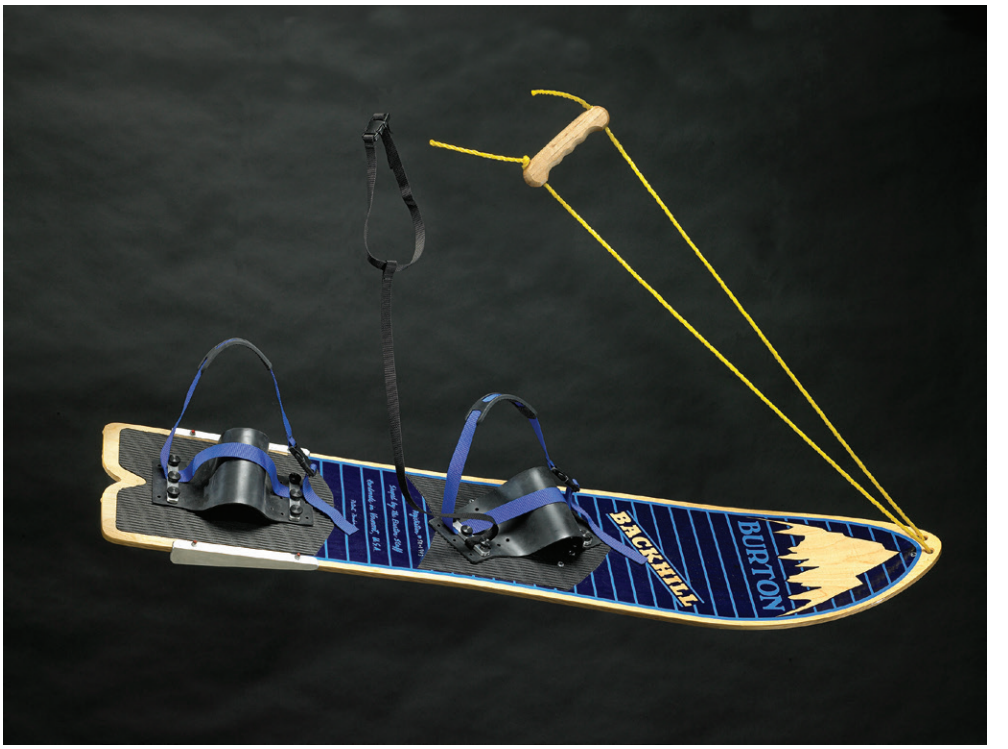

FIGURE 2.3. The Backhill was Jake Burton Carpenter's first commercial snowboard, with aluminum fins and waterski bindings mounted to a board wider than a Snurfer. Credit: Hugh Talman, National Museum of American History, Smithsonian Institution.

It had more flex, making the ride smoother and the board easier to maneuver. Carpenter added fins made of aircraft-grade aluminum alloy on either side of the tail. Functioning like a skeg on a boat, the fins improved stability, made it easier to track straight in the snow, and provided greater control in tight turns. On the top side, he also attached rubber foot straps that were soon replaced with adjustable water ski bindings.[29] The bindings enabled riders to stay on the board, a big challenge when hurtling downhill at 50 miles per hour. They also allowed riders to adjust their stance and efficiently transfer energy and movement to the board, helping them maintain control and minimize muscle fatigue.

Sales of the Backhill were negligible until early 1979, when Carpenter showed up in Michigan at his first official snow surfing competition. While most athletes planned to use Snurfers, he was among several competitors who wanted to race boards of their own design. Since the rules did not limit equipment to Snurfers, event organizers created an open division, which Carpenter won with his Backhill. Media coverage of the competition conveys the disruption caused by Carpenter and his invention. Although his time was reported by one newspaper as being within a second of all but one of the other winners, another reported that the JEM Corporation was "relieved that Carpenter's time on the Burton Board was seconds slower than all five winners in the Snurfer competition."[30] Riders saw through the spin, took note of the Backhill, and Carpenter had no problem selling his inventory of boards the following winter.

The creation of an open division accelerated the influx of better-designed boards in competition. It was a turning point that changed the trajectory of the sport. The Snurfer

soon fell out of favor for racing, while the designs of Burton and other fledgling snowboard manufacturers, such as SIMS and Barfoot, helped launch the sport we now know as snowboarding. Whereas the Snurfer remained essentially unchanged, Carpenter and other entrepreneurs continued to improve their board technology. Higher-performing snowboards enabled athletes not only to gain a competitive advantage, but also to push the boundaries of the sport.

While improving the equipment was crucial to evolving the sport of snowboarding, it only solved part of the problem. Carpenter spent the 1980s developing his company, improving his board technology, sponsoring athletes and competitions, and promoting the sport. Advocates, notably Carpenter and rival snowboard manufacturer Tom Sims, worked to convince ski resorts to open their slopes to snowboarders for recreation, training, and competition. Success was slow and hard won. In the early 1980s, communities of snow surfing and snowboarding enthusiasts began taking root near a handful of ski resorts. Ski Cooper and Berthoud Pass in Colorado, Soda Springs Ski Bowl in California, and Suicide Six and Snow Valley in Vermont held competitions. The tipping point came in 1985 when Carpenter persuaded the major Vermont resort Stratton to allow snowboarding on their slopes and to host the U.S. Open Snowboarding Championship. Meanwhile, the JEM Corporation had ceased production of Snurfers and the Midwest competition was canceled that year due to "an unusually low number of entries."[31]

By 1990, most ski resorts in the United States allowed snowboarders on the slopes, which, along with better performing equipment, enabled the sport to grow. As with skateboarding, early competitions mimicked alpine skiing events. However, as the board technology and other equipment improved, athletes developed tricks and techniques that led to the creation of terrain parks. Snowboarders learned from each other, and the competitions that Carpenter and other enthusiasts helped organize became places of invention. "The actual events themselves seemed to stimulate innovations," Sims recalled.[32] The competitions also helped stoke public interest, resulting in the inclusion of snowboarding at the 1998 Winter Olympics.

Since the early 1980s, Burton has been a leader in the snowboarding industry. The company has continued to innovate, developing boards with new and different materials, designs, and construction methods to meet the changing performance needs of competitors. In 1995, Burton collaborated with Olympic contender Shannon Dunn to develop a women's board. It wasn't the first board marketed to women athletes, but the Dolphin 44 pioneered the design of smaller, lighter snowboards that catered to the needs of women riders.[33]

Originally made from all-wood laminates, snowboards now include materials such as aluminum, fiberglass, carbon fiber, and Kevlar to optimize certain performance qualities.[34] Board shapes also vary to enhance performance. Those used for slalom events are long, stiff, and narrow, with a square back and nose for quickly maneuvering through course gates. Snowboards used for half-pipe, slopestyle, and big air are shorter, more flexible, and wider, with an upward-curving nose and tail to enable athletes to ride in both directions, execute tricks, and maintain stability. Not surprisingly, boards used for snowboard cross, a hybrid event, combine features of both alpine and freestyle designs.[35]

Carpenter has long been regarded as the "father of snowboarding." When he passed away in 2019, *Forbes* reported on his legacy and the success of his ingenuity, noting that Burton had a 32% share of the market and an estimated $400 million in revenues for 2018. The family-owned company was valued at approximately $700 million.[36] Although his early life was anything but auspicious, Carpenter made an astute and consequential

observation: "I always felt there was an opportunity for [the Snurfer] to be better marketed, for serious technology to be applied to it," he told *Sports Illustrated* in 1997, "so Snurfing could become a legitimate sport instead of a cheap toy."[37] Through motivation, focus, inventive thinking, and perseverance, Carpenter's pipe dream not only turned a Snurfer into a snowboard, but helped transform a recreational activity into a highly competitive Olympic sport.

Arielle Rausin: 3D-Printed Wheelchair Racing Gloves

Another seemingly unlikely inventor is wheelchair marathoner Arielle Rausin. Born in 1993, Rausin grew up in Florida and was not initially interested in sports. At age 10, a car accident left her paralyzed from the waist down and confined to a wheelchair. Rausin became even less enthusiastic about sports but was forced to take gym class in middle school. Luckily, a teacher devised ways for Rausin to participate, and her reluctance subsided. She reflected, "It was the first time I realized sports could be really fun even though I was in a wheelchair."[38] Rausin began to race in her standard wheelchair as part of the middle school cross-country team. Upon entering high school, she upgraded to a racing wheelchair. Offered a coveted spot on the University of Illinois's wheelchair track team, Rausin headed to Urbana-Champaign to take advantage of the university's superior coaching and facilities, which had recently been designated an official U.S. Paralympic training site.

Having decided to major in business, Rausin did not expect that a 3D-printing assignment for a "Digital Making" course in 2015 would become a transformative experience. She recalled, "I racked my brain for a while trying to come up with an object that was interesting and maybe held potential to be turned into something more."[39] At the time, wheelchair track-and-field athletes wore gloves purely for safety reasons, to protect against hand injuries and to minimize vibrations that can cause excessive muscle fatigue and nerve damage. Expensive, commercially available gloves made of leather and rubber provided inadequate protection and wore out quickly, however, so Rausin and other elite athletes spent hours painstakingly molding their own gloves by hand from melted plastic pellets. The resulting hard plastic gloves helped reduce injury and were more durable, but they were time-consuming to make and impossible to perfectly replicate.

Rausin discussed the 3D-printing assignment with her coach, Adam Bleakney, who suggested trying to print one of her pellet gloves. Neither had prior experience in design, engineering, or 3D printing, so they did not expect that the assignment would produce a useable printed glove. Nevertheless, Rausin gave it a try. She later recalled, "I brought a glove [to the visualization lab], and we spent the whole class time scanning [it]. The scan came out and then we cleaned it up on some software." Rausin uploaded this digital file to the 3D printer, which laid down thin strips of plastic, layer by layer, to manufacture an exact physical copy of the glove (Figure 2.4). She was surprised that the production process was so quick and efficient. Rausin had her doubts about the printed gloves, but began to wear them during training. "I kept waiting for something to go wrong, like the gloves weren't going to be strong enough, the material wasn't going to be good. But it worked so well."[40]

Rausin's 3D-printed gloves were easy to make, durable, cheap, and provided a custom fit like the handmade pellet gloves. Unexpectedly, she also found that they were ultralightweight, a result of the internal honeycomb lattice structure that 3D printing makes possible. "My glove made by hand is 174 grams. One printed glove is 70 grams. When

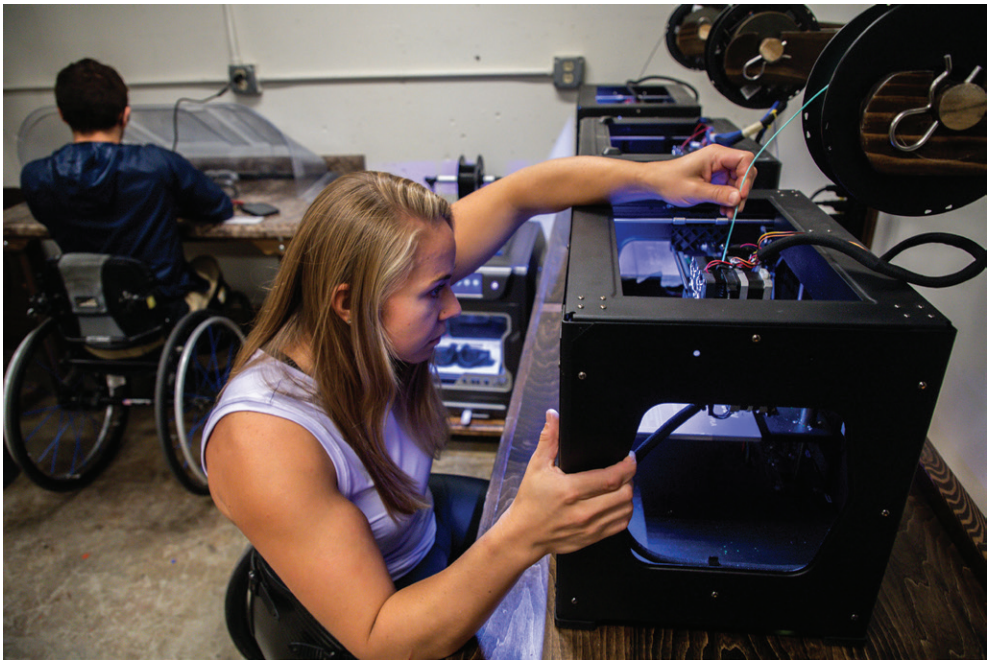

FIGURE 2.4. Arielle Rausin 3D-printing a lightweight yet durable wheelchair racing glove. Credit: *Chicago Tribune* / TCA.

you're pushing for 26 miles, that 100 grams shaved off could definitely make a difference." The lighter weight of the gloves significantly reduced the impact and fatigue to Rausin's arms and hands. In addition, the hardness of the printed plastic enabled her to grip and propel her wheelchair with greater force and go faster. Not too many inventors can boast of an initial prototype that works as envisioned, but Rausin raced in the 2015 Boston Marathon using the first gloves she printed (Figure 2.5). "After the first marathon I pushed using the gloves," she recalled, "I felt a little proud, but mostly relieved that both the gloves and I survived in one piece!"[41]

Word of Rausin's gloves spread quickly. By fall 2015, the enterprising Rausin received funding from the University of Illinois Student Senate for a new 3D printer and supplies.[42] Soon, she was filling requests to print gloves for her fellow athletes. Rausin recalled that "it felt good to help out my teammates, many of whom had taught me so much about wheelchair racing over the years. Making gloves for them felt like a nice way to return the favor."[43] Bleakney was quick to realize the significance of digital scanning and 3D printing, which enabled rapid reproduction of bespoke race gloves. "The ease of replication—that's the real key point to what Arielle is doing," he observed. "You could have a dozen gloves ready to go, with different pushing surfaces to address different environments that you have to train and compete in. And that doesn't exist now. We don't have that in our sport. So, this really, truly is groundbreaking."[44] Prior to Rausin's invention, wheelchair athletes did not have the luxury of owning multiple pairs of gloves suited to different racing conditions. As a result, according to Rausin, "some of the world's best athletes lose important races because they weren't prepared for the conditions, whether it was rain, humidity, or even snow."[45] Through 3D printing, it was possible for athletes to optimize their performance regardless of race-day weather.

FIGURE 2.5. Arielle Rausin competing in the 2015 Boston Marathon using her mitt-like 3D-printed prototype gloves. Credit: Ali Engin.

Rausin channeled her empathy for other wheelchair athletes and interest in 3D printing to help others in the sport. Despite minimal experience in design engineering or additive manufacturing, Rausin launched Ingenium Manufacturing in 2016 to commercialize her racing gloves. She collaborated with four-time Paralympic medalist Gyu Dae Kim and his coaches to further improve the glove design for elite athletes. Together, they developed a custom glove printed with a groove to guide athletes in pushing the rim of their wheelchairs.[46] This feature improved the efficiency of the athletes' stroke technique, enabling them to go faster. Rausin also devised a glove with a universal design that could be used by all racers.[47]

Creating her prototype and additional free gloves for her teammates was serendipitously trouble-free, but Rausin found that commercializing her invention was more complicated. "One of the biggest obstacles I faced was determining a design that was both able to accommodate different hand shapes and still feel like it was a good fit." Rausin struggled to find a middle ground between bespoke gloves and one-size-fits-all. Testing and tweaking with input from customers helped resolve the problem. "After about a year or so of orders we determined five unique sizes were enough," she recalled.[48] Like other inventors and entrepreneurs, Rausin by necessity had to acquire new skills. For example, she relied on YouTube videos to learn how to maintain the company's computers and 3D printers and troubleshoot their problems.

In giving back to athletes who had mentored her, Rausin recalled her own challenges in pursuing wheelchair racing. "When I first joined the University of Illinois team, I was very inexperienced and slow. It took five years of training every day to make my first national team and become a funded athlete, so I learned that good things take a lot of time and effort. There were times with racing and glove making that I wanted to give up. But

when I would take a breath, pause, and reflect on how far I came, it helped me see my envisioned future as more achievable."[49] Rausin derived tremendous satisfaction from making her invention available to others. "Seeing my friends win Paralympic medals using the gloves was a great feeling," she commented. "But at the Move United Junior Nationals [in July 2022], where at least 75% of the kids there were pushing with Ingenium gloves, that actually made me teary-eyed. The sport of wheelchair racing gave me so much. If my gloves can be a tiny part of that for another kid with big dreams, that means the world to me."[50]

Chris Yu and Mark Cote: The Low-Speed "Win" Tunnel

Athletes turned independent inventors and entrepreneurs generate many of the technologies that enhance performance. However, major sports equipment companies who wish to stay ahead in the marketplace are also significant sources of invention. Specialized Bicycle Components was founded in 1974 by Mike Sinyard and based in Morgan Hill, California. Sinyard coined the mantra "innovate or die" to stimulate improvements in his company's processes and products.[51]

To that end, in 2012 Specialized brought on graduate student Chris Yu, initially as an intern. Born in California in 1985, Yu had always been fascinated by airplanes and flight. As a seventh grader, he designed a wind tunnel that was displayed at a county fair.[52] Yu subsequently attended the California Institute of Technology (CalTech) and then Stanford University to study aerospace engineering. He soon applied his growing knowledge and skillset to sports. For example, Yu used a campus wind tunnel to study the aerodynamics of the controversial Adidas Jabulani ball used during the 2010 FIFA World Cup. He later used a water tunnel to design and optimize a sailboat keel for racing.[53]

Yu assumed his career would entail designing planes, but the internship at Specialized was ideal because it merged his passion for cycling with his academic background. The opportunity could not have been more timely. Scientific studies conducted in the decade prior found that almost 90% of elite cyclists' total power output was being used to overcome aerodynamic drag.[54] Consequently, Specialized eagerly sought creative solutions to this problem. Yu joined mechanical engineer Mark Cote in the company's efforts to improve cyclists' aerodynamics.

Born in Maine in 1985, Cote grew up participating in triathlons. In high school, he became interested in finding physical solutions to technical problems; he was particularly fascinated by design issues related to sports equipment. He studied mechanical engineering at the Massachusetts Institute of Technology (MIT), where he conducted cycling aerodynamics research using the university's Wright Brothers Wind Tunnel. After two internships at Specialized, Cote was hired full-time to help design engineers and product developers create more aerodynamic gear and apparel.

To improve and expand Specialized's portfolio of more streamlined products, Yu joined Cote in overseeing the wind tunnel testing underlying the firm's aerodynamic research and development. Although Specialized prioritized innovation, the company did not have its own wind tunnel testing facility. Instead, the two engineers rented time at commercial and institutional wind tunnels. This was problematic because these facilities were equipped with fans designed to simulate the air resistance experienced by vehicles moving at speeds considerably faster than bicycles. Yu, Cote, and the product developers at Specialized were frustrated that these tunnels—designed to develop and test cars, planes, and rockets—could not be adapted to accurately measure the drag forces that impact cyclists.

Specialized's ability to innovate was stymied by the cost and inefficiency associated with infrequent access to unsuitable rented testing facilities. "We would have to book time out six months in advance," Yu explained. "We would be constrained in the number of things that we would test. So we would have say 100 questions, and based on the three days we could book, we'd have to cut that number of questions down to thirty."[55] Given these limitations and the complexity in coordinating different tests at multiple facilities with various product teams, Yu and Cote realized the company was "starting to miss things."[56] The engineers determined that Specialized needed its own wind tunnel, and successfully pitched the idea to senior leadership.

In 2013, Yu and Cote led a project team to create a low-speed testing facility at the company's Morgan Hill headquarters. The facility would later be called the Win Tunnel. First, the team used a whiteboard to diagram what the facility needed to be able to do. From there, they created a design brief that outlined the tunnel's technical specifications. They listed essential features, plus some "nice to haves" that would enhance the testing experience. They also listed some unhelpful tunnel characteristics that would detract from the kinds of testing they wanted to perform.[57]

Must-have features of the tunnel included fans both capable of slower speeds and built in a configuration optimized to better regulate air flow; increased tunnel size to accommodate multiple athletes and simulate race-day conditions (e.g., the main pack of riders making up the peloton); a sensitive but robust force balance machine placed on a revolving platform capable of precisely measuring aerodynamic forces on equipment or athletes at any angle; and computer software that could better identify the areas of cyclists' bodies and their equipment that create the highest drag.[58] After soliciting input from racers with whom they conducted performance testing at rented tunnels, Yu and Cote recognized the need for better communication and feedback. They added an audio system to connect those in the tunnel with those in the control room, video and photo overlays to compare different positions and test configurations, and a means of projecting performance data in the testing area for athletes to assess and make adjustments in real time.[59]

Their wish list in hand, Yu and Cote consulted with their former aerospace engineering professors, including MIT's Alexander Slocum, whose expertise and mentorship helped them avoid expensive mistakes.[60] To avoid irreversible design errors, the engineers ran numerous computer simulations to determine which fan configurations pulled the air slowly but uniformly through the tunnel.[61] They also assembled an ad hoc team of in-house experts. Chris D'Alusio had previously built "aero" bikes streamlined for speed in road races. As a design engineer and product developer, he provided critical input on the proposed testing facility. Chuck Teixeira, a mechanical engineer who headed Specialized's machine shop, oversaw the design, construction, and installation of the force balance used to accurately measure the forces and torques that affect aerodynamics.[62] Alfredo Echauri coordinated the materials and construction logistics. Friends of Sinyard with experience overseeing large programs at NASA advised Echauri on project management.[63]

Specialized's inventive strategy, Yu explained, was "fast iterative design—basically learning and adjusting as we went."[64] Once he and Cote had ordered off-the-shelf electrical and control components, they divided the work of developing the software needed to operate the tunnel. As they coded the software, their teammates designed the custom force balance and began construction of the tunnel. Yu commented, "The number one lesson I learned from [constructing] the tunnel [was] progress over perfection."[65]

FIGURE 2.6. Using Specialized's Win Tunnel, Chris Yu evaluates ultradistance cyclist Christoph Strasser to improve his aerodynamics, mechanical power, and efficiency. Credit: © Lex Karelly.

Seventeen months in the making, Specialized's Win Tunnel—an industry first—was fully operational in 2013 (Figure 2.6). The company quickly reaped the benefits of unlimited access to a custom-built testing facility. "Now we can see right away if something works aerodynamically," Cote remarked. "Instead of assuming something is more aero, we can check the validity of a design immediately."[66] In 2015, Specialized debuted an aerodynamic suite of gear—bike frame, tires, race suit, shoes, and helmet—that the company claimed could save cyclists up to five minutes over a 40-kilometer (25-mile) race.

Yu described the testing and tweaking process of the "Five Minutes Faster" project as "death by a thousand cuts, or speed by a thousand gains." For example, the bike developers of the new Venge Integrated Aero Solution (ViAS) made minute modifications based on Yu's and Cote's aerodynamic testing and evaluation, tweaking various parts—the handlebars, the stem, the head tube—millimeter by millimeter. "Each one of those changes represented a gram or two of force improvement, so pretty small," Yu explained. "In the aggregate, there was a big ultimate gain in the entire frame system."[67] Specialized partnered with McLaren to test the new products using the Formula 1 racecar manufacturer's proprietary performance simulation software; the tests confirmed Specialized's findings.[68] Then, to dispel skepticism regarding Specialized's improbable performance claims, the company invited journalists to test their new aero products; the journalists again confirmed the significant performance benefits of the gear and came away especially impressed by the ViAS (Figure 2.7).[69]

Specialized has used the Win Tunnel not only to improve the performance of elite cyclists, hand cyclists, and triathletes, but also that of athletes in non-cycling sports such as alpine skiing, skeleton, and bobsled. Like those competing in cycling sports, these athletes have benefited from a comprehensive scientific analysis at the facility to assess their unique

FIGURE 2.7. Chris Yu adjusts the Venge ViAS prototype during wind tunnel testing. Credit: Specialized Bicycle Components, Inc.

physiological strengths and weaknesses, and to improve their biomechanics, techniques, race-day strategies, and equipment. For example, following the U.S. speed skating team's poor performance at the 2014 Olympics—blamed on the uniforms—sports apparel company Under Armour partnered with Specialized. Comprehensive testing in the Win Tunnel and direct input from the athletes led to the development of a more aerodynamic bodysuit.[70] Making the facility available for non-Specialized products and sponsored athletes has provided the company with additional income that can be reinvested in Specialized's own research and product development.

Specialized's investment in building the Win Tunnel has driven innovation and increased profits. "The learnings were and continue to be intense!" Cote commented. "Having a custom-built, cycling-specific speed tunnel right next to Specialized design and engineering headquarters was an amazing advantage that I believe has changed the shape of bikes today. Suddenly categories like 'aero bikes' were eclipsed by the entire line of bikes becoming aero!"[71] Although Yu and Cote assumed that other major bike brands would soon build their own tunnels, that has not been the case. Indeed, few sports companies could match Specialized's financial resources, expertise, and commitment to research and development. "If you don't have the entire back investment and capability to drive innovation and utilize the insights from something like the tunnel, then it doesn't make sense to do it," Yu explained. "There are very few brands that are at the intersection of that within the cycling industry, that are at that scale and have that investment in R & D."[72]

Specialized's testing facility remains an innovation and an industry outlier, and the company continues to benefit from its use. "How do you make the range of an e-bike longer? Or how do you make a commuter helmet quieter? These are things that we've

never spent tunnel time on in the past," Cote commented. "Where we would have had maybe five guys at a test [at a third-party wind tunnel], now we'll sometimes have maybe 50. Many of our designers and engineers come over and see with new eyes and bring new ideas—and those are the great ideas."[73]

Why Invent to Win?

Athletes are motivated by the desire to win. To achieve a competitive edge, they increasingly partner with scientists, engineers, and inventors, whose creative problem solving helps individuals and teams achieve their potential and gain recognition for their memorable performances. As demonstrated in the cases discussed here, athletes and inventors channel this desire and frustration with the status quo into inspiration and ingenuity, creating new and better technologies to enhance athletic performance. Their inventive strategies may vary. Many inventors, for example, borrow expertise from domains and technologies developed for other purposes to solve problems within sports. For some, like Mike Schultz, adapting existing technologies like bicycle shocks for new purposes is a winning strategy. For others, such as Arielle Rausin, applying a different material or new process like 3D printing first solves a problem, then proves to be a game changer. And for Jake Burton Carpenter, Chris Yu, and Mark Cote, creative rethinking of current products and technologies is the key to improving them.

With varying levels of scientific sophistication, sports inventors engage in a process of experimentation, testing and tweaking ideas, and learning from both successes and failures. Some inventors, like Jake Burton Carpenter and Mike Schultz, lack technical knowledge and need to spend time acquiring new skills to create a viable solution. In contrast, Chris Yu and Mark Cote were able to apply their extensive knowledge of aerospace engineering to competitive cycling and other sports. However, even with their training, they relied on former professors to help minimize mistakes, especially those that would be costly to correct. An outlier, Arielle Rausin was able to use her cursory knowledge of 3D printing to rapidly and easily create a prototype that was not only race-ready but performed significantly better than existing technology.

While this inventive activity may initially be intended to privilege certain athletes, those who innovate for sports can also be altruistic. At first, Mike Schultz's and Arielle Rausin's inventions improved their own athletic performance. However, driven by empathy for their fellow athletes, they quickly became motivated to start companies so that others could benefit from their creations. Of course, striving for profits and leadership in the marketplace provides immense pressure and incentive to innovate, as exemplified by the industry leaders Burton Snowboards and Specialized Bicycle Components. From their founding, both companies have invested heavily in research and development, not solely to help athletes succeed but to increase sales.

Why is such inventing necessary? For athletes, playing sports becomes a worthwhile and meaningful endeavor when competition is strong and individuals and teams perform to their highest potential. Especially for athletes in peak form—when the difference between winning and losing can be measured in hundredths of a second or mere millimeters—even a slight edge in competition can be crucial. New technologies and innovative thinking often provide athletes with a marginal but substantive performance benefit. As athletes approach their physical limits in many sports, invention—in all its varied forms—will enable them to extend those boundaries. With higher-performing gear and apparel, unconventional

techniques and strategies, advanced technologies and tailored regimens to expedite recovery and prevent injury, athletes will continue to triumph over their rivals, break records, and redefine human potential.

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Chapter 3

Health and Safety

Jeffrey L. Brodie

The human body is the most important piece of "equipment" in the playing of any sport. All athletes, whether elite professionals or more casual weekend warriors, challenge themselves to maximize their bodies' potential, push beyond their previous limits, and outperform the opposition. Playing sports offers fun, enjoyment, and enhanced health benefits, but all athletes risk serious injury when they push their bodies to their physical limits and expose themselves to the impacts and collisions common to many sports. Athletes suffer frustration and disappointment when their performance suffers due to fatigue, or injuries keep them off the field. In response, inventors, often working in collaboration with athletes, create technologies that help participants in both competitive and recreational sports maximize health and fitness, reduce the risk of injuries, and stay safe when playing their favorite games.

It is often presumed that inventors must possess specialized skills and knowledge to create new technologies and solve complex problems. A common assumption is that only those with an expert knowledge of the human body—orthopedists, biomechanical engineers, or physiologists, for example—can invent new health and safety technologies for sports. However, this chapter finds that motivation, passion, personal experience with the problem to be solved, and the ability to collaborate are just as important as engineering or medical expertise. As a motivation, there is nothing more universal than wanting to understand the human body, promote its well-being, and protect it from harm. Because the experiences of fatigue and injury are common, inventors of health and safety equipment come from a range of backgrounds. Some inventors are former athletes, while others traveled more unusual and serendipitous paths to invention. These unexpected inventors did not consciously decide to pursue invention as a career, but they had the determination to solve an important problem in their lives. Drawing upon personal experience and connections, they adapted ideas and technologies from disparate industries and applied them to sports. Collectively, the inventors of health and safety equipment challenge traditional stereotypes and expand our understanding of who can be an inventor.

Shawn Springs and Crash Cloud Technology

Shawn Springs demonstrates that the pathway to becoming an inventor is open to anyone with the drive and passion to improve the world around them. In 2011, following his career as an All-Pro cornerback in the National Football League (NFL), Springs founded Windpact to address perhaps the greatest contemporary invention challenge in sports—namely, protecting players from head injuries in American football.

Though immensely popular, football is exceptionally violent. The game is played by strong and imposing athletes moving at tremendous speeds; collisions, fractures, and concussions have long been regarded as inevitable and even celebrated in ways that retrospectively appear very cavalier.[1] Throughout its history, football has struggled with the physical risks to athletes, including paralysis and death, and looked to new safety rules and inventions to protect players. During the first decade of the 2000s, the medical study of concussions advanced substantially and the prevalence of brain injuries in football drew significant new attention. Following years of well-documented physical and mental decline, Hall of Fame center "Iron Mike" Webster died in 2002 from a heart attack. A post-mortem diagnosis clearly determined that he suffered from chronic traumatic encephalopathy (CTE) resulting from repeated concussions, which launched the issue into national consciousness.[2] The subsequent highly publicized deaths by suicide of well-known former NFL players, such as Junior Seau in 2012, further publicized the long-term effects of repeated concussions, including CTE, dementia, and depression. These repeated tragedies galvanized inventors, scientists, engineers, doctors, athletes, and, later, the National Football League, to search for solutions to protect players and save the game of football.[3]

Although Springs's career as a world-class professional athlete differentiates him from most other inventors, it was his personal experiences off the field that provided crucial additional motivations for action. Springs was born in Williamsburg, Virginia, in 1975, and later moved to the Washington, D.C., area. His father, Ron Springs, played football at Ohio State University and later professionally for the Dallas Cowboys. Like his father, Springs was a college football standout at Ohio State, and was drafted in 1997 by the Seattle Seahawks as the third overall pick in the NFL draft. Living in Seattle in the late 1990s would prove a formative experience, as the city was emerging as a hot spot of innovation where companies like Amazon, Starbucks, and Microsoft were disrupting traditional industries. Paul Allen, cofounder of Microsoft and owner of the Seattle Seahawks, served as an important mentor to Springs. When Springs asked Allen what it was like to earn a billion dollars, Allen advised Springs to think less about the money and more about doing something special to change the world. "That stuck with me," Springs said, "and I started to see the world differently at that point."[4]

Following his time in Seattle, Springs played for NFL teams in Washington, D.C., and New England. He completed a 13-year career in 2009 with 33 interceptions, but never stopped thinking about Paul Allen's challenge to make a larger impact beyond football. As a player, Springs was painfully aware of the injuries caused by the violent impacts in football; he believes he may have suffered as many as fifteen concussions (mostly undiagnosed) during his career. One collision knocked him unconscious, resulting in his hospitalization. Springs also saw firsthand in his father and other former players the long-term impacts of playing football, as their cognitive abilities deteriorated later in life. Springs began to wonder why the technology of his helmet was no different and no better than that of the helmet his father had used 20 years earlier. Surely, he believed, something could be done to improve helmet technology and performance. In 2011, Springs and cofounder Maurice Kelly assembled a team of scientists, designers, and engineers and founded the start-up company Windpact (Figure 3.1) to invent new technologies that would help prevent concussions.[5]

The origins of Windpact's Crash Cloud technology were rooted in a chance meeting with Ken Duffy, an executive at Safety 1st, a manufacturer of advanced car safety seats. Duffy gave Springs car seats for his young children, and Springs was impressed with the technology. Springs had always equated playing football to experiencing a series of car

FIGURE 3.1. Shawn Springs with helmets produced by Windpact. Credit: Ricky Carioti, Getty Images.

crashes and was already considering how to adapt technology from the automotive industry to improve football helmets. A life-changing event in 2012 convinced Springs that car seat technology could be modified to prevent brain injuries in football players. Driving home on Interstate 95 late at night with his three children in his SUV, Springs struck a stopped car in the middle of the highway. Though Springs's vehicle was mangled, its safety features and the Safety 1st car seats enabled Springs and his family to escape the accident with only minor injuries. Springs was shaken to the core by his experience and all that he could have lost. In his own words, he was "awestruck" by the ability of the car seats to protect his children and he was convinced that the same technology could also prevent head injuries in football.[6]

With support from Safety 1st, Springs's Windpact team of scientists, engineers, and designers began working to adapt car seat technology for use in football helmets. Like the Safety 1st car seats, the key innovation in Windpact's patented "Crash Cloud" technology (Figures 3.2 and 3.3) is its ability to combine energy absorption and dispersion in one system to mitigate impacts to the head and brain during a collision. Described as "an airbag for the brain,"[7] the padding and foam in the Crash Cloud absorbs energy from sudden impact. Simultaneously, a plastic air chamber sitting atop the foam padding compresses to absorb additional energy and disperses it by releasing air through tiny vents. Windpact next created Crash Clouds of differing sizes and shapes and arrayed them in a matrix inside the hard outer shell of the football helmet. The Crash Clouds decrease the amount of energy and force transferred to the brain during a collision, thus reducing the potential for concussion and brain injuries.

Just as Springs looked outside football to adapt automotive safety technology to help address football's biggest challenge, the success of Windpact's Crash Cloud technology fueled ideas for uses beyond football. Windpact, in Spring's vision, is not a helmet

FIGURE 3.2. The Crash Cloud acts as a miniature airbag for the brain during collisions. Credit: Kaarta Maron, Windpact.

technology company; it is a materials science company that seeks to reduce the impact of collisions in any aspect of daily life. The company has modified the size and shape of Crash Clouds to protect different parts of the body in a variety of sports, including in a helmet for men's hockey, a catcher's mask for women's softball, a face mask for baseball umpires, a helmet for girls' lacrosse, and chest protectors for baseball and softball players. Windpact also developed a proprietary database with the performance characteristics of a variety of foam and plastic materials and can 3D-print pads in a variety of matrices, changing the size, shape, and performance characteristics to suit different applications. In recent years, Windpact also developed products for a range of applications beyond sports, including military helmets, protective gear for first responders, and padding that can be worn by older adults susceptible to slips and falls. Coming full circle, Windpact's technology is also being used to improve automobile safety.[8]

As a professional football player, Springs enjoyed exceptional opportunities and advantages, including the chance to meet and network with influential people. But his path to invention was anything but easy. In addition to contending with obstacles common to aspiring inventors and entrepreneurs, Springs faced and still faces racial discrimination in attracting investors to fund his company. "As a young CEO and a minority," Springs has said, "I believe intelligence and diligence are colorblind. [But] I have been that guy in the boardroom who had to announce he is the CEO of a company [to be treated as an equal at the table]." These experiences have led Springs to adopt an expanded metric to measure the success of his enterprise. Beyond protecting people from injury and selling his invention, Springs also seeks to inspire others to be creative and inventive, especially minorities and

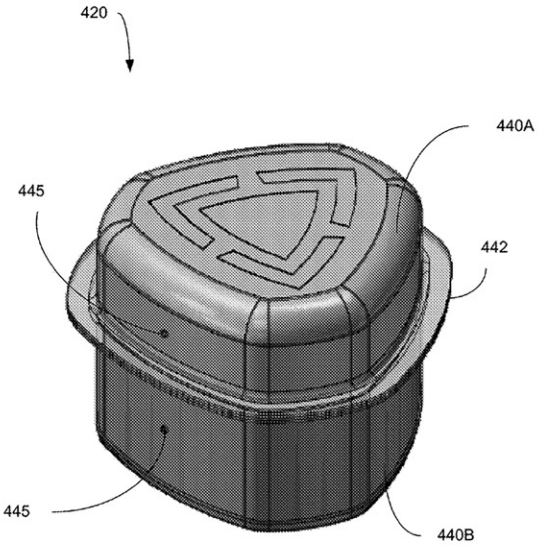

FIG. 4C

FIGURE 3.3. Crash Cloud patent. Credit: United States Patent and Trademark Office.

children from inner cities. Since the early 2020s, Springs has been working with national organizations including the United States Patent and Trademark Office, the Smithsonian Institution, Project Invent, and the Washington Commanders as a role model, public speaker, and mentor to increase the inclusion of minorities and women in invention.

Robert Hubbard and the Hans Device

Auto racing fuses athletic prowess, judgment, split-second decision making, and risk-taking with ever-advancing technology, materials science, and engineering to propel athletes and machines at speeds exceeding 240 mph. Since the first organized auto races in the 1890s, inventors have engaged in parallel efforts to engineer fast cars while also keeping drivers alive and safe. Generally, the technology to increase speed has outpaced the technology to protect the athlete, with drivers paying the ultimate price. Every generation of racing has suffered fatalities, including to some of its best-known superstars. Yet, to this day, auto racers accept this risk as they push themselves and their cars to the limit.

Despite many advances in safety technology, the 1970s and early 1980s were marred by deaths of auto racers who suffered head injuries in crashes, particularly when their cars

struck the racetrack wall headfirst. On 17 August 1980, auto racer James R. Downing was in first place during the Molson 1000 in Toronto when he crashed his Mazda RX-7.[9] Fortunately for Downing, his car turned backward before striking a concrete barrier at the base of a hill. The future driving champion felt lucky to have survived the crash and believed a head-on collision would have been fatal. The following year, on 9 July 1981, fellow driver Patrick Jacquemart died following a crash during a test session at the Mid-Ohio Sports Course. When the cause of death was determined to be a skull fracture, Downing wondered, "Why can't something be done about head injuries?"[10]

Downing posed his question to his brother-in-law, Dr. Robert Hubbard, a biomechanical engineer and professor at Michigan State University. A former track athlete, Hubbard earned his PhD in 1970 from the University of Illinois in theoretical and applied mechanics, focusing on the biomechanical properties of the skull. His research included working with the Highway Safety Research Institute at the University of Michigan to study methods for protecting the safety of automobile drivers and passengers. After graduation, Hubbard became a research engineer at General Motors Research Laboratories where he led the design and validation of the head model used for the Hybrid III line of crash test dummies. As an engineering professor at Michigan State, Hubbard taught a product development class focused on identifying problems, inventing solutions, and bringing products to market.[11] So when his brother-in-law asked him why auto racers were dying from skull injuries in crashes, Hubbard took the question seriously and, with Downing's help, set out to invent a solution.

Critically, Hubbard sought to understand precisely why drivers were dying from head injuries. In the 1970s, prevailing wisdom held that racing fatalities were the result of fate or some kind of fluke. But in fact, drivers were dying from basal skull fractures, a specific injury suffered in head-on collisions. "We didn't know that dying from basal skull fractures was a common way for drivers to be killed," Hubbard explained. "People weren't keeping track."[12] From his research, Hubbard understood that basal skull fractures occurred when forces unleashed on the body during crashes caused the head and neck to move in different directions from the body. When a race car struck the wall head-on, the body was restrained by the seat belt and harness, but momentum on the unrestrained head placed extreme tension on the tendons and ligaments holding the skull to the torso. The impact stretched and broke vertebrae in the spine, and/or severed the basal artery that provides blood to the brain, resulting in death.[13]

To invent a device that could prevent basal skull fractures, Hubbard needed to understand and measure the forces unleashed on the body during a head-on collision, as well as the amount of these forces the body could tolerate before injury or death occurred. So Hubbard developed new tests using crash test dummies and an array of sensors to quantify the forces the body absorbed during head-on impacts. He drew upon biomechanical testing conducted by the U.S. Air Force, the Transportation Safety Institute, the Society of Automotive Engineers, and other groups during the late 1970s, which provided critical data establishing the threshold of traumatic force the head and neck can tolerate in an impact before injury or death.

Hubbard reportedly created his first mockup of a safety device at his kitchen table in East Lansing, Michigan, using manila folders, scissors, duct tape, and glue. His vision was to create a wearable device that would restrain the head and neck relative to the torso during impact, even as it transferred and dissipated the impact forces to other parts of the body. In 1984, Hubbard began collaborating with Jim Downing to produce a prototype, with Jim

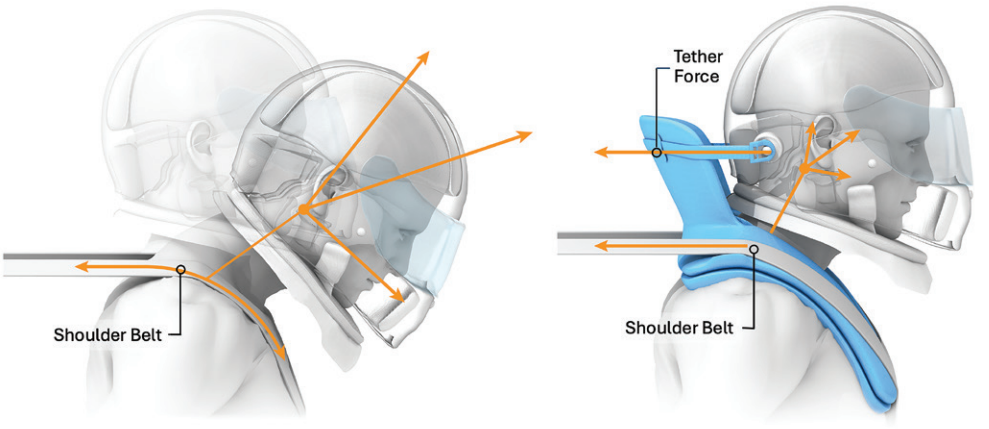

FIGURE 3.4. The HANS (head and neck support) device keeps the head and torso aligned during a crash. Credit: Artwork by Bryan Christie Design.

serving as the first human model for the new device. Hubbard made a plaster cast of Jim's body seated in a race car, then constructed a wood-block mold from the cast. In an era before 3D printing, Hubbard worked with machinists, woodworkers, and other technicians to create early prototypes. In the summer of 1985, Hubbard made an improved prototype out of fiberglass that featured a high collar with three tethers attached to a helmet. The one-piece yoke was connected across the chest by a breastplate; a strip of rubber ran over the shoulders of the yoke to prevent the device from slipping and moving during a crash.

Drawing on his experience as a champion race driver, Downing suggested adjustments to the size of the collar that resulted in a better fit and improved comfort without inhibiting performance. Other tweaks were the inclusion of carbon fiber to strengthen the device, and removal of the breastplate, which could ride up in a crash and injure the larynx. In the redesigned "HANS" (head and neck support) device, tethers held the head and neck in alignment with the body during a crash and transferred forces from the head and neck to the torso, which was supported by seat restraints (Figure 3.4).[14] By 1986, the HANS Model I had evolved beyond the early prototypes and Jim Downing wore it for the first time in competition. Hubbard was awarded patent 4,638,510 for his device in early 1987.

Hubbard and Downing believed that their invention protected racers from basal skull fractures, but could they prove it? In 1988, Hubbard founded Biomedical Design Inc. to continue iterating the HANS prototype while developing empirical tests that would validate its effectiveness. Sled tests "driven" by Hubbard's advanced crash test dummies recorded frontal, angled, and side impact forces as well as rotational torque forces on the neck.[15] The testing confirmed that the dummies wearing only helmets incurred forces well above fatal thresholds. In contrast, the HANS device reduced impact forces by more than 75%, a potentially life-saving reduction.[16] Downing and Hubbard jointly created HANS Performance Products in 1991 to manufacture and distribute their invention.[17]

Despite the impressive test results, drivers initially refused to use the HANS device. Entrenched cultural attitudes led many drivers to reject the invention, since using it might suggest they were afraid of the dangers in their sport. Other drivers argued that Hubbard should make the device smaller and provide more flexibility for drivers to turn their heads. Hubbard and Downing also identified the need for a one-size-fits-all model, partly to avoid

the expense and time of individual fitting and fabrication, but also to entice more drivers to use their invention.

Meanwhile, technological advances, particularly in traction control, encouraged drivers to race at even greater speeds, leading to disastrous crashes. During the 1990s, all major racing circuits—Formula 1 (F1), the Indy Racing League (IRL), the Championship Auto Racing Teams (CART), and National Association for Stock Car Auto Racing (NASCAR)—lost drivers to basal skull fractures, including the famous racers Ayrton Senna (F1), Greg Moore (CART), Scott Brayton (IRL), and Adam Petty (NASCAR).[18] Racing fans often witnessed these fatal crashes in real time at racetracks and broadcast on live television. To save their drivers and respond to mounting public criticism, the CART series mandated the use of head and neck support devices by all drivers in 1999.

But it was the death of NASCAR legend Dale Earnhardt that shocked the racing world into acceptance of the safety device. Known as "The Intimidator" for his ruthless and courageous driving style, Earnhardt steadfastly refused to wear the HANS device. On the final lap of the 2001 Daytona 500, Earnhardt lost control of his car and ran straight into the wall. Fox Sports commentator and racing champion Darrell Waltrip immediately feared the worst. "TV does not do that justice. That is an incredible impact," Waltrip gravely observed. "Those are the kind of accidents that are absolutely frightening." Tragically, Waltrip's fears were realized, and Earnhardt died from a basal skull fracture before a live viewing audience of more than 17 million people.[19] In October 2001, NASCAR mandated the use of head and neck support devices for all racers; Formula 1 followed suit in 2003.[20]

After their initial resistance, drivers adjusted to wearing the HANS device. Dale Earnhardt, Jr., a racing superstar following in his father's footsteps, noted, "when I first put it on it was a pain in the ass. It took a while to get used to but now I wouldn't think of running without it."[21] In the week following the accident at Daytona, HANS Performance Products sold more HANS devices than they had in the previous 10 years. Many drivers credit the HANS device for saving their lives; Jeff Gordon claims it saved his life twice.

Hubbard's initial motivation was to invent a safety solution that might save his brother-in-law's life. In 2010, the HANS device did just that, when Downing's race car flipped and the roof structure collapsed. Today, there are more than 275,000 HANS devices in use throughout the world and deaths from basal skull fractures have been eliminated from professional racing. "If you set out to solve a problem and the problem disappears," Hubbard observed, "then you have achieved your goal."[22]

Arthur Ehrat and the Break-A-Way Basketball Rim

In 1975, Randy Albrecht, the assistant basketball coach at St. Louis University, had a problem: when his players practiced the newly popularized slam dunk, they bent and damaged basketball rims. Even worse, they sometimes injured their wrists on the inflexible rim or sustained cuts from flying glass when the backboard shattered. Randy shared the problem with his uncle, Arthur Ehrat. "It just shouldn't be that way today," Ehrat remarked. "We ought to figure out some way to make that better."[23] Ehrat would soon apply his expertise as a farmer to solve a problem on the basketball court. This ability to transfer knowledge and technologies across different domains is a valuable skill and a hallmark of many successful inventors.[24]

Ehrat was not a basketball player, or even a fan. While growing up near Shobonier, Illinois, during the Great Depression, he learned various farming skills; he could milk

cows, bale hay, and operate and repair heavy equipment such as threshing machines. Later, Ehrat managed the grain elevator at Farmers Elevator Co. in Lowder, Illinois, for nearly 30 years. As Ehrat recalled, "if something broke, you fixed it . . . you couldn't go to town. We were doodlers [and] tinkerers." His nephew Randy appreciated his uncle's ingenuity and hoped Ehrat could help fix his problem with basketball rims.[25]

In 1967, the National Collegiate Athletic Association (NCAA) outlawed dunking in basketball. Many considered the dunk an "unskillful" shot and unsportsmanlike. Some critics cited safety concerns; they believed the ban protected athletes who might injure themselves while dunking. Others considered the ban an underhanded attempt to limit the dominance of taller, stronger players like Lew Alcindor (Kareem Abdul-Jabbar). But just a few years later, Julius "Dr. J" Erving—who played professionally for the American Basketball Association's Virginia Squires—reignited excitement about basketball with his high-flying slams. The NCAA reinstated the dunk to college basketball in 1976. Soon, the slam dunk became enormously popular and exciting for fans, but league officials needed a solution to protect players from injury and basketball hoops from destruction.

In studying the problem, Ehrat determined that the solution would involve a hinged rim that could fold down under pressure when the player slammed the ball, to prevent the backboard from shattering (Figures 3.5, 3.6). But the key was inventing a rim that would snap back to its original position following the dunk, allowing play to continue. Ehrat thought a simple spring might suffice to snap the rim back into place.

Starting with a flimsy $20 basketball rim, Ehrat began to build, test, and refine prototypes of a new invention. Ehrat tried door springs, car valve springs, and trunk springs bolted to two plates; one remained fastened to the backboard while the other flexed down under pressure. He eventually tried more than 20 different kinds of springs, but "none of them seemed too [very] strong." Ehrat's longtime friend Jim Beatty recalled, "we'd find the springs, and we'd cut the springs, and we'd grind the springs. And then we'd find out that wasn't going to work. There was a lot of trial and error." Ultimately, Ehrat drew upon his familiarity with maintaining heavy equipment and began experimenting with a larger and stronger spring from a John Deere field cultivator. "It had strength and quality," he said, "that's why I tried John Deere. I thought it would be the right size and diameter." The new spring worked, and Ehrat presented his nephew with a prototype. Randy Albrecht, in turn, shared prototypes of Ehrat's new basketball rim with his fellow coaches. According to Albrecht, "once the coaches realized that [Ehrat's rims] don't bend, players have less injuries, and the backboards don't break, it got to be a pretty popular idea."[26]

Ehrat took advantage when the NCAA hosted the 1978 Men's Final Four Basketball Championship in St. Louis. During the tournament, he conducted demonstrations showcases of his new "Break-A-Way" basketball rim for administrators and coaches, including legendary coaches John Wooden, Bobby Knight, and Dean Smith. More locally, Ehrat built somewhere between 30 to 40 prototypes of his new rim and gave them to high schools to start to market his invention.[27] Ehrat applied for a patent in 1976 but was rejected twice and had to defend his invention claim in court before finally receiving a patent in 1982.[28] Once his invention was secured, he sent letters to more than 60 sporting goods companies to solicit interest.[29]

Despite Ehrat's efforts, it would take several years before the Break-A-Way rim achieved widespread use. However, in the late 1970s and early 1980s college and professional players like Darryl "Dr. Dunk" Dawkins repeatedly broke basketball rims and

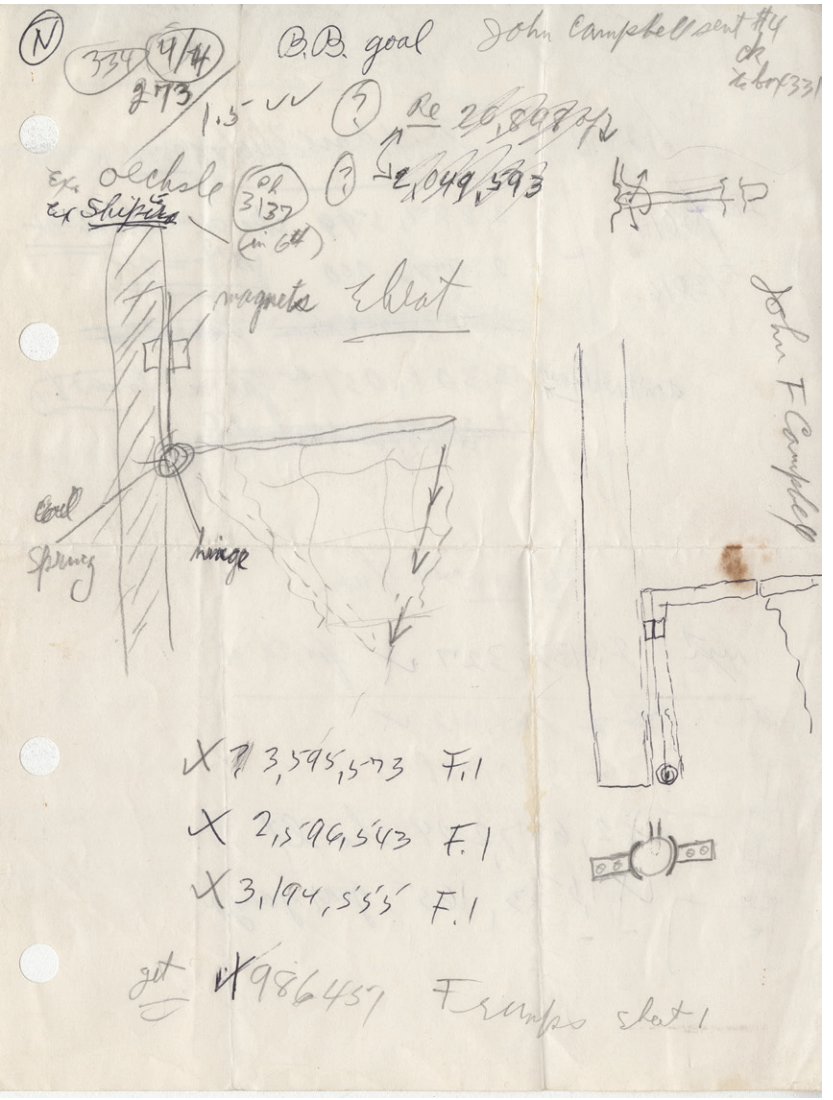

FIGURE 3.5. Arthur Ehrat's concept sketch for a collapsible basketball rim, circa 1975. Credit: Arthur Ehrat Papers, Archives Center, National Museum of American History, Smithsonian Institution.

shattered glass backboards with their dynamic power. This excited fans and players but led to injuries and long delays to replace the rim and backboard. After testing Ehrat's Break-A-Way rim and some other competitors' models, the National Basketball Association (NBA) eventually adopted Kenneth Mahoney's "Toss Back" device for the 1981–1982 season.[30] Despite that disappointment, Ehrat eventually signed licensing agreements that permitted Huffy, Lifetime, and several other sports equipment companies to manufacture and sell his collapsible rim design, which spread his invention worldwide.[31]

Ehrat's family and friends have provided additional insights on the unusual case of a farmer whose invention became crucial to basketball, even though he had only a passing interest in the sport. His daughter Sharon observed, "My father is an amazing man. He really is amazing and it's interesting for a normal person to do something that had such an

FIGURE 3.6. Arthur Ehrat with a "Break-A-Way" rim prototype, which he created using a John Deere cultivator spring. Credit: *Farmweek* photo.

impact in sports." His friend Jim Beatty added that "farmers, as a rule, they got to think outside the box. And he could really do that and had a lot of good ideas."[32] But Ehrat himself downplayed his role as an inventor who impacted the game, even suggesting that "the basketball thing was really a fluke."[33] "Honest to pieces, I know practically nothing about the damn game," he told the *Chicago Tribune* in 2005. "I pay attention to the dunk. That's the only thing I wait for."[34]

Dr. Robert Cade and Gatorade

The ability to apply your personal knowledge and experiences to new problems is a hallmark of creativity and a critical skill used by successful inventors. A chance encounter and an odd question set Dr. Robert Cade—a nephrologist who specialized in the

functioning of the human kidneys—on a new path to invent a solution to protect athletes from dehydration.

James Robert Cade was born in San Antonio, Texas, in 1927. After serving in the Navy, he attended the University of Texas (1948–1950), married Mary Strasburger in 1953, graduated from the University of Texas Southwestern Medical School in 1954, and joined the faculty of the University of Florida Medical School in 1961. There he established the university's Division of Renal Medicine and brought the first kidney dialysis machines to the state of Florida.

As a professor and teacher, Cade trained nephrologists to study the functioning and diseases of the kidney. Outside the laboratory, Cade was highly regarded as a Renaissance man of wide and varied interests. He restored antique Studebakers, grew and hybridized roses, and played the violin and viola in the local orchestra. To entice staff and students to come together to discuss kidney disease, he even used laboratory alcohol to invent new cocktails. "You just never knew what he was going to do next," said his long-time colleague and co-inventor, Dr. Dana Shires. To his daughter Phoebe, "he was an artist, and he viewed science as an art."[35]

In August 1965, University of Florida assistant football coach Dewayne Douglas (a former UF linebacker) was also employed as a security officer at the university's Shands Hospital, where he often had coffee and lunch with Dr. Cade and his colleagues.[36] Douglas reportedly asked them an unexpected question: Why don't players wee-wee [urinate] after a game?[37] Douglas explained that the Florida heat and humidity was taking its toll on the football team. Some players lost as much as 18 pounds during a single practice and more than 20 players were sent to the infirmary for heat exhaustion and dehydration during one notorious weekend.[38] Reflecting on the extremely hot and grueling practices, wide receiver Gene Peek dryly observed that "there was always the threat of dying."[39] Douglas said, "we gotta do something" and looked to Cade and his team for help.[40]

As a nephrologist, Cade focused on the functioning of the human kidneys and how they clean the blood of toxins and waste and remove them from the body as urine. Now Cade faced a different kind of problem. In hot and humid weather, athletes were sweating so much in the attempt to keep cool that they were suffering from dehydration. Many players simply drank water or took salt pills to replace the fluids, electrolytes, and sugars lost through sweat, but these conventional methods were sometimes ineffective and even hazardous to the athlete's health. Drinking too much water too quickly could cause body cramps and limit the athlete's performance; taking too much salt could lead to high blood pressure, heart disease, and stroke. As a result, many football coaches of that era stressed that "tough" players did not ask for water breaks; indeed, playing for as long as possible without a water break was considered proof of a player's toughness. Despite the resistance of players and coaches, Cade believed there must be some kind of drink that could prevent dehydration in athletes. With his colleagues Dr. Dana Shires, Dr. James Free, and Dr. Alex de Quesada, Cade was determined to invent something that would enable athletes to "get the most out of what God gave [them]."[41]

At the time, it was well-understood that sweat was composed of water and salts such as potassium, magnesium, and sodium. Losing too much of these salts through perspiration shifted the chemical balance in the body and disrupted normal muscle and nerve function, resulting in headaches, cramps, dizziness, and heatstroke. However, the specific composition and amount of each substance lost through sweat was unknown. A successful rehydration invention would have to replace the specific amounts of fluids, salts, and sugars lost by

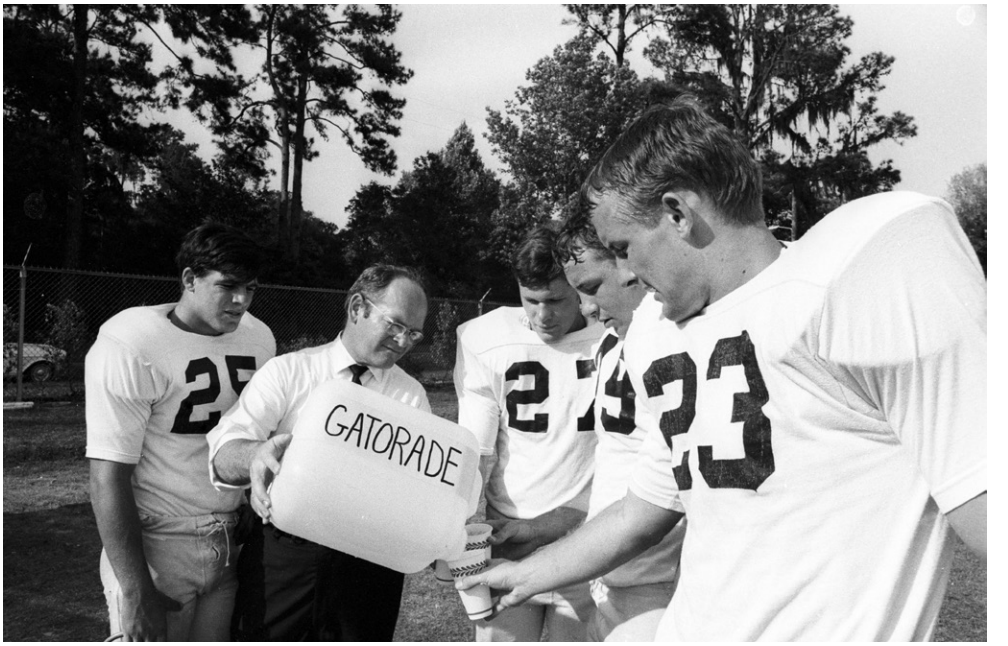

FIGURE 3.7. Dr. Robert Cade serving Gatorade to football players during practice, 1968. Credit: Lynn Pelham via Getty Images.

the body through sweat. It would also need to be quickly and easily absorbed by the body without impeding athletic performance. To create the precise solution for this complex problem, Cade and his team transformed the football field into a laboratory and initiated a study to understand dehydration among athletes.

Starting with the freshmen team—head football coach Ray Graves initially refused to let his varsity team be used as "guinea pigs"—Cade and his team designed a research study and began collecting data. They analyzed blood and urine samples taken before and after practice and determined the volume of sweat lost through perspiration by weighing athletes pre- and post-practice. Players wore rubber gloves up to their elbows to collect sweat for analysis. Doctors even injected blue dye into the players to measure their blood volume. This field testing enabled the research team to understand how much of each electrolyte the players were losing through sweat and how that loss affected the chemistry of their blood.[42]

Cade knew that glucose, a simple sugar, was easily absorbed into the body through the intestines and could carry water with it. He hypothesized that a glucose solution offered the best potential to replace electrolytes (salts), prevent dehydration, and improve athletic performance. Cade and his team began mixing solutions of water, electrolytes, and glucose, and within six weeks began giving it to the freshmen football players.[43] The players, immersed in a culture of toughness, were initially reluctant to try Cade's solution. Their skepticism was exacerbated by the drink's horrible taste. The players variously described Cade's solution as putrid, horrific, "sweat in a bottle," toilet bowl cleaner, and bad enough "to choke a maggot." Cade's wife Mary came to the rescue when she suggested adding freshly squeezed lemons to the mix, leading to a more palatable drink (Figure 3.7).[44]

While Cade and his team continued testing and collecting data from the players, a different and perhaps more important test took place on the football field. On 1 October

1965, Florida's freshmen team played in the "Toilet Bowl," a game against Florida's "B" team. Being older, more experienced, and more physically developed than the freshmen, the B team was always favored in these matchups. But after jumping ahead in the first half, the B team—which drank only water during the game—withered in the heat and humidity during the second half. In contrast, the freshmen drank the new magical elixir, replaced their lost electrolytes without cramping, and as Dr. Dana Shires put it, "whipped [the B-team's] ass."[45]

Now convinced, Coach Graves allowed Cade to test his formula with the varsity team. Because the solution was designed to help the University of Florida "Gators" football team, Dr. Jim Free suggested "Gatorade" as the name for their new invention. News of Gatorade soon spread widely, especially as the University of Florida football team improved and rival teams wanted to know what the Gators were drinking on the sideline. The mystique reached a crescendo in 1967 when the Gators beat Georgia Tech to win their first Orange Bowl. After the game, Georgia Tech Head Coach Bobby Dodd told Coach Graves that "we didn't have Gatorade. That made a difference."[46]

Much has been written about Gatorade's commercial success. Initially licensed to Stokely-Van Camp in 1967, Gatorade created a new market—sports drinks. By 2020, Gatorade held a 70% market share, with sales topping $6 billion in the United States.[47] The statistics are astounding, but they do not speak to Cade's passion and purpose as an inventor. Gene Peek, a Gator wide receiver in 1967, recalls that "Dr. Cade was looking at a much bigger picture."[48] Even as he was studying the players of the Florida football team, he was thinking about how he could apply what he learned to other life-threatening problems.

Indeed, Cade set out to help the hundreds of babies admitted to Shands Hospital who were suffering and sometimes dying from dehydration. At the time, there was no safe and effective oral treatment for dehydrated infants. Cade prescribed the same formula he gave to the football team and opened new avenues of research to treat pediatric dehydration. Co-inventor Alex de Quesada is most proud of Gatorade's impact off the field. "Babies used to die because of severe diarrhea and vomiting. There was little they could do. With the use of Gatorade, that problem was solved," he said. "We never claimed that Gatorade is a medication. But it can be used to hydrate people who need hydration for medical reasons. That is one of the great satisfactions I have, . . . how many lives it has saved."[49]

Cade's passion and curiosity about the world, combined with his unassuming nature, are a reminder that anyone can create game-changing inventions. On the occasion of Gatorade's fiftieth anniversary in 2015, Cade's former colleagues and family reflected on his legacy: "What Bob Cade did for the university was make everyone else realize that someone working quietly on something interesting and relevant to them could make a difference," said Dr. Nikolaus Gravenstein, the Jerome H. Modell, MD, professor of anesthesiology in the UF College of Medicine. "You could be in any discipline and make a difference."[50] Or, as daughter Phoebe Cade Miles stated, "You never know when that small idea is going to take wings and literally change the course of history."[51]

Nathan Ramasarma and Smart Apparel

Many inventors get their start when they encounter a problem that needs a solution. Nathan Ramasarma took the opposite approach. "I had no clue about what I was going to do," he freely confesses, "but I wanted to do something that would touch a lot of people's

lives, and that was my core." Ramasarma was driven by a desire to invent something with the potential to improve the quality of life for everyone worldwide. He referred to this bold aspiration as his "invention moonshot."[52]

As an electrical engineering student in India, Ramasarma studied personal area networks and his work attracted the attention of researchers at the MIT Media Lab. Coming from a middle-class family, Ramasarma did not have the financial resources to pursue his research in graduate school. However, with help from his contacts at MIT, he received a scholarship offer to study electrical engineering at Virginia Tech from Charles W. Bostian, regarded as the "father of satellite communication." At the time, Ramasarma had never traveled by airplane, but he soon found himself in the United States studying wireless communications with top engineers. After earning his master's degree in 2002, Ramasarma moved to San Diego and began working at Qualcomm, an industry leader in semiconductors, software, and wireless technology.

During nearly a decade at Qualcomm, Ramasarma witnessed a revolution as smartphones enabled mobile communications and wireless data transfers. Ramasarma contributed to the development of smart watches, smart glasses, and "media flow," which enabled smartphones to stream live content. Ramasarma recognized the growing potential for technologies worn directly on the body to change the quality of life. He left Qualcomm in 2014 to start his own company, create his own products, and seek a "moonshot" that would have global impact.[53]

Initially, Ramasarma struggled to identify a problem common to billions of people worldwide, but achy knees soon pointed him in a productive direction. Like many recreational runners in their late 30s, Ramasarma was developing debilitating pain in his knees. Though he tried physical therapy, rest, support tape, braces, and even acupuncture, nothing helped. As an engineer, Ramasarma wanted to discover the root cause of his problem; he contemplated that his knee pain was tied to how his body moved as an integrated, complex system. Because he lacked supervised coaching, Ramasarma had never learned the proper body form to run long distances without injuring his body. Though Ramasarma was experiencing pain in his legs, he understood that he needed to evaluate, measure, and correct the kinesthetic functioning of his entire body to heal and prevent future injury.

Moreover, if proper biometric form could prevent pain and injury in athletes, Ramasarma reasoned that maintaining proper form would be useful for everyone, everywhere, no matter what activity they were performing. Ramasarma had identified his "moonshot" proposal: to help people worldwide measure and maintain proper biometric form to prevent injuries and live better lives. But what kind of new technology would enable people worldwide to record and measure proper "form"?[54]

Ramasarma drew inspiration from automotive technology. "The human body is much like a car," he observed. "Each is a complex system with thousands of parts." A malfunction in one part of the system has cascading effects and can cause dysfunction elsewhere. But he also observed that, unlike an automobile, the human body does not have as many signals or alarms when something is wrong. "If you have low pressure in your tires," Ramasarma explained, "you have the signal from a sensor that will tell you that. If you are low on gas, there is a sensor that measures your gas tank. There are multiple sensors gathering information that your car needs to know and then communicate to you through a dashboard. But unfortunately, our body, even though it's much more complex than a car, [has] only one signal—pain. At that point, it's too late."[55]

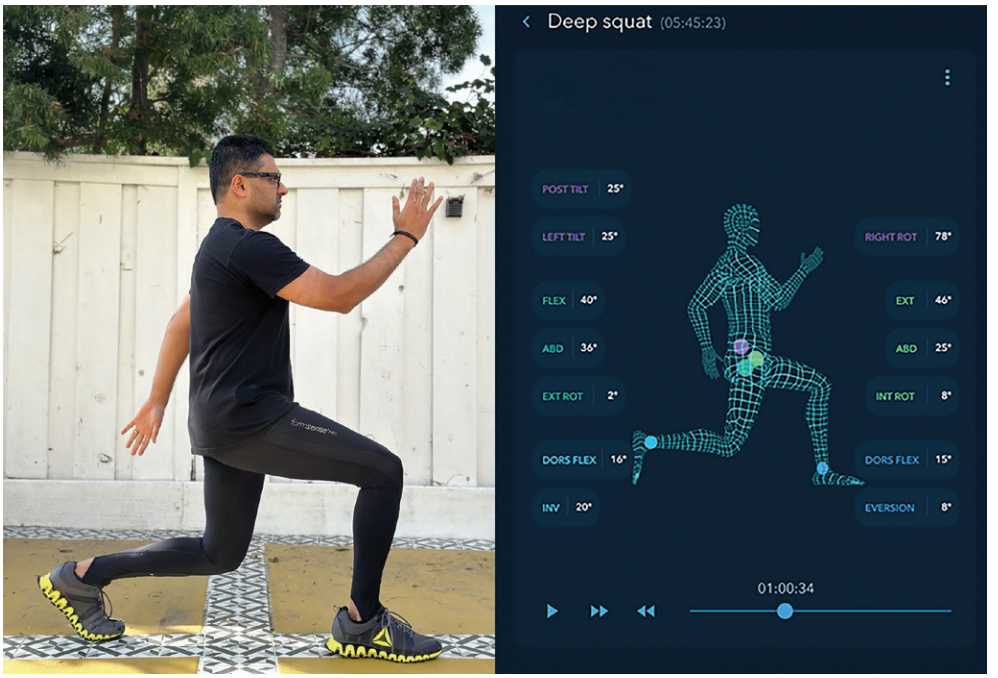

FIGURE 3.8. The formsense Smart Legging creates a real-time virtual avatar of the user with data overlays. Credit: Nathan Ramasarma, formsense.

Ramasarma set out to develop high-precision sensors capable of measuring an individual's form in three dimensions. He had previously developed a sensor that could be attached to skateboards and bicycles to analyze tricks; he modified this to be worn directly on the human body. A network of the sensors could be combined to collect data and create a virtual avatar of the user's body in motion performing different activities (Figure 3.8). Once an individual's form was recorded and analyzed, it could be compared against reference datasets of optimal form for various exercises or sports, as established by exercise physiologists and biomechanical engineers. By detecting areas of stress or weakness in the body, or any deviations in movement from established norms, Ramasarma's network of sensors would enable the user to correct improper form and prevent future injuries.

Ramasarma named his new start-up formsense. Intentionally spelled with a lowercase "f," formsense's first design was a ring embedded with the new sensor, worn on the finger. Early prototypes affirmed the effectiveness of the sensors to measure body movement, specifically motor control and dexterity of the fingers. Though not useful for analyzing full body movement, the formsense rings were able to help identify the slight hand tremors that emerge during the early stages of neurological conditions such as Parkinson's disease.[56] Further advancements reduced the size of the sensors, enabling formsense to attach them to straps worn on different parts of the body. But more than 20 straps and sensors were required to record the full movement of the body, and requiring anyone to work or play while wearing that many straps was not realistic. So Ramasarma and his team challenged themselves: "Why not try to figure out if those sensors could be miniaturized so they can be woven into swatches of fabric?"[57]

In 2019, formsense was able to miniaturize its sensors and integrate them into a comfortable base layer of shirts and leggings. The new "smart garments" were designed

to be comfortable, sturdy, functional, washable, and easy to use, so that users would record and analyze their form consistently over time. Only then could the invention provide the continual feedback users needed to maintain proper form, especially for those unable to benefit from the direct observation of coaches, trainers, and therapists. In mid-2022, Ramasarma launched Tune Smartwear as a direct-to-consumer brand backed by formsense technology that would help users record data and improve their individual bodies and experiences. Regular use of the formsense smart garment, Ramasarma noted, allows users to draw unique personal insights from their own form that "are only relevant to them."[58]

Ramasarma does not know if his smart garment will prove to be the ultimate solution to the problem of "form" or whether it will have the worldwide impact he seeks. But it is a challenge he eagerly accepts. "To solve a problem for one person, that is fine. But to identify a problem and [create] a product that can help whether you are a 15-year-old kid or a 75-year-old person . . . that is the real 'moonshot.'"[59]

Inventing for Health, Safety, and Beyond

The desire to protect athletes' health and safety is an incredibly powerful motivation for invention. We play sports to have fun, to enhance our physical and mental health, to test ourselves against competitors, and to explore the boundaries of what our bodies can do. But playing sports and pushing our physical limits carries the risk of injury, pain, and sometimes even death. Despite the high stakes, the inventors described in this chapter and throughout the book offer an encouraging and optimistic future. Their case studies offer the key insight that anyone can employ to identify problems and invent creative solutions.

First, the inventors who create technologies to protect the body represent a diverse array of knowledge and experiences. Certainly, Dr. Robert Cade's training in nephrology and Dr. Robert Hubbard's background in biomechanical engineering were instrumental to their solving complex medical challenges. But deep subject matter expertise and advanced professional training are not required for successful invention. Shawn Springs, Arthur Ehrat, and Nathan Ramasarma drew on their personal experiences and applied their knowledge from fields beyond sports and medicine to invent safety equipment for athletes.

Second, the inventors in this chapter asked and listened to important questions—even seemingly silly ones—to identify critical problems. Springs asked himself why he and his family survived their automobile crash unscathed when the car itself was destroyed. When a University of Florida assistant football coach asked, "Why don't players urinate during a game?" Cade took the question very seriously. When race car driver Jim Downing asked rhetorically, "Why can't something be done about head injuries?" his brother-in-law Hubbard began searching for an answer. Ramasarma asked himself, "Why do my knees hurt?" and then set about investigating proper running form. Each of these questions prompted a reexamination of the status quo and sparked new inventions.

Third, the invention stories in this chapter highlight the importance of exploring problems from different perspectives and transferring knowledge and expertise from one area to another.[60] Ehrat knew nothing about basketball but a great deal about farm equipment. He used that knowledge to invent a breakaway rim to protect basketball players worldwide. Springs instinctively equated football collisions to car crashes and understood that the safety technology in car seats could help prevent concussions in football. He then

adapted Windpact's Crash Cloud technology to help prevent injuries among military personnel, construction workers, and older adults. Ramasarma applied his knowledge of personal area networks to create an array of mobile, wearable sensors that help athletes measure, track, and maintain proper form to prevent injuries. Now, that same formsense technology is being adapted to monitor patients recovering from strokes and living with Parkinson's disease. Dr. Robert Cade transferred his scientific laboratory procedures to the football field. Although he invented Gatorade to help athletes improve their performance, he subsequently adapted the sports drink to treat infants suffering from dehydration.

Finally, and most importantly, the examples in this chapter debunk the myth of the lone inventor. No inventor works alone. As these vignettes demonstrate, collaboration is essential to successful invention. Asking questions, exchanging ideas, debating strategies, and sharing information are inherent to the invention process, because no single person possesses all the ideas, knowledge, and skills required to invent something new. In sports, and in every other context, invention thrives as a team sport.

Innovations in health and safety equipment provide ample reasons for optimism, but there are still major obstacles to overcome. The ability to protect athletes sits in constant tension with forces that continue to expose athletes to injury. In many cases, cultural traditions and gender stereotypes of masculine "toughness" can inhibit athletes from using safety technologies, lest they be perceived as afraid or weak. Likewise, athletes' legitimate concerns about privacy and improper use of biometric data could make them reluctant to use helpful smart garments and other wearable fitness technologies. But the greatest tension preventing the adoption of safety technologies is the threat to overall performance. Inventions that protect an athlete's body can also limit their mobility, vision, and comfort. And to maintain a competitive edge, athletes may forego their own safety when huge salaries and prize money are on the line.[61] Finally, despite the tremendous advances in safety technologies, no technology-based solution can provide athletes with complete protection against death or serious injury. There will always be safety risks when competitors push themselves to their limits, or beyond.

Despite myriad technological advances, there are always more opportunities to change the game, in sports and in other aspects of everyday life. Fortunately, anyone can practice an inventor's most important skills. Anyone can ask good questions, collaborate with others, and draw on their personal experiences to identify and solve important problems. So, what will you invent? How will you change *your* game?

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61. Perceived threats to a sport's culture and traditions are one of many reasons why some sports inventions are rejected. For an analysis of these tensions, see Chapter 6.

Chapter 4

Fairness and Accuracy

Eric S. Hintz

No one cheers for the referees, but most sports fans agree that athletic contests are only meaningful if they are officiated honestly, equitably, and accurately. This chapter examines how inventors help sports officials keep time, keep score, and ensure accurate results.

We often take officiating technologies and protocols for granted, but they are fundamental to most sports. Game clocks and scoreboards define the duration of contests while tallying the points. Stopwatches and automated timing systems—in the pool, on the track, and on the ski slopes—measure an athlete's performance to the millisecond and help determine the winner. Photo-finish cameras, slow-motion instant replays, and ball-tracking systems enhance human perceptions and help officials make the right calls.[1]

What motivates inventors to develop improved officiating technologies? The inventors featured here—Bill Parkinson, Doug DeAngelis, Tony Verna, and Paul Hawkins—were frustrated by bad calls, officiating errors, and the limitations of existing technologies. They channeled their frustrations into developing solutions that could improve the fairness and accuracy of officiating. However, their innovations sometimes introduced unintended ironies and uncertainties. Notably, "automated" officiating technologies still require significant human intervention, which introduces new—if less visible or traceable—sources of error. Moreover, sports leagues have tended to deploy new officiating technologies in somewhat arbitrary and uneven ways. Indeed, some sports federations have decided that perfect officiating is unnecessary and even unwelcome, since the occasional missed call or controversy tends to make the games more interesting and entertaining. Together, these cases suggest that there are no objective standards for fairness and accuracy in technological officiating—they are always negotiated.

Bill Parkinson, ChampionChip, and Fully Automatic Timekeeping

Timekeeping is fundamental to most modern sports. Clocks define the duration of many athletic contests; a WNBA basketball game, for example, is made up of four 10-minute quarters. Timekeeping is also essential in certain contests—like bobsled or alpine skiing—where only one competitor is on the course at a time. To determine the winner in these events, the officials must record and sort each individual's time. Timekeeping is also crucial for long-term record keeping. This is how we know that Usain

FIGURE 4.1. Judges at the finish line, Stockholm Olympics, 1912. Credit: Erik Bergvall, ed., *The Official Report of the Olympic Games of Stockholm 1912* (Stockholm: Wahlström & Widstrand, 1913), plate 120.

Bolt—clocked at 9.58 seconds in the 100-meter dash in 2009—is the fastest sprinter ever, 0.62 seconds faster than Jesse Owens, who was timed at 10.2 seconds in 1936.[2]

For most of human history there was no timekeeping at sporting events; determining the order of finish was an official's primary responsibility. Through the 1960s, even in elite Olympic sports, it was common to see a group of officials congregating near the finish line to visually determine the order of finish during sprints, swim races, and cycling events (Figure 4.1). As cheap, mass-produced mechanical stopwatches became widely available in the 1850s (Figure 4.2), officials began reporting the racers' finishing times as a supplement to the order of finish. According to historian Allen Guttmann, this "numeration of achievements" was a key characteristic that differentiated modern sports from their earlier function as religious and political rituals.[3]

But almost immediately after its invention, observers understood that the stopwatch's greatest limitation was its human operator. When using a handheld stopwatch, a sports official can introduce reaction-time errors at both the start and end of the race. To account for differences in individual reaction times—and differences in the precision of individual stopwatches—sports federations started to require multiple handheld time observations for the same athlete. In Olympic swimming, for example, three timers with handheld stopwatches were assigned to each lane. The official result was determined by reporting the average of the three times. Swim timing in most youth leagues still follows this approach.[4]

At the turn of the twentieth century, cognitive psychologists conducted stimulus-response experiments that quantified human reaction time at approximately one-tenth of a second (0.1 sec).[5] Yet, the winners and losers in elite competitions are sometimes separated by mere hundredths of a second, which is beyond the limits of human perception and well within the margin of error for human reaction time. A notable controversy ensued after a razor-close finish between Australia's John Devitt and the USA's Lance Larson in the

FIGURE 4.2. The horse-racing craze of the 1850s inspired the American Watch Company to invent the first mass-produced mechanical stopwatch. Credit: Dane Penland, National Museum of American History, Smithsonian Institution.

men's 100 m freestyle swimming final at the 1960 Rome Olympics. The three first-place judges were split, with two favoring Devitt, and one favoring Larson. The second-place judges saw the opposite; two believed Devitt placed second, while one believed Larson finished second. Thus, the six finishing judges were split three–three on the winner. The stopwatch timers recorded times of 55.0, 55.1, and 55.1 seconds for Larson and 55.2, 55.2, 55.2 seconds for Devitt—an apparent victory for Larson. However, the chief judge ruled Devitt the gold medalist by virtue of the two-to-one margin among the first-place finishing judges. Swim officials lamented the difficulties in visually judging the order of finish given the splashing and underwater touches. The controversial result spurred calls for reform and greater automation in timekeeping.[6]

Enter inventor Bill Parkinson, a physics professor at the University of Michigan and occasional timer at the Wolverines' collegiate swim meets. In 1954, Parkinson began experimenting with an automated swim-timing system that eliminated human timers entirely. The system had an electronic starting gun that automatically and simultaneously started separate electronic clocks assigned to each lane; this eliminated human reaction-time errors at the start. Parkinson also designed an underwater pressure pad that could automatically stop the lane clock when a swimmer touched the pad.

Inventing the touchpad presented several interesting design challenges. First, electricity and water do not mix; Parkinson had to engineer a waterproof pad that would not electrocute swimmers. Second, a pad resting even a few inches underwater could be triggered by the water pressure itself. So how would a pad distinguish a swimmer's touch from the hydrostatic pressure? Finally, in a multi-lap race, how would the touchpad distinguish between a mid-race flip turn and the final touch to stop the clock at the end of the race?[7]

FIGURE 4.3. A swim-timing touchpad from Colorado Time Systems. Credit: Jaclyn Nash, National Museum of American History, Smithsonian Institution.

With help from his wife, Martha, Parkinson sewed copper wire in a zigzag pattern, mounted it a few millimeters from an aluminum plate, and then sealed everything inside a waterproof rubber mat. When a finishing swimmer pressed the mat, the copper wires would touch the aluminum plate, complete an electric circuit, and stop the clock. Parkinson also filled the pad with non-conductive silicone oil. This neutralized the hydrostatic water pressure and kept it from inadvertently closing the circuit, while still offering minimal resistance to a hand touch. Finally, to prevent an accidental mid-race clock stoppage, Parkinson rigged up a hinge that allowed officials to drop the touchpad into the water just before the finish. Parkinson secured U.S. patents 3,230,325 (1966) and 3,302,007 (1967) for his inventions.[8]

During the late 1960s and early 1970s, firms including Omega (Switzerland), Seiko (Japan), and Colorado Time Systems (USA) introduced commercial timing systems and swim touchpads inspired by Parkinson's designs (Figure 4.3). In 1969, swimming's governing body, the Fédération Internationale de Natation (FINA), eliminated finish judges and adopted fully automatic touchpad timing to determine the order of finish.[9]

As Parkinson's story suggests, the specific requirements of different race environments—such as a racetrack or swimming pool—can inspire new timekeeping innovations. For large races run on city streets, including 5Ks, 10Ks, marathons, and triathlons, the timing challenge was a problem of scale, not speed. With tens of thousands of competitors in larger races, it was difficult for officials to manually collect and report so many finish times. Staggered starts presented another challenge. Several minutes would elapse before racers at the back of the pack even crossed the starting line, making it difficult to report a racer's true finish time with any precision. Given these uncertainties, most participants recorded their own unofficial race times by clicking their chronograph wristwatches as they crossed the start and finish lines.[10]

In 1993, a Dutch start-up company called ChampionChip set out to automate these manual timekeeping processes for the Zevenheuvelenloop (Seven Hills Run), a local road race in Nijmegen, Netherlands, that annually attracts 25,000 participants. ChampionChip deployed a wireless radio-frequency identification (RFID) system composed of thousands of individual transponders, one assigned to each racer, along with receiver antennae positioned at the starting line, finish line, and various "split" locations along

FIGURE 4.4. ChampionChip RFID timing chip worn by Patrick Walsh during the 2010 Boston Marathon. Credit: Jaclyn Nash, National Museum of American History, Smithsonian Institution.

the racecourse. Each racer affixed an RFID timing chip to their shoelaces (Figure 4.4). When a racer passed an antenna, the central race computer recorded a timestamp for that specific timing chip. After the race, the organizers collected the RFID chips to use again. Meanwhile, by subtracting the precise elapsed time when each racer crossed the starting line, the computer automatically calculated each racer's net finish and split times and quickly sorted the race results. On 25 September 1994, the Berlin Marathon was the first major race event to deploy the commercial ChampionChip system. In 2008, ChampionChip merged with another Dutch firm, AMB i.t., that had similarly used radio transponders to time remote-control car races. The merged company, called MyLaps Sports Timing, now embeds the tiny RFID transponder chips directly into each racer's bib.[11]

Doug DeAngelis and the Digital Photo Finish

Accurately capturing the finish is the holy grail of fully automatic timing. Since the 1880s, inventors have deployed photography to help determine the winner. "I venture to predict," legendary photographer Eadweard Muybridge told *Nature* in 1882, "in the near future that no [horse] race of any importance will be undertaken without the assistance of photography to determine the winner of what might otherwise be a so-called 'dead heat.'"[12]

Muybridge was an expert in equine photography; his series of stop-motion photographs definitively proved that all four of a horse's legs left the ground when at a full

gallop, resolving a long-standing debate. His prediction was prescient. By 1889, an article in *Outing* magazine noted that Ernest Marx, a racetrack photographer from Plainfield, New Jersey, "has been successful in determining the winner in close or 'dead heats' of racing contests."[13] Over the next several decades, photographers and inventors such as John Hemment, William Barber, and Gustavus T. Kirby advanced the art and science of "instantaneous photography" to determine the order of finish in horse races and other contests.[14] More than a century after Muybridge's prediction, a competitive runner and engineer named Doug DeAngelis made another significant advance when he introduced the FinishLynx digital photo-finish camera and timing system.

DeAngelis grew up in the small town of Orrington, Maine, where he was a four-year member of Brewer High School's track and cross-country teams. At the University of Maine, DeAngelis studied electrical engineering, ran the middle distances for the Division I track team, and captained the cross-country team in his senior year. After graduating in 1988, DeAngelis moved to Boston and took a job with the engineering firm Honeywell Bull, where he worked with the firm's proprietary computer operating system and set up a local intra-office network. In the evenings, DeAngelis attended graduate courses at the Massachusetts Institute of Technology (MIT). On the weekends, he ran competitively for a local track club, the Greater Lowell Road Runners. At one competition, the meet director—a fellow University of Maine alumnus named Steve Vaitones—asked DeAngelis to help him operate an Accutrack photo-finish camera to time an important race.[15]

In 1974, inventors from the Specialty Instruments Corporation of Grand Prairie, Texas, had secured a U.S. patent for an improved photo-finish camera. Their "Accutrack" system deployed quick-developing Polaroid film and was used widely at track-and-field events through the early 1990s. Instead of a traditional shutter, the Accutrack camera featured a narrow aperture that was aligned with the finish line. Using a technique known as "strip photography," the camera advanced a strip of Polaroid film horizontally behind the vertical slit at approximately the same speed as the racers crossing the finish line. In addition, the Accutrack imprinted a time stamp directly on the film every tenth of a second as it passed across the aperture. The result was a continuous time-stamped image of the finish line captured over a period of a few seconds.[16]

According to media historian Jonathan Finn, the slit camera marked a paradigmatic shift in how we conceptualize, capture, and interpret the photo finish. A traditional photographic camera, Finn writes, "captures a distinct space in a moment in time," whereas "a slit camera captures the passing of time through a small, fixed space."[17] Thus, a photo-finish image from a slit camera looks vaguely like a standard photographic image, but it must be interpreted differently. In photo-finish images from slit cameras, writes Maarten Vanvolsem, the finish line is "everywhere in the image. What changes when one looks from left to right and from right to left is time, not space."[18]

DeAngelis became an accidental expert in race-timing equipment as Vaitones and other meet directors continued recruiting him to operate the cumbersome, unreliable Accutrack. The system used a combination of a spring-powered motor, cables, and pulleys to advance a 3 × 5-inch strip of Polaroid film across the narrow aperture. When the mechanism jammed, the film failed to advance, leaving officials without a photo or finishing times. Moreover, DeAngelis observed, the Accutrack "had a short piece of film, and when it ran out, you got no more times." This was no problem for a sprint, when the finishers were tightly clustered, but a major problem for longer distances. In one 5,000-meter race "there were 14 guys that went under 14 minutes," DeAngelis told

SportsTravel magazine, "but I only got the top five guys on film, and being a distance runner, I thought that was really sad." Over the next several months, DeAngelis grew increasingly frustrated with loading and unloading the film for every race, clearing jams, and interpreting the Accutrack's poor-quality images.[19]

In 1990, at age 24, DeAngelis sketched out a two-page idea for what he called a "digital Accutrack." Instead of capturing finish-line images on Polaroid film, his concept would capture *digital* images of the finish line, time-stamp them, and store them on a computer. The computer would then immediately reassemble the time-stamped images on a high-resolution screen. By moving the cursor along the photo-finish image to the position of each finisher, an official could automatically record each runner's race time and placement in a matter of seconds.[20]

In the spring of 1991, DeAngelis enrolled in Eric von Hippel's *New Enterprises* graduate course at MIT and developed a business plan to commercialize his idea. Von Hippel had recently developed his theory of "user innovation," and DeAngelis proved to be an archetypal "lead user" whose frustration with the Accutrack motivated him to develop a superior solution.[21] He changed the product's name from "FinishLink" to "FinishLynx" and featured the fast cat in his branding. DeAngelis kept his job at Honeywell Bull and recruited two MIT students, Kirk Sigel and Mike Ciholas, to help him build the prototype on evenings and weekends. To build the digital camera, they harvested a charge-coupled device from a fax machine, which converted the optical images into digital pixel data. A quartz timer time-stamped the digital images and then passed the data to a standard desktop computer, where it was compressed and stored on the hard drive. A software interface reassembled the digital images and time stamps on screen. When an official placed the red-line cursor on each runner's chest, it immediately reported the finish time, accurate to 1/100 of a second (0.01 sec) as required by the track federations.[22]

In February 1992, DeAngelis (Figure 4.5) debuted the FinishLynx prototype at the U.S. Indoor Track and Field Championships and garnered immediate interest from the community. Unlike when using Accutrack, with FinishLynx there was no waiting around for the Polaroid film to develop; the new system provided reliable, near-instant race results. DeAngelis incorporated Lynx System Developers in March 1992 and soon received a $1,500 down payment from his first customer, an influential track coach named Tom Jennings. DeAngelis barnstormed the country that summer to demonstrate FinishLynx at several track meets. By December 1992, he had secured five orders which he fulfilled by assembling the systems in his bedroom.[23]

DeAngelis eventually resigned from Honeywell Bull to lead Lynx full-time and continued working with Sigel and Ciholas to improve the product. The second-generation FinishLynx system featured an integrated camera and timer that produced color images at 2,000 frames per second, for race times accurate to 1/1,000 of a second (0.001 sec); today's top-of-the line camera captures up to 20,000 frames per second. DeAngelis and his team have also expanded Lynx's product line to include an automated false-start detection system and a laser distance measurement tool for shot put, javelin, and other field events.[24] Most ambitiously, Lynx developed the IsoLynx player-tracking system. During a successful test at a National Football League (NFL) game in December 2012, DeAngelis's team placed tiny RFID tracking tags in the shoulder pads of every player, which relayed real-time location data to receivers arrayed around the stadium. A central computer converted the data into a series of statistics and graphics. IsoLynx became the first system to track the movement of every NFL player in real time. Elements of this system are used under license

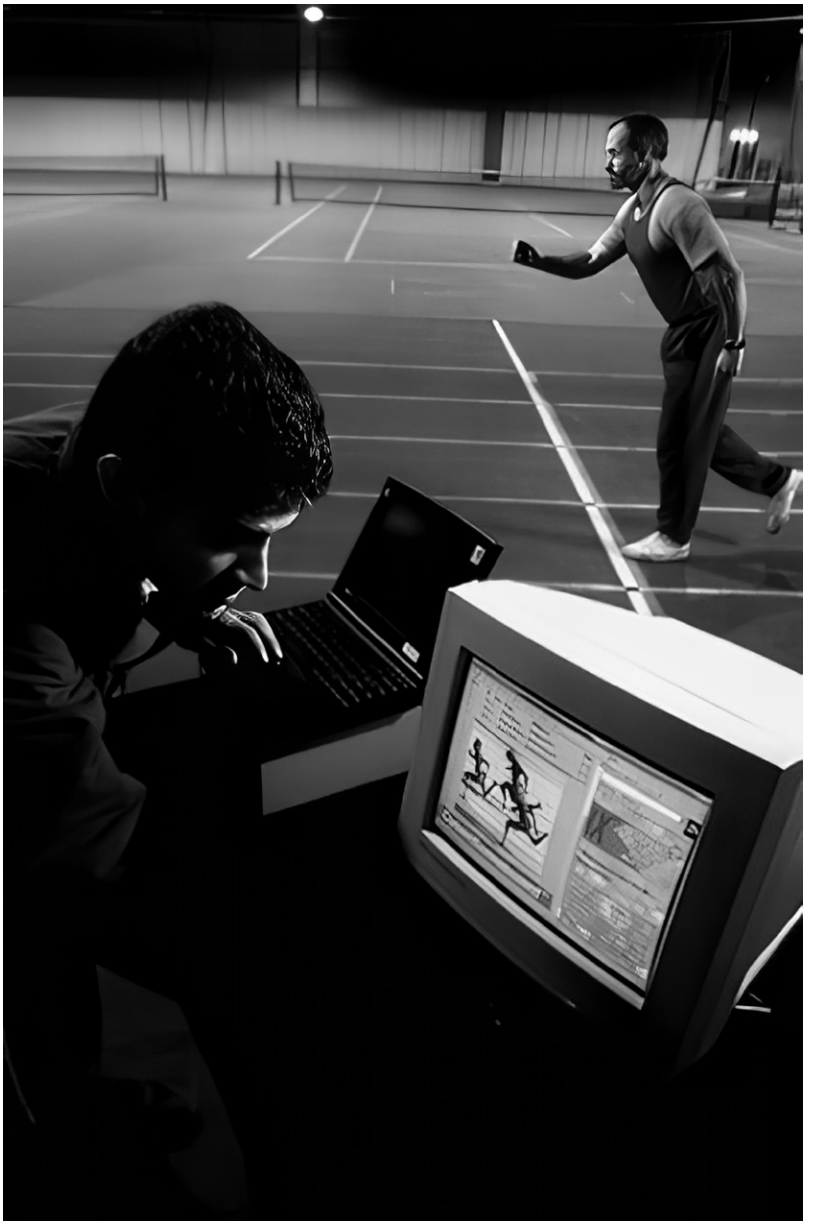

FIGURE 4.5. Inventor Doug DeAngelis (left) and a colleague test the FinishLynx digital photo-finish system at MIT in 1992. Courtesy Lynx System Developers, Inc.

to help the NFL provide "Next Generation Stats," such as the number of yards a defender traveled to make a tackle.[25]

In 2021, Lynx Systems Developers, Inc. merged with Nevco Sports, a manufacturer of scoreboards and LED video displays that integrate with Lynx's products.[26] FinishLynx remains the most popular photo-finish and race results system in the world, with thousands of customers in 135 countries. It provides results for any sporting contest with a finish line, including the U.S. Olympic Track and Field trials, the Kentucky Derby, NASCAR, and the Tour de France. FinishLynx, DeAngelis observes, remains popular because it was "not

just a replacement for film-based photo-finish." Rather, it was "a comprehensive software package that would make the operations of running an athletic event a much simpler and faster thing."[27]

Tony Verna and Instant Replay

With modern photo finishes, an official must interpret a single image that captures one moment: the racers crossing the finish line. Yet officials in many sports have the exquisite challenge of interpreting motion pictures—instant replays—of many different in-game situations. Did the basketball player release the shot before the clock expired? Did the base runner beat the shortstop's tag at second base? Compared to photo finishes, which first emerged in the 1880s, instant replay is a relatively recent innovation, first perfected by CBS television producer Tony Verna in 1963.

Verna's innovation drew on several years of technological development in audio and video recording equipment led, in part, by Bing Crosby. During the 1930s and 1940s, entertainers like Crosby always performed and broadcast their radio shows live. In fact, to satisfy advertisers, they sometimes had to do two live performances—one at prime time on America's East Coast, and then a second live show three hours later during prime time on the West Coast. To remedy this burden, Crosby proposed pre-recording his shows, editing out any mistakes, and re-broadcasting them, but the networks balked. Although phonographs and gramophones had been around for decades, those recording technologies played back with unacceptable hissing and crackling. In 1946, Crosby established the Electronics Division of Bing Crosby Enterprises (BCE) and began to search for a high-fidelity recording solution that was indistinguishable from a live feed.[28]

Crosby found the solution at a San Francisco Bay Area start-up called Ampex, founded by Alexander M. Poniatoff in 1944. Ampex had reverse engineered the German AEG Magnetophon K4 magnetic recording machines that Army Signal Corps officer Jack Mullin had brought back to the United States as one of the technological spoils of World War II. Crosby hired Mullin, who used two high-fidelity Ampex 200A magnetic audio recorders to record and play back each episode of Crosby's radio show during the 1948–1949 season.[29]

As Crosby moved into the emerging medium of television, he encouraged engineers at Ampex and BCE to develop a similar magnetic recording solution for videotape. In the early 1950s, BCE developed a series of prototype video recorders, the Mark I, II, and III. However, Ampex's VR-1000 (the Mark IV) was a superior solution. It used a set of four video heads mounted on a rotating drum that scanned transversely across a 2-inch-wide reel of magnetic tape. In April 1956, Ampex successfully demonstrated the VR-1000 at the National Association of Radio and Television Broadcasters convention in Chicago and orders poured in at $50,000 per unit.[30] Just seven months later, on 30 November 1956, CBS used the Ampex VR-1000 to record and air the first "tape delayed" television broadcast, an episode of *Douglas Edwards and the News*. Ampex's engineers won a technical Emmy award for their innovation in 1957.[31] A few years later, CBS's Tony Verna would use the VR-1000 to shrink the three-hour West Coast tape delay into an "instant" replay.

Verna (Figure 4.6) was born in Philadelphia in 1933. After briefly studying engineering at West Point and the University of Pennsylvania, he dropped out to pursue a career in live television production. He worked on *American Bandstand* for Philadelphia's ABC affiliate WFIL before moving across town to help produce Ed McMahon's morning show at CBS's

FIGURE 4.6. CBS television producer Tony Verna in 1966. Credit: CBS Photo Archive.

WCAU. Verna rose quickly through the ranks at CBS, and by 1955—though just 21 years old—he was directing national baseball telecasts.[32]

Beginning in 1956, CBS and other networks used Ampex's VR-1000 to provide taped replays of exciting sports plays, but these highlights were usually only shown at halftime or after the game. Even with the machine's numerical counter, it was difficult to cue the tape to the precise location of the desired play. During playback, the VR-1000 initially showed 7 to 10 seconds of distorted "video hash" before the magnetic heads warmed up and showed a clear image. Verna had encountered these problems during CBS's coverage of the 1960 Summer Olympics in Rome. Verna would review the tapes from the day's action and insert voice cues on the tape's secondary audio track, which viewers could not hear. For example, he would tell the directors to watch for sprinter Wilma Rudolph crossing the finish line, then put the tapes on an airplane bound for CBS Studios in New York City. However, his voice directions were disregarded because they also got distorted in the video hash as the tape heads came up to speed. CBS's directors told Verna not to waste his time.[33]

Later, as a lead director for football games, Verna became frustrated by the long periods of inaction between plays. He also wished he could show viewers the multiple different camera angles he was seeing on his monitors, which helped explain the action. "I remember times when [receiver] Tommy McDonald of the Philadelphia Eagles would miss one of quarterback Norm Van Brocklin's bombs, but we didn't see why he wasn't there for the catch," Verna wrote in his memoir, *Instant Replay.* "I wanted to show them what I had seen on another monitor, that McDonald had been tripped leaving the line of scrimmage." By 1963, Verna had begun to experiment with an "instant" in-game replay that would allow his television audience to experience more of what he was seeing in the director's booth.[34]

Verna selected the 1963 Army–Navy football game to attempt the first instant replay, but it was a somber occasion. The game had been postponed to December 7—Pearl Harbor Remembrance Day—following President John F. Kennedy's assassination on November 22. On November 24, Jack Ruby had shot and killed the suspected assassin, Lee Harvey Oswald, on live television. It had taken the networks approximately nine minutes to replay the video tape.[35]

To prepare for the game, Verna borrowed a VR-1000 machine and several rolls of magnetic tape from CBS's New York studios and trucked everything down to Philadelphia. He believed his best chance for a replay would be to isolate one camera on the game's dynamic quarterbacks, Army's Rollie Stichweh and Navy's Heisman Trophy winner, Roger Staubach. However, Verna warned his play-by-play announcer, Lindsey Nelson, not to promote the experiment during the telecast because he was not sure if it would work.[36]

To achieve instant replay, Verna thought back to the voice cues he had recorded on the tapes from the Rome Olympics. Verna's key insight was to use 440-cycle audio tones—not the human voice—to cue each play on the secondary audio track. These tones, even if slightly garbled, were still recognizable through the distorted video hash as the Ampex machine came up to speed. Verna instructed a technician to mark each play, inserting one beep when the offensive team broke the huddle and two beeps when the quarterback reached the line of scrimmage. If a key play occurred, Verna believed he could play back the tape starting at the first beep and allow the 10 seconds of video hash to clear as the machine warmed up; then the play itself would replay starting at the second set of beeps. Verna, like many innovators, had envisioned a new way to use an existing set of technologies.[37]

During the game, Verna and his production team struggled with their equipment. The 1,200-pound VR-1000 machine, full of fragile vacuum tubes, had been jostled on the 90-mile road trip from New York to Philadelphia and was operating inconsistently. Moreover, CBS had not given Verna a brand new $300 reel of tape. Instead, he received a previously used reel that contained an old *I Love Lucy* episode and several Duz soap commercials. Early in the game, when Verna attempted an instant replay, he found that the Ampex machine was not fully recording over the old footage and Lucy was bleeding through![38]

By the fourth quarter, Navy led Army 21–7, and Verna's team had made about 30 unsuccessful attempts to record and rebroadcast a play. In his memoir, Verna remembered the excitement that followed Stichweh's touchdown as his production team assembled the first successful instant replay:

> It was an isolation that George Drago had caught on his camera and which John Wells immediately rewound and then hit the play button. And during the seven to ten seconds while the

> pre-roll played back, I heard the tones strengthen, and lo and behold, clean video came up. My technical director Sandy Bell, punched it up while I shouted into Lindsey [Nelson's] ear, "This is it!" And when that Instant Replay hit the screen, we got Lindsey's famous on-air shout, "This is not live! Ladies and gentlemen, Army did not score again!"

Given the production's team's hesitancy to promote instant replay during the telecast and the nation's somber mood, Verna's innovation initially went unnoticed. Magnetic tape was expensive, and per CBS policy at that time, the 1963 Army–Navy game tape was eventually reused for some other event and taped over. Ironically, there is no surviving footage of the first instant replay.[39]

Nevertheless, instant replay was quickly perfected and readily adopted. By fall 1964, CBS production crews used instant replay for most of its National Football League (NFL) games, and the technique quickly became a staple of sports broadcasting. Over time, with improved recording and playback equipment, viewers were eventually able to see dramatic slow-motion replays that highlighted the athleticism and close calls in many different sports.[40]

Occasionally, an instant replay would show that an official had missed a call on the field. For example, during an April 1977 baseball game between the Houston Astros and the Atlanta Braves, the stadium scoreboard operator showed a replay that clearly highlighted the umpire's errant call. When fans booed, the four-man officiating crew left the game in protest. Despite initial resistance from referees, the National Football League (1986), the National Hockey League (1991), the National Basketball Association (2002), and Major League Baseball (2008) eventually adopted instant replay to aid officiating.[41]

Players, coaches, and fans hold a range of views about the use of instant replay as an officiating tool. Advocates promote instant replay to ensure accurate officiating and just outcomes; detractors argue that instant replay slows down the games and removes the spontaneity, unpredictability, and controversy that make sports so exciting and entertaining. But there is little argument about the entertainment value of instant replay, which helps fans understand and enjoy the action. "Replay completely changed how we all look at sports," observed legendary sportscaster Dick Enberg. Indeed, sportswriter Patrick Saunders has noted that "fans have come to not only expect but demand that they can sit on the couch, watch a flat-screen, high-def TV and revisit close calls."[42]

As for Verna? He went on to pursue a decorated career directing live sports and entertainment events, including five Super Bowls, twelve Kentucky Derbies, the NBA championships, the Stanley Cup finals, the 1984 Los Angeles Olympics, and the 1985 Live Aid concerts to raise money for African famine relief. In addition to two Emmy awards, Verna earned a Lifetime Achievement Award for Sports Direction from the Directors Guild of America in 1995. Verna battled leukemia and died in January 2015, but his legacy is assured. "What should it say on my tombstone?" Verna mused in a 2013 interview. "Son of Italian immigrants. Invented Instant Replay."[43]

Paul Hawkins and the Hawk-Eye Ball-Tracking System

With instant replay, multiple cameras capture the action, but a human official must interpret what happened and make the call. With ball- and player-tracking systems like

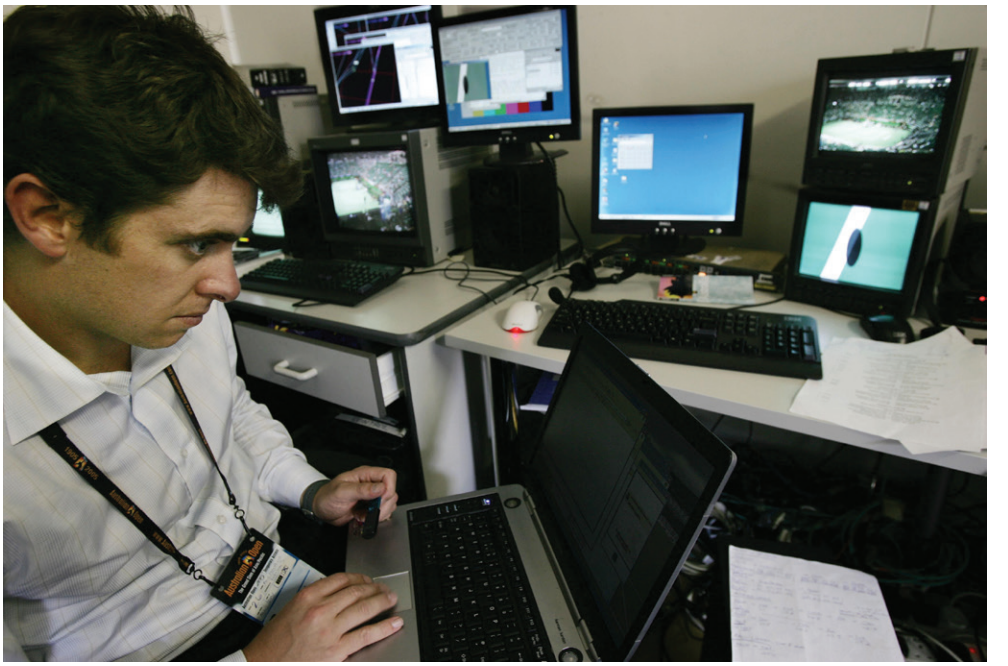

FIGURE 4.7. Paul Hawkins oversees the Hawk-Eye ball-tracking system at the 2005 Australian Open. Credit: The AGE / Joe Armao.

Hawk-Eye, multiple cameras capture the action, but sophisticated software algorithms interpret what happened and automatically make the call without human intervention.[44]

Hawk-Eye was conceived and developed by British inventor Paul Hawkins (Figure 4.7). Hawkins grew up in Hampshire playing all kinds of sports, including cricket and table tennis. As an undergraduate at Durham University, he rowed on the university crew team while studying mathematics and computer science. Hawkins stayed at Durham in the mid-1990s to pursue a PhD in artificial intelligence and continued to play semi-professional cricket. For his thesis project, Hawkins built a natural language processing system that used contextual clues to make better sense of written and spoken words, the kind of technology that subsequently enabled digital assistants such as Siri and Alexa.[45]

While at Durham, Hawkins once drove 300 miles to play in a cricket match, where he fell victim to a bad call. While Hawkins was at bat, an official called him out after determining that the ball hit his leg but otherwise would have hit the wicket. "It was a very long drive back to my university after the match ended," Hawkins told *Slate* in 2012, "and I had plenty of time to stew and think." The so-called "leg before wicket" has always been a subjective judgment call, but Hawkins believed it did not have to be. Drawing on his experience in artificial intelligence, Hawkins realized that if a system of high-speed cameras could track a ball's flight path and velocity, then a software algorithm could use that data to predict the ball's likely future path. When a ball hit the cricket batter's leg, his imagined system could determine—based on the ball's predicted trajectory—if it otherwise would have hit the wicket unimpeded.[46]

Hawkins finished his PhD in 1999 and took a job at Roke Manor Research Ltd., a consulting firm that provides research and development (R & D) services to industrial and government clients. Hawkins was assigned to work on a missile defense system and again

imagined applying those same trajectory-prediction principles to a cricket ball. After making a successful pitch to Roke's leadership, Hawkins and his coworkers were encouraged to build a prototype ball-tracking system.[47] In January 2000, Hawkins and Roke's David Sherry applied for British patent 2357207A, describing a "ball tracking and trajectory prediction" system, and in January 2001, Roke helped spin out a new start-up firm, Hawk-Eye Innovations Ltd. By August 2001, Hawkins had arranged for a live competition test at the Ashes cricket series. The Hawk-Eye team trained their cameras on the pitch between the bowler, batter, and wicket, then set up their computers and monitors in the broadcast booth. In the beginning, Hawk-Eye was purely an entertainment tool. It enhanced the Channel 4 broadcast by showing the speed and trajectory of the bowler's pitches and suggested the likely path of the ball in "leg before wicket" situations, but officials did not initially use Hawk-Eye to aid their decision-making.[48]

Hawk-Eye's breakthrough would come in tennis, where it was used to judge in/out line calls. In February 2002, Hawkins and his team conducted Hawk-Eye's first public tennis tests during BBC's Davis Cup coverage. By using 10 cameras arrayed around the stadium capturing 340 frames per second, Hawk-Eye could cover the ball and every line on the court, including the baselines, sidelines, and service lines. As with cricket, Hawk-Eye was initially used only to enhance tennis broadcasts. However, fans and players soon began calling for Hawk-Eye as an aid to officiating. The International Tennis Federation conducted successful tests of the Hawk-Eye system in 2005, then introduced a challenge-and-review system. Players were given three challenges per match and could ask the umpire for a review of a human linesman's call. They had to accept Hawk-Eye's replayed trajectory as the final arbiter. The major grass- and hard-court tournaments, including the Australian Open, the U.S. Open, and Wimbledon, soon adopted Hawk-Eye. However, the French Open and other clay-court tournaments demurred, since players and officials could easily review marks the ball left in the dirt.[49]

Reaction to Hawk-Eye was mixed. Fans and broadcasters generally embraced the technology. During a challenge, spectators often clapped rhythmically in anticipation as the Hawk-Eye official cued up the replay. "As a player, and now as a TV commentator," Pam Shriver noted, "I dreamed of the day when technology would take the accuracy of line calling to the next level. That day has now arrived."[50] Yet one of the sport's biggest names, Roger Federer, was a frequent and vocal critic. Facing Rafael Nadal in the 2007 Wimbledon men's final, Federer appeared to win the first set until a set-point call was reversed by Hawk-Eye. "It's killing me today," Federer lamented later in the fourth set as multiple Hawk-Eye calls went in Nadal's favor. Federer eventually rallied to win the championship in five sets.[51]

After Hawk-Eye's breakthrough in tennis, Sony acquired Hawk-Eye Innovations in 2011. In 2012, FIFA approved the use of Hawk-Eye goal-line technology to adjudicate goal/no goal calls in soccer.[52] And because Hawk-Eye's cameras can track the precise location of the ball and every player on the pitch, FIFA began conducting tests in 2021 to automate onside/offside calls. When a goal is scored, Hawk-Eye can determine if an offensive player was offside and relay that information to the on-field referee in about half a second.[53]

But Hawk-Eye is not perfect. In 2020 and 2022 the firm made public apologies to teams and fans after its goal-line technology made errant calls when players congregated near the goal line and obstructed the camera's view of the ball.[54] Critics have also raised more fundamental, epistemological concerns. In their 2016 book *Bad Call*, researchers

Harry Collins, Robert Evans, and Christopher Higgins reminded readers that Hawk-Eye is not a replay of the ball's actual position; rather, it produces a prediction of a trajectory, based on the ball's prior velocity and position. Overall, the authors lament that Hawk-Eye and other "decision aids" have undermined the authority of referees and umpires, which they consider bad for the games.[55]

Nevertheless, officials have continued to rely on Hawk-Eye; it now aids referees and broadcasters in twenty sports worldwide, from badminton to Gaelic football. Paul Hawkins remains proud of what he and his teammates have achieved since he fumed over that bad cricket call. "I was young and entrepreneurial and took a risk," Hawkins told the BBC in 2014, "to follow my passion which is sport and technology."[56]

Conclusion

Although we sometimes take them for granted, officiating technologies are fundamental to most competitive sports. Game clocks, stopwatches, photo-finish cameras, instant replays, and ball-tracking systems are crucial for helping officials and leagues keep time, keep score, and ensure accurate results.

What motivates sports inventors to develop officiating technologies? In short, inventors like Bill Parkinson, Doug DeAngelis, and Paul Hawkins become frustrated by missed calls and officiating errors, then develop technological solutions to improve the status quo. Frustration also motivated director Tony Verna, who invented instant replay after lamenting that television viewers could not enjoy all the different camera angles he saw in the control booth.

The specific requirements of different sports environments often inspire new officiating innovations. A single high-speed photo-finish camera works well to determine the winner of any contest with a finish line. However, a multi-camera system like Hawk-Eye is better suited to tracking the ball and players in sports like tennis, soccer, and baseball. Similarly, an RFID tracking tag attached to each runner is a better solution than a traditional photo-finish camera for road races that use staggered starts to accommodate thousands of competitors. To time swim races, a pressure-sensitive touchpad works best for marking the splashy, underwater finishes that a photo-finish camera would be unable to discern.

Athletes and fans have typically embraced advanced officiating technologies because their accuracy and precision exceed the limits of human perceptions and reflexes. Fully automatic timing systems have eliminated the variability and reaction-time errors of human timers with handheld stopwatches. Similarly, with photo finishes and instant replays, the camera's eye can see things that the human eye cannot. And yet—paradoxically—even as these visual technologies exceed the limits of human perception, we still require fallible human officials to interpret photo-finish images and review instant replays.

Indeed, human stakeholders—including athletes, officials, the media, and fans—ultimately determine whether to use, limit, or ban officiating technologies. For example, officials did not immediately adopt instant replay or Hawk-Eye to aid with officiating; initially, they were strictly media technologies used to enhance sports telecasts. Timekeeping standards are still applied unevenly, even arbitrarily, across various sports. In boxing, for example, officials use a clock to precisely measure the duration of the 3-minute rounds, but the referee uses a hand count—not a stopwatch—to measure a 10-second knockdown. Likewise, today's photo-finish cameras are accurate to 0.0001 seconds, yet most sports federations report results in increments of only 0.01 seconds. Some observers would suggest

that perfectly accurate sports officiating is not required or even desirable. They argue that human imprecision, missed calls, and the occasional controversy spark conversations among fans and make sports more fun and entertaining.[57]

Athletic contests are only meaningful if they are officiated honestly, equitably, and accurately. Thus, officiating technologies tend to provoke spirited debates among athletes, officials, the media, and fans. Collectively, these stakeholders determine what is fair, how accurate is accurate enough, and how we should govern technologies like FinishLynx, Hawk-Eye, and instant replay. Ultimately, there are no purely objective standards for fairness and accuracy in technological officiating; they are always negotiated.

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Chapter 5

Fun for Everyone

Monica M. Smith

Humans are innately playful and social creatures, as evidenced by toys and games found among artifacts of civilizations from ancient history to the present. Depending on the geography and available resources, people of many different cultures develop new sports or tweak existing ones for activity on land and in water; in hot climates and cold; indoors and outside; with balls, disks, and sticks; and on boards, wheels, cleats, and blades. People play sports for a range of reasons—physical fitness, community or religious ritual, team building, military training, personal or national glory—but mostly they play for fun.

Traditionally, however, not everyone has been allowed to play. For much of sports history, participation was limited to non-disabled boys and men. Beginning in the late nineteenth century, social, cultural, legal, economic, and technological changes enabled more women and people of all backgrounds to take part in sports. In the United States, these changes accelerated after World War II. Greater inclusion followed the civil rights movement of the 1950s and 1960s, and the years since have seen important federal legislation, including 1972's Title IX, which addressed gender inequities; the 1990 Americans with Disabilities Act, which protected people with disabilities from discrimination; and the ongoing Equal Rights Amendment (ERA) battle.[1] These movements made it possible for more—and more diverse—people to enjoy both the fun and competition of sports.

Like the love of sports, the impulse to invent is shared across time and cultures. Indeed, inventors, entrepreneurs, and their sports technologies have long played an important yet underappreciated role in helping more people have fun, while also more recently democratizing and diversifying access to sports. Having fun may be the simplest motivation for sports innovation; just ask any group of kids who invent playground games. Even though a small number of sports dominate global media, people continue to create new sports such as pickleball, spikeball, and ultimate frisbee. Motivated by inadequate equipment or their limited participation, frustrated athletes also adapt existing technologies to help themselves and their friends take part in their favorite pastimes.

This chapter presents five case studies of diverse and inspiring athletes, inventors, and entrepreneurs whose work expanded participation, access, and fun in sports. The cases describe Lisa Lindahl, Hinda Schreiber Miller, and Polly Palmer Smith, who co-invented the first sports bra, called the "Jogbra"; Howard Head, who improved skis and tennis rackets; Marilyn Hamilton and her partners Don Helman and James Okamoto, who invented the Quickie sports wheelchair; Frank Nasworthy, who developed urethane skateboard wheels; and Tahira Reid, who designed an automated double Dutch jump-roping machine. These inventive individuals exemplify how people continue to adapt,

tweak, and improve athletic equipment, clothing, and other sports-related technologies to make sports more fun and inclusive for themselves and a growing community of athletes worldwide.

Lisa Lindahl, Hinda Miller, Polly Palmer Smith, and the Jogbra

Enactment of Title IX of the Education Amendments in 1972—the same year Congress passed the Equal Rights Amendment (although it was not subsequently ratified by the states)—led to a revolution for women's participation in both recreational and professional sports. Title IX reads: "No person in the United States shall, on the basis of sex, be excluded from participation in, be denied the benefits of, or be subjected to discrimination under any education program or activity receiving federal financial assistance."[2] The legislation specified that all public and private elementary and secondary schools, school districts, colleges, and universities receiving any federal government funds must comply with Title IX, which required those institutions to provide equal opportunity in athletics. While it also covered other areas, Title IX is best known for its impacts on increased participation, more scholarships, better funding, and higher public profiles for female athletes.

Lisa Lindahl was one of millions of women who began taking part in sports in the wake of Title IX. Lindahl started jogging, a sport that took off as part of the 1970s fitness boom. Yet the available sports clothing was designed mostly by and for men. "I loved [running] except for the discomfort that my bouncing breasts created," she recalled. "The options were going braless or wearing a regular bra that was one size too small, which still gave me issues with discomfort and straps slipping."[3] Since she could not find a supportive athletic bra, Lindahl decided she would ask her friends to help her design one. In 1977, after earning a bachelor's degree in education from the University of Vermont, Lindahl approached her childhood friend Polly Palmer Smith, who was working with Hinda Schreiber Miller, also a jogger, to design costumes for the Champlain Shakespeare Festival in Burlington, Vermont. They joined forces to create the first commercial sports bra.

The coinventors started by conducting market research and testing. They visited lingerie stores, tried on various bras, surreptitiously jogged in place in dressing rooms, and then purchased the best models to evaluate while running outside. Smith knew from her clothing-design experience that bras are difficult to make, so she deconstructed and reverse-engineered the bras to better understand the key features. The women wanted to design a sports bra that was extra supportive, to limit breast bounce while also offering comfort and minimizing chafing.

While discussing ideas, Lindahl recalls, "My sister was the one who said, 'Guys have jockstraps. Why isn't there a jockstrap for women, just applied to a different part of the body?' We thought that was hilarious. But I also thought it wasn't a bad idea."[4] As they had done with the bras, they deconstructed two jockstraps and again studied the designs. Then Smith chose a relatively new material, a cotton/poly/Lycra blend—cotton for comfort, polyester for strength, and Lycra for support—with plush-lined elastic for the rib band and straps (Figure 5.1).

Smith constructed some prototypes by sewing the seams on the outside of the garment to avoid irritating the skin and passed them along to Lindahl and Miller to test. According to Miller, "After work that afternoon, Lisa and I went to the track. She ran forwards, I ran

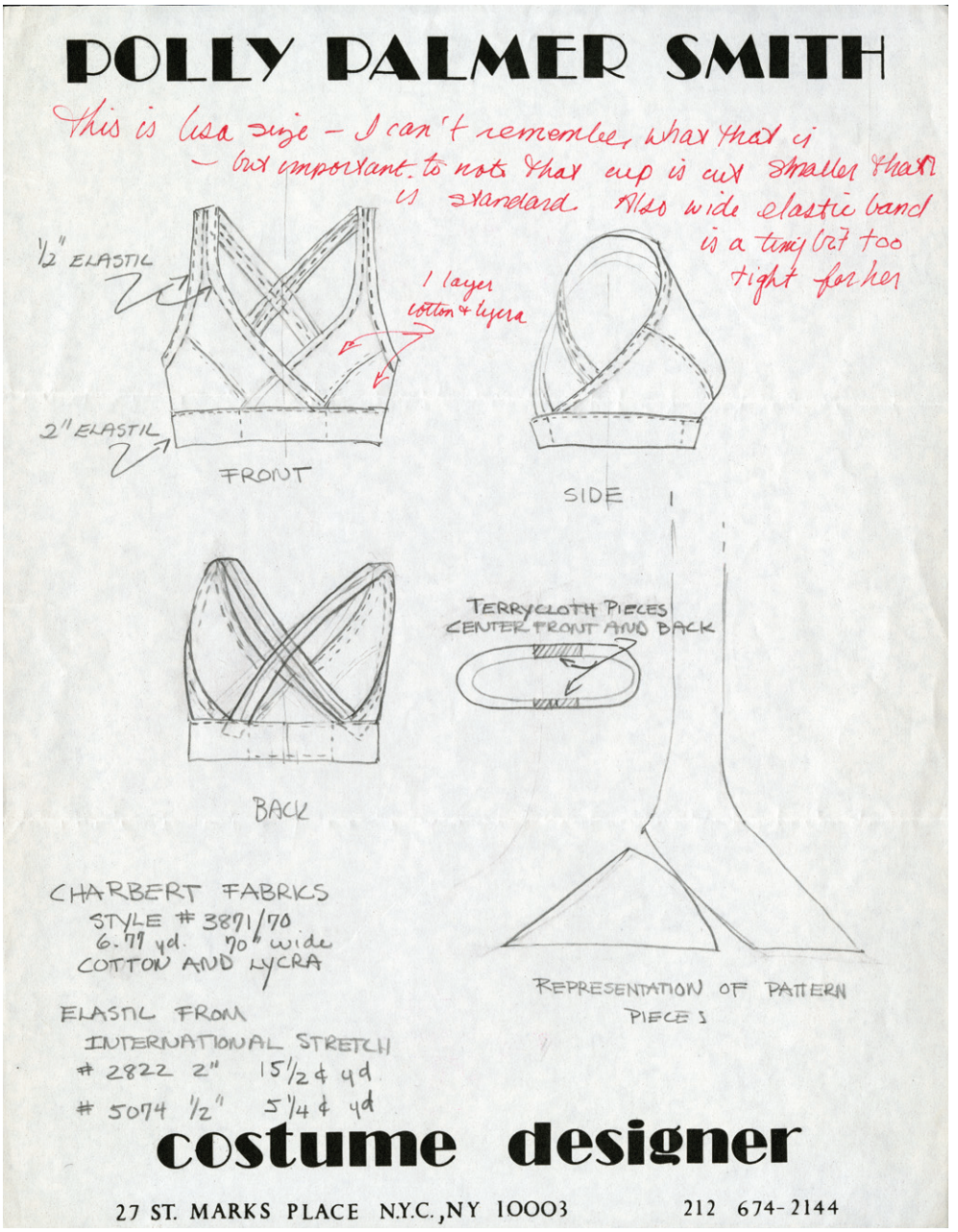

FIGURE 5.1. Early Jogbra design sketch. Credit: Jogbra, Inc. Records, Archives Center, National Museum of American History, Smithsonian Institution.

backwards in front of her, watching her breasts. They did not move as much as before. 'This is it! This really works!' We were tickled."[5] They quickly realized they had invented an effective solution, not only for themselves, but also for other women athletes.

They first nicknamed their creation the "jock bra" because of the jockstrap inspiration. However, concerned about negative feedback, they changed the name to "Jogbra" and quickly set about protecting and commercializing their invention. They established

Jogbra, Inc., then renamed the firm SLS, Inc. (for Smith, Lindahl, and Schreiber) in 1978, before settling on JBI, Inc., in the early 1980s. They secured U.S. utility patents 4,174,717 in 1979 and 4,311,150 in 1982 for their "athletic brassiere" inventions as well as design patents. Smith almost immediately stepped away from the venture; she joined The Jim Henson Company in 1978 as a professional costume designer and eventually won seven Emmy Awards for her work on *Sesame Street*. Lindahl and Miller became the co-owners and faces of JBI, Inc.

Initially, Lindahl and Miller had difficulty convincing the male owners of sporting goods stores to carry the Jogbra. Men did not understand the need for a sports bra or see the product as a piece of sports equipment.[6] Because business*men* did not believe it would sell, Lindahl and Miller marketed the Jogbra directly to female customers through mail-order forms attached to ads in sports journals (Figure 5.2). Their direct-to-jogger strategy helped the business expand quickly.

At first, the target market was narrowly defined to adult female runners, and the Jogbra was offered in only small, medium, and large sizes. However, based on consumer response, Miller and Lindahl realized they needed to tailor their invention to accommodate women of a greater range of sizes and engaged in distinct types of physical activity. Working with exercise physiologist Dr. Lajean Lawson, they devised what they called the Motion Control Requirement (MCR)—a trademarked system resulting from research into athletes' bodies—which helped consumers customize their purchases based on "breast size, sport activity, and personal preference; whether it be classic compression, underwire, natural shaping or shoulder comfort."[7]

Over time, the company expanded the list of Jogbra-wearing activities from high-impact aerobic activities like running (high MCR) to basketball, bicycling, golf, gymnastics, soccer, tennis, volleyball, and more (a mix of high and low MCR activities). As the company grew, the inventors continued to adopt the latest fibers and technologies to broaden the aesthetic options and incorporate wicking and antimicrobial features. When Lindahl and Miller sold JBI, Inc. to Playtex Apparel in 1990, they became co-presidents of the new Jogbra Division. A year later, Playtex sold the division to the Sara Lee Corporation and Lindahl left the business. Miller continued to serve as president and eventually became the CEO of Sara Lee's Champion Jogbra Division. After being co-branded with Champion, the Jogbra was finally carried in department stores.

Miller eventually left the company in 1997 to pursue other interests. She was elected and served as a Vermont state senator from 2002 to 2013, and then became president of DeForest Concepts, a consulting firm that supports small businesses and promotes women entrepreneurs. Lindahl, who had been diagnosed with epilepsy at age four, moved on to become senior vice president of the board of directors of the Epilepsy Foundation from 1991 to 2000. In 2001, with Dr. Lesli Bell, Lindahl cofounded the Lightning2 Company (doing business as Bellisse) to design and market their patented Compressure Comfort Bra. Their innovation offered a special compression garment for women suffering from lymphedema—tissue swelling caused by an accumulation of protein-rich fluid that is usually drained through the body's lymphatic system, which often occurs as a result of cancer or cancer treatment.[8]

Together, Lindahl, Miller, and Smith were inducted into the U.S. National Inventors Hall of Fame in 2022. Their patented sports bra may seem simple, but it completely changed the game for female participation on the playing field. Prior to their product, bras had been primarily designed by men to enhance the appearance of women's breasts. Invented in 1977, the Jogbra provided a whole new category of athletic apparel, reclassifying the bra as a piece

FIGURE 5.2. Jogbra advertisement, late 1970s. Credit: Jogbra, Inc. Records, Archives Center, National Museum of American History, Smithsonian Institution.

of athletic gear, not lingerie. It was quickly recognized as necessary sports equipment for women. Thanks to Title IX, school-age girls and young women had been given equal access to educational opportunity, and consequently began participating in sports like never before. Independent-minded adult women looking to lead full, more physically active lives also started jogging in significant numbers just as the Jogbra hit the marketplace. This helped fuel the jogging craze and the aerobics boom that followed.

Jogbra's signature cultural moment came in 1999. During that year's Women's World Cup, American soccer star Brandi Chastain famously whipped off her jersey—exposing her Nike sports bra—after kicking the winning penalty shot for a United States win over China. Touted in headlines as "the bra heard 'round the world," the image of Chastain kneeling in her white shorts and black bra screaming with joy became a visual icon across the world, exemplifying women's empowerment.[9] A year later, FIFA, soccer's world governing body, formally banned the removal of jerseys by players during goal celebrations. Yet, the powerful image of Chastain proudly showing off her sports bra continues to carry great significance for women of all ages. As Lindahl noted, "Jogbra was the right product at the right time. Women and girls were venturing forward—on the field and in life—experiencing the empowerment that being strong and athletic could provide. At Jogbra, we were both crafting and sharing that journey."[10]

Howard Head's Improved Skis and Tennis Rackets

Inventors regularly draw on their own knowledge and expertise to solve technical problems, but sometimes they benefit by also incorporating insights from other professional fields. This interdisciplinary curiosity helps inventors creatively break out of accepted approaches or products to invent new things.[11] Lifelong tinkerer Howard Head combined his aircraft engineering knowledge about structural principles and materials with his personal interests as a dabbler in sports to create revolutionary ski and tennis racket designs for both amateur and professional athletes.

Born in Philadelphia in 1914, Head started at Harvard University as an English major but switched specialties to earn a bachelor's degree in engineering sciences. In 1936, he started working as a riveter and then became an engineer at the Glenn L. Martin Company near Baltimore, Maryland. There he helped build and design B-26 bombers and PBM-3 flying boats during World War II.[12] In his spare time, Head liked to try different sports. When he first attempted downhill skiing, Head felt frustrated because he kept falling. He partially blamed his heavy, awkward wooden skis for making the sport so difficult for him, and began to think about how to design lighter, more flexible skis using metal-sandwich construction, which was common in the aircraft industry.

Head first envisioned skis made with two lighter layers of aluminum bonded to walls of thin plywood and filled with honeycomb plastic. Having learned to ski in Stowe, Vermont, in 1947, he asked instructors there to evaluate early versions of his skis out on the slopes. His prototypes often broke, but he learned something new with each experiment, making tweaks and designing new iterations. Throughout the following winter, Head constructed test skis for pro skier Neil Robinson, who supplied helpful feedback. After several design refinements, Head eventually developed metal skis that were just as strong as wooden skis, but much lighter.

While these later iterations were more durable, they still were not perfect. To strengthen the skis further, he swapped out the honeycomb plastic for plywood cores. Also, the aluminum bottoms of the skis froze up and snow would build up on them; Head responded by covering the bottoms of the skis with plastic that could be waxed. Eventually, he also covered the tops and sides of the skis with the same plastic. Head also found that the skis' aluminum edges dulled quickly. To harden them, he capped the waxed aluminum with razor-hard steel edges bonded to the skis.

FIGURE 5.3. Howard Head with a prototype composite ski, ca. 1950. Credit: Howard Head Papers, Archives Center, National Museum of American History, Smithsonian Institution.

To start a business to commercialize his skis, Head supposedly used $6,000 in poker winnings and money borrowed from friends. He quit his job and founded the Head Ski Company in Baltimore with three part-time associates with whom he had worked at Martin: Frank Kaminski, Petrus ("Peter") Meyer, and Joe Waldych. "Not knowing any better, I assumed I could make a better ski out of aircraft materials than wood," Head recalled. "It turned out to be harder than I thought. It took 40 versions before I came up with the ski that would work, and that was three years later."[13] In 1950, his company introduced the new Head ski commercially, and it was soon dubbed "the cheater" because its lighter, more flexible design made the skis more maneuverable. Compared to clunky wooden skis, it was notably easier for more people to carve turns and safely ski downhill (Figure 5.3).

In 1954, Head earned U.S. patent 2,694,580 for his "composite wood and metal ski having plastic running surface." His sandwich-construction skis quickly became the standard for the sport, both for recreational and professional skiers. At the 1964 Winter Olympics in Innsbruck, Austria, American skier Jean Saubert captured three medals using a pair of Head skis. Within two years, most top racers had abandoned wooden skis in favor of Head's design. He also capitalized on the popularity with recreational skiers. By 1966, the Head ski company was selling 300,000 skis a year and grossed $25 million annually in global sales.[14] In 1969, Head sold his company to AMF, Inc., for a reported $16 million.

But Head was not finished with either sports or inventing. After retiring from his eponymous company, Head started a new hobby, tennis, and again found he was not particularly good at it. Head partially blamed the equipment—small, wood-framed rackets—for his difficulties. "I found the racket twisted in my hand on an off-center hit," he

explained. "And since I couldn't improve my own athletic skills, I set out to design a racket that would be more stable to an off-center hit."[15]

Head had purchased an automated tennis-ball machine from the Prince Manufacturing Co. in New Jersey to help him practice, but as with the wooden rackets, he was dissatisfied with how it worked. Instead of just tinkering with the racket or the ball machine, he became board chair of Prince and later bought the company. But Head was not content to sit in the boardroom. He returned to the workbench, challenging the company's engineers not only to help him design an improved racket for his own use, but also to "optimize it so it works well for everyone who wants to play tennis."[16] Mirroring his experience with skis, Head channeled his frustration with traditional wooden equipment to develop a stronger, lighter, and more flexible aluminum tennis racket that was easier for recreational players to use. Head leveraged his earlier sports equipment development and commercialization expertise to develop what became the Prince Classic tennis racket.

In 1975, Head applied for a patent on his oversized aluminum tennis racket design, with ownership assigned to Prince Manufacturing, Inc. Sports purists initially derided the Prince racket as a "snowshoe" for its oversized head. Standard rackets of that period had heads with 78 square inches of surface area, while Head's racket had 108 square inches (Figure 5.4).[17] But recreational tennis players loved it thanks to its light weight, large sweet spot, and increased power.

The Prince Manufacturing Company received a huge boost when African American tennis pioneer Arthur Ashe adopted a Prince tennis racket around 1975. During that year, Ashe defeated Jimmy Connors to win Wimbledon, won the World Tennis Championship against Björn Borg, and received the inaugural Association of Tennis Professionals (ATP) Player of the Year Award. Professional players took additional notice in 1978 when 16-year-old Baltimore native Pam Shriver shocked the tennis world by using a Prince racket to beat top-seeded Martina Navratilova in the semifinals before finishing a close second to Chris Evert in the finals of the U.S. Open.

The company continued to innovate and boosted its profits by introducing tennis rackets made of other materials, including magnesium, graphite, and boron. According to a company history, "Prince captured its 30 percent US market share and nearly 25 percent global market share during the late 1970s and early 1980s by selling its line of 108-inch rackets. Besides generating more sales from rackets than any other manufacturer in the world, Prince was able to generate some of the highest profit margins in the industry because its rackets were protected by US patent 3,999,756."[18]

Head retired in 1982 after Chesebrough-Pond's bought Prince (of which he owned a controlling share) for about $62 million. He spent his later years doing philanthropic work in Baltimore and Aspen, Colorado, with his second wife, Martha ("Marty") Becker Fritzlen. For his innovative contributions to sports, Head was elected to the U.S. National Ski and Snowboard Hall of Fame in 1979 and was inducted posthumously into the U.S. National Inventors Hall of Fame in 2017.

Describing his inventive motivations, Head said, "I invent when it's something I really want. The need has to grow in your gut. People who go around trying to invent something generally fall on their tails. The best inventions come from people who are deeply involved in trying to solve a problem."[19] Head was a mediocre athlete, but a brilliant inventor and entrepreneur. His inventions helped democratize skiing and tennis so that professionals and weekend warriors alike could enjoy and excel at sports.

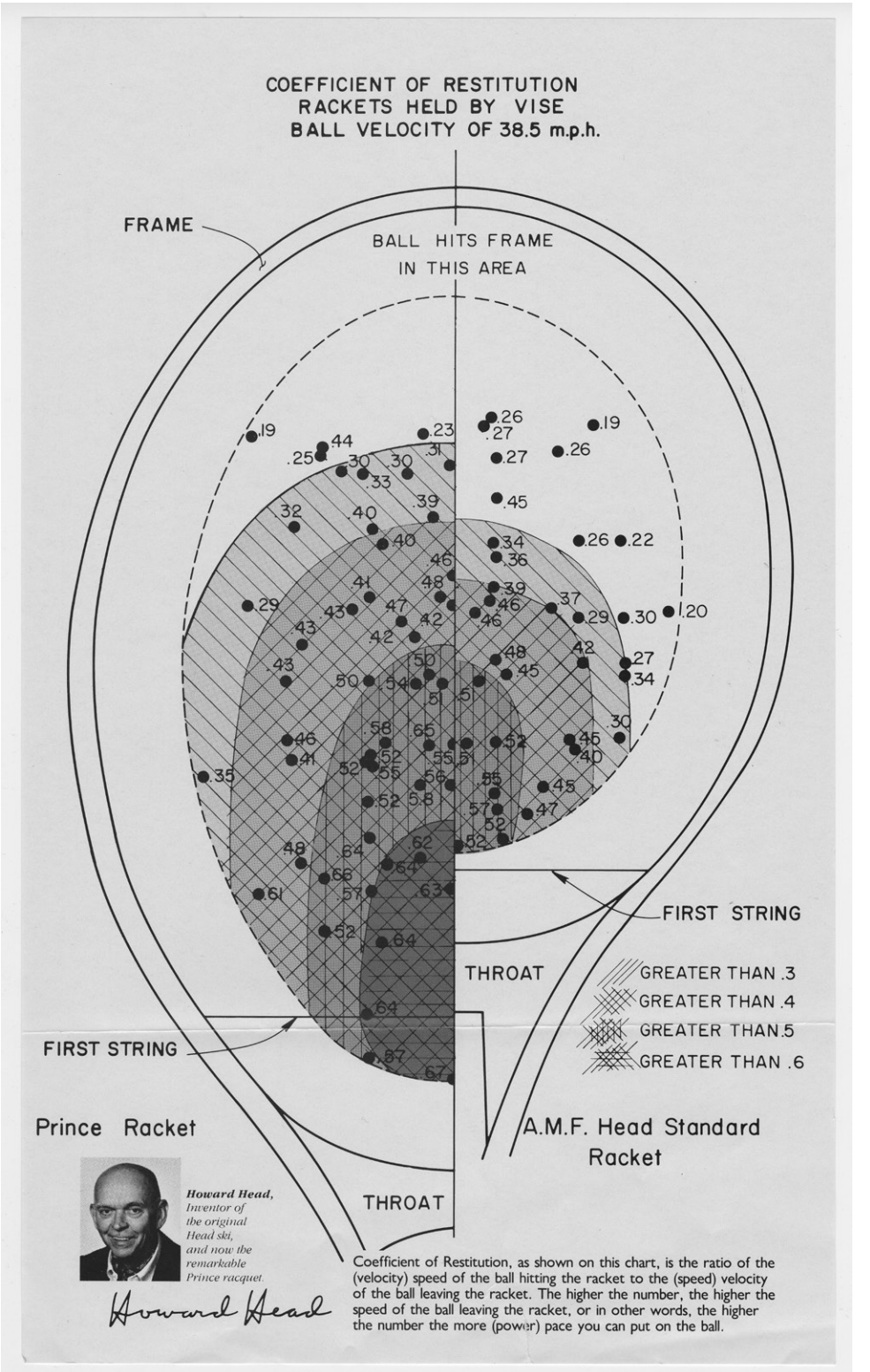

FIGURE 5.4. Prince tennis racket product information, circa 1975. Credit: Howard Head Papers, Archives Center, National Museum of American History, Smithsonian Institution.

Marilyn Hamilton and the Quickie Wheelchair

Wheelchair-based sports are a relatively recent phenomenon as societies have become more understanding, welcoming, and supportive of people with disabilities. During and after World War II, as injured soldiers, marines, airmen, and sailors returned to the home front, medical professionals in the United States and Europe introduced wheelchair-based sports as part of their patients' rehabilitation processes. Veterans engaged in sports not only for physical activity but also as a competitive outlet. Initially, they used standard hospital wheelchairs, but soon sought out lighter-weight designs to allow more agile movement and provide options for diverse types of physical disabilities. Over time, the idea of wheelchair-based sports broadened beyond medical rehabilitation to recreational and Paralympic athletes, who proactively engaged in technical modifications to their chairs.[20]

Born in 1949, Marilyn Hamilton would become an inventor of a more maneuverable sports wheelchair in the 1980s. Hamilton grew up in San Joaquin, California. An active child, she enjoyed waterskiing, snow skiing, go-kart racing, and tennis. She also loved building model airplanes with her brother. A self-described "adrenaline junkie," Hamilton took up hang gliding in her twenties, using a glider customized for her size, weight, and range of motion. In August 1978, she made a split-second mistake at takeoff and suffered a crash landing that left her paralyzed from the waist down. Despite her life-changing accident, Hamilton was determined to continue her active, athletic lifestyle. She was buoyed by her uncle, who had suffered his own spinal injury and used a wheelchair. He assured Hamilton that she could do anything she put her mind to.

However, her heavy stainless-steel hospital wheelchair made it difficult for Hamilton to stay active in sports. Even the sports wheelchairs available at that time offered little advantage in performance. So Hamilton reached out to two friends who were hang-glider designers, Don Helman and James ("Jim") Okamoto, to collaborate on inventing an easy-to-maneuver wheelchair made with the same lightweight aluminum tubing and components used in hang gliders and racing bicycles.

From the beginning, they imagined a fully adjustable commercial wheelchair that could be customized for a broad range of users, not just Hamilton. They assessed different materials and components, then rode around in test chairs. As she described it, "The configuration of the wheelchairs changed with the material. We were able to add adjustability to it. We could put brackets on the side and sit higher or lower. We could put on different kinds of wheel locks [and] change the size of the front castors. So, we really tried to look at how to create more function and usage with adaptability that could also be manufactured in a streamlined way."[21]

In 1979, Hamilton cofounded Motion Designs with Helman and Okamoto to manufacture and commercialize the wheelchair—one of the first ultralight designs—which they later marketed under the brand name Quickie (Figure 5.5). Helman and Okamoto worked on the manufacturing and tooling while Hamilton worked primarily in sales. For the first three years, the cofounders did not take a salary. Sales grew primarily by word of mouth as people saw Hamilton and others using the chair.[22] It helped that Hamilton had a high profile as a competitive tennis player, winning the U.S. Open Women's Wheelchair Championship in 1982 and 1983.

Although Quickie wheelchairs were more expensive than most other wheelchairs on the market, consumers and physical therapists saw the product's potential for better comfort and

FIGURE 5.5. Marilyn Hamilton with her Quickie cofounders Don Helman and Jim Okamoto, 1984. Credit: Courtesy of Marilyn Hamilton.

performance. The company helped change the non-disabled population's perception of people in wheelchairs by using colorful marketing materials that featured active people. They also called their products fun names like "Breezy" for chairs aimed at seniors and "Zippie" for chairs used by teens.

The Quickie had a significant impact beyond sports. It was one of the first ultralight designs that revolutionized chair mobility in the early 1980s. Previously, wheelchairs were designed mainly for indoor use, and users often needed assistance. Their heavy and bulky designs quickly exhausted anyone trying to push them independently. Hamilton's versatile and ultralight sports wheelchairs provided thousands of people with disabilities the opportunity to independently engage in a variety of sports and other activities. As Hamilton proudly stated, "Some of my greatest joys were changing people's lives with freedom of movement."[23]

As Motion Designs grew steadily, the firm's infrastructure needs eventually outpaced what the founders were willing to manage. In 1986, Hamilton, Helman, and Okamoto sold the company to Sunrise Medical, where Hamilton stayed on as vice president for global strategic planning. Hamilton later moved on to pursue her interest in mechanical stimulation devices that help people with neural conditions improve their muscle and bone health and performance. Reflecting in 2021 on the Quickie wheelchair, Hamilton noted, "I truly loved seeing our athletes do things that they'd never done before. I love being part of a world-class team of rebels, revolutionizing mobility change, pioneering sports, and helping people go after their dreams."[24]

Frank Nasworthy and Urethane Skateboard Wheels

In looking for new ways to have fun, inventors sometimes tweak existing products by seeing possibilities where others do not. In the 1960s, skateboarding had its first mini boom of popularity. The sport originated with surfers, who used wooden skateboards as a fun way to exercise and train on land when the waves were not good. The first mass-produced skateboards became popular among pre-teens and teens as a fun way to play and get around town.

The rapidity of the sport's initial rise and fall was notable. According to a 1975 *Sports Illustrated* article, "Ten years ago, there were a lot of warehouses full of skateboards. The fad had started in Southern California's beach towns, where kids nailed roller-skate wheels to miniature wooden surfboards and whiled away the days when the ocean was flat, scaring the daylights out of passing motorists. The craze swept inland and . . . fifty million skateboards were sold [in 1965]. Manufacturers sponsored teams . . . and orthopedists coined a term, 'skateboard fracture,' for a shattered elbow. And by January 1966 it was all over. The skateboard was a good toy, but the kids had pushed it to its limits and were on to new challenges."[25]

For the sport to ascend again, skateboarders needed improved equipment. George Franklin ("Frank") Nasworthy, Jr., was a surprising catalyst for a skateboarding renaissance. Born in 1951, he moved around often as a Navy kid. Frank learned to surf in the Atlantic Ocean when his father, Captain George Franklin Nasworthy, Sr., was stationed in Norfolk, Virginia. Frank Jr. also rode around Norfolk on a skateboard with then-typical clay wheels. The family eventually moved to Annandale, Virginia, outside Washington, D.C., where his father served as the NATO liaison for the joint chiefs of staff. "At this point in my life," Nasworthy recalled, "I was a land-locked surfer having moved from the Virginia Beach area." While skateboarding was fun, surfing remained his main passion.

After graduating from Annandale High School in 1969, he began studying engineering at Virginia Tech. In the spring of his first year at college, Nasworthy was suspended for civil disobedience after protesting the infamous Kent State shooting and the Vietnam War. Abruptly out of college, he found his old skateboard in his parents' garage and spent time with his friend Bill Harward skating all over Washington, D.C., and Northern Virginia. He said, "We skated on weekday nights and went to Cape Hatteras surfing on the weekends until the winter's cold prohibited it."[26]

During the summer of 1970, Harward took Nasworthy to visit a friend, Richard Heitfield, at Heitfield's father Vernon's nearby Creative Urethanes factory. Among other things, this Virginia factory produced polyurethane roller-skate wheels.[27] (Vernon Heitfield invented them initially so his younger son Tom could roller-skate indoors in the basement

of their house.) While at the factory, Nasworthy and Harward saw some rejected wheels, which they were allowed to take away with them. They put the urethane roller-skate wheels on their skateboards at home and immediately realized what an improvement they were over standard metal and clay wheels. With the standard wheels, one small rock or rise in the road would cause the board to stop. With the new urethane wheels' smoother ride and better traction, a skateboarder could continue rolling through an extended tight turn, where a metal or clay wheel would fishtail and skid out. This chance visit to Heitfield's Creative Urethanes factory would change Nasworthy's life—and skateboarding's trajectory.

The following summer, Nasworthy traveled with Harward to California to visit friends. While Harward stayed in Los Angeles, Nasworthy moved south to Encinitas in San Diego County. Nasworthy surfed the Pacific and would ride his skateboard when the waves were low. Given the frustrations he had experienced using clay wheels, he asked his father to send his old skateboard with the Creative Urethanes wheels that he had left behind in Virginia. As Nasworthy explains it, "The Creative Urethane wheels were unique in that they used a 'loose ball bearing assembly' [which allowed easy removal and replacement]. This loose bearing wheel allowed the Creative Urethane wheels to be easily adapted to the old skateboards lying around in everyone's garage. This was a fundamental aspect that connected the Creative Urethanes wheel to the existing base of old skateboards."[28]

Nasworthy asked Creative Urethanes for more of the roller-skate wheels, but they said they couldn't sell wheels that were already promised to other customers. So Nasworthy placed his own order with Creative Urethanes for 1,000 urethane wheels, which he specified should be two inches in diameter and one inch wide (plus 16 loose ball bearings per wheel to attach them to the board).[29] Then he formed the Cadillac Wheels Company. The name signaled that his skateboard wheels provided the smoothest possible ride, just like the luxury car company.

To get the word out, Nasworthy tried peddling the skateboard wheels to surf shops up and down the California coast, but few showed interest as they did not understand what was special about the urethane wheels. Plus, at $8 a set, they were more expensive than the standard clay version. So Nasworthy turned to word of mouth; he began by giving hundreds of wheels away for free, then started selling them through ads placed in surfing magazines. The news spread like wildfire among the skateboarding community; the new Cadillac Wheels provided a better, quieter ride with greater traction and more flexible movement (Figure 5.6).

Nasworthy's urethane wheels had an important impact even beyond skateboarding. "The wheels gave skateboarders the ability to do things on land that surfers only dreamed of doing in the ocean," Nasworthy explained. So surfboard designers started making their classic longboards shorter and more maneuverable. According to Nasworthy, skateboards with his urethane wheels "literally drove the surfboard industry to design [shorter surfboards] to allow surfers to act out on water what they now were able to do on land using their skateboards."[30] This rapid success meant other skateboard companies quickly jumped into the business of selling urethane wheels. Even Creative Urethane expanded its sales from roller-skating to skateboarding.

The Cadillac urethane skateboard wheels inspired other inventions and innovations in skateboarding. These improvements included sealed bearings, which made the wheels dirtproof as well as more reliable and safer; the development of more responsive "trucks,"

FIGURE 5.6. Frank Nasworthy with a red K-73 Cadillac Wheel. Credit: Courtesy of Lucia Griggi.

or axles that attach the wheels to the skateboards; the use of fiberglass and molded plastic decks instead of wood; and the addition of "kicktails" to the board design, which helped skaters spin and pivot. Collectively, these innovations helped drive a second resurgence in skateboarding during the 1980s. Both boys and girls used their boards to have fun, express their creativity through new tricks, and travel from place to place with a mode of transportation that was cheaper and easier to carry around than a bike.[31]

Without the financial wherewithal to mass-produce his urethane wheels, Nasworthy decided to license his version to Bahne and Co., which packaged Bahne skateboards with Cadillac wheels. At their height, they were selling up to 20,000 skateboards a month. Then, as described in a 1984 magazine article: "The smell of money attracted all kinds of predators, and they began circling the industry, hungry for some of the spoils . . . . Fly-by-nighters were buying second-quality wheels from reputable manufacturers and selling them for more than top-quality wheels were worth."[32] The skateboard industry was changing. Soon premium precision wheels replaced Cadillac's style of loose-bearing wheels, and skateboard materials and designs entered a period of rapid innovation.

Nasworthy decided it was time to move on. He returned to college, earning a bachelor's degree in applied mechanics at the University of California San Diego in 1984. Then he pursued a different kind of inventive career as a mechanical engineer, primarily at Hewlett-Packard, where he helped to develop the first wide-format thermal inkjet printer. Nasworthy was listed as a co-inventor on multiple patents, including one related to paper handling.

Now retired, Nasworthy continues to surf, but leaves skateboarding to his children and grandchildren.[33] Nevertheless, Nasworthy's legacy in the sport is secure. As the Skateboarding Hall of Fame and Museum summed up for his 2012 induction, "Frank Nasworthy paved the way for the second great boom of skateboarding in the 1970s. The

new Cadillac Wheels made for more traction and opened the door to the low-slung surf style and further evolution into banks and soon thereafter vertical pool riding. In effect, Frank Nasworthy helped to evolve skateboarding by quite literally reinventing the wheel."[34]

Tahira Reid and Her Automated Double Dutch Jump-Roping Machine

When asked what inspired them to become inventors, many adults tell stories about playing as children and how they adapted those playful skills and habits of mind for their work in adulthood.[35] In some cases, they revisit childhood interests and develop them further using skills and experience gained later in school or on the job. One such example is Tahira Reid. As a child growing up in the Bronx, New York, in the early 1980s, Reid loved the popular neighborhood game of "double Dutch" jump-roping. This fun sport eventually propelled her to become an inventor and engineering professor.

Brought to the United States by immigrants from the Netherlands, the two-rope style of double Dutch jump-roping became popular after World War II, especially among Black girls living in urban areas. On streets, in playgrounds, and in parks, two people would turn two ropes simultaneously, in opposite directions, while one or more players jumped. "Using clotheslines, braided rope, and even super-long phone cords, they took it onto the streets . . . . The *tick-tack-tick-tack* of the ropes laid down a beat not only for jumping but for rhymes and game songs that became part of the tradition. Often turned away from guy-dominated sports, girls doing double Dutch transformed jump roping into a realm all their own."[36] In the 1970s, the game evolved alongside the rise of hip-hop music and culture in New York City. Soon "street double Dutch" extended to cities such as Baltimore, Chicago, Philadelphia, and Washington, D.C. It continues to be enjoyed primarily by Black girls and women as a tradition handed down from generation to generation.

Young Reid would occasionally become frustrated when she could not find two friends in her Bronx neighborhood to turn the ropes when she wanted to practice for her school's double Dutch team. Fortunately, she was always encouraged by her Jamaican-born parents and public-school teachers to put her math, science, and creative skills to work. So while only in third grade, Reid drew a picture of an automated double Dutch jump-roping machine for an elementary school poster contest and won first prize (Figure 5.7).[37]

Reid's idea lay dormant for years. Later, as an undergraduate at Rensselaer Polytechnic Institute (RPI), she was inspired to return to her childhood idea while taking a popular class called "Introduction to Engineering Design" with Professor Burt Swersey. Swersey had also grown up in the Bronx and was familiar with double Dutch. He encouraged Reid—a self-described shy sophomore and one of only two Black students in the class—to develop the idea further. "I opened my notebook and had all these drawings, all this stuff sketched out. It was pretty clear that I was sort of 'a mad scientist' with all the writing I had in my notebook and all the work I had put in to think about it."[38] With Swersey's guidance, Reid shared her idea and sketches with her team, and they decided to build an automated double Dutch jump-rope machine as their final group project for the course.

FIGURE 5.7. Tahira Reid's third-grade sketch of a double Dutch jump-rope machine, 1986. Credit: Courtesy of Tahira Reid.

With her teammates, Reid built, evaluated, and tweaked her invention. After designing multiple iterations using different materials, Reid developed her idea into a working prototype at RPI (Figure 5.8), leading to two grants from the National Collegiate Inventors and Innovators Alliance (now VentureWell). In 1999, Reid and her classmate Andrew Burdick earned U.S. patent 5,961,425 for their automated jump-roping device. The next year, Reid demonstrated the double Dutch machine at the Smithsonian's National Museum of American History as part of the Lemelson Center's symposium "The Playful Mind." Reid, Swersey, and several co-inventors received a second patent for the jump-roping device in 2003. As the patent states: "The jump rope device employs DC [electric] motors to rotate two ropes and a dual closed loop control system is used to synchronize the turning movements of the two ropes. An infrared beam is used to monitor the use of the machine by a user. Each motor housing is mounted on a tripod support which can be collapsed for ease of transportation and storage."[39]

Thanks in part to Swersey's advocacy, Reid's story gained national acclaim.[40] In 2000, Reid was featured on NBC's *Today* morning show, where she explained the machine's operation to cohosts Katie Couric and Al Roker, as young jump-ropers showed off their skills outside the show's New York City studio. Reid's story was also featured in two children's books and in the Spring 2017 New York State English and Language Arts exam for fourth graders.[41] Reid has spoken about her story serving as an inspiration for other young aspiring inventors—especially women and young people of color, who are often overlooked and undervalued in the innovation ecosystem.[42]

FIGURE 5.8. Tahira Reid testing the double Dutch machine prototype, with Professor Burt Swersey in the background, 2000. Credit: Courtesy of Tahira Reid.

Reid received both her bachelor's and master's degrees in mechanical engineering from RPI, and in 2010 she earned a PhD from the University of Michigan in design science, with a focus on mechanical engineering and psychology. In 2012, she joined Purdue University's faculty as an assistant (later associate) professor of mechanical engineering and founded the REID (Research in Engineering and Interdisciplinary Design) Lab. In 2023, she moved into a new role at Penn State as professor of engineering education and associate department head for inclusive research and education.

Her research interests include product and conceptual design, decision-making and behavioral psychology, and integrating human judgments into the design process. With her lab team, Reid applies her expertise to tackle engineering challenges in sustainability, healthcare, and medicine that will specifically benefit the Black community. For example, Reid and her team study the thermal properties of curly hair. Reid notes that "the persistent lack of information on exactly how much heat can be applied to hair without causing permanent damage had long been a source of frustration for Black women, limiting their choices of styling options and products."[43] Reid and her colleagues developed novel techniques to measure the thermal properties of hair and to develop evidence-based guidelines for the usage of flat irons.

Reid did not initially see a clear path to commercialization and set aside her jump-rope device for a few years as she grew busy with her academic career. However, she has not given up on it completely; she hopes to one day manufacture and sell the invention. Regardless of whether it achieves commercial success, Reid's story shows how creative

childhood ideas can become realities. It also shows how aspects of Black culture, or any culture, can provide inspiration for new inventions.[44]

Now a professor, Reid loves mentoring young aspiring inventors, especially Black women like her, to help them achieve their dreams. "Jamaican people say: 'Nothing beats a try but a failure.' If you feel it, go for it; just try. Don't give up, don't let anyone talk you out of what you're thinking," Reid says. "And if you fail, just keep trying again. Don't underestimate your ideas and chase them!"[45]

Conclusion

Humans are intrinsically playful, social, and inventive. As exemplified in this chapter's case studies, people from diverse backgrounds can use their personal and professional knowledge and experiences to solve problems in sports and other aspects of daily life. Indeed, the inventors showcased here created more fun for themselves, even as their inventions broadened access to help others get into the game. For each of these inventors, frustration, perseverance, and a desire to help others became powerful motivators for invention and innovation.

Lisa Lindahl, Hinda Miller, and Polly Palmer Smith combined their costume-design skills and love of running to develop the Jogbra, the first sports bra that truly revolutionized women's participation in sports. Howard Head successfully used his aircraft engineering knowledge and experience to improve equipment for skiing and tennis to help recreational players, himself included, have more fun. After a hang-gliding accident, Marilyn Hamilton and her co-inventors Don Helman and James Okamoto developed the Quickie wheelchair to help people with disabilities more easily take part in their favorite sports. Amateur skateboarder Frank Nasworthy's chance visit to a roller-skate wheel factory changed his sport's trajectory with the introduction of urethane wheels. And college student Tahira Reid turned her childhood dream of an automated double Dutch jump-roping machine into a reality.

What do these inventive people have in common? Each of them faced some challenge with a favorite sport or hobby, including frustration with inadequate equipment, frustration with their own mediocre athletic abilities, or frustration because gender discrimination, physical differences, or other socio-cultural limitations hindered their ability to participate fully in their chosen activity. Rather than succumbing to their frustrations and quitting, they drew on their skills and training to solve the problems they identified. They also borrowed know-how and materials from fields beyond their particular sport, including aerospace engineering and plastics. Each of these inventors also collaborated with business partners and mentors, who shared their expertise, support, and encouragement. Along the way, these inventors took risks, failed, secured patents, raised money, and expended significant effort to promote their inventions. Finally, many of them benefitted from good luck, coming up with the right sports technology at the right time.

Regardless of their commercial success, the diverse inventors featured in this chapter provide an intriguing glimpse into the innately human creative drive to have fun and share that fun with others. Why does this matter for society? While having fun is important, improvements in technology can also be catalysts for building community and technological competencies. People love playing sports, and tapping into that interest can help foster creative and inventive activity among young people like Frank Nasworthy, Black people like Tahira Reid, people with disabilities like Marilyn Hamilton, women like the Jogbra

inventors, and weekend warriors like Howard Head. Thus, inventing for sports has the potential to diversify our invention ecosystem, where women, young people, people of color, and people with disabilities remain underrepresented.[46]

More specifically, why is it important to invent for fun and inclusion? Because sports offer physical, intellectual, emotional, and social benefits for anyone who participates. According to the Aspen Institute's 2021 Project Play report: "The most common motivations students reported for playing high school sports were having fun (81%), exercise (79%), learning and improving skills (66%), playing with and making new friends (66%), and competing (64%). Black students (50%) were more likely than other races and ethnicities to cite college scholarship opportunities as a motivation for playing. Males showed higher motivation rates for winning games and competing than females. Females were more likely to report mental health and well-being, belonging to a group, and improving their college applications as the reasons they play sports."[47] In short, playing sports offers many positive benefits for individuals and communities, so any inventions that expand access and participation will benefit everyone.

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Chapter 6

Fair or Foul? The Governance of Sports Technology

Arthur A. Daemmrich

New sports technologies spark controversy when sportswriters, fans, coaches, and athletes raise questions about fairness, and associations, leagues, and governments regulate their use. Though not new, these controversies intensified during the first decade of the 2000s, when journalists began characterizing the use of breakthrough sports inventions as "technological doping." Drawing an explicit analogy to banned performance-enhancing drugs, the term soon entered common parlance to describe the seemingly unfair advantages conferred by a range of newer technologies used by both competitive and recreational athletes. In 2006, the World Anti-Doping Agency (WADA) updated its 2003 anti-doping code to include mention of technological doping, defining it as using technologies that are harmful to health, enhance performance beyond training alone, or undermine the "spirit of sport."[1] WADA's regulatory remit, however, remained focused on performance-enhancing drugs, leaving each sport's governing body with little guidance to assess and make decisions about technological inventions like the ones described throughout this book.[2]

This chapter explores how key constituents—inventors, engineers, and athletes—interact with decision-makers—leagues, international associations, and governments—to negotiate the governance of performance enhancement, safety, fairness, and participation as sports inventions come into wider use. Specifically, it examines the emergence and resolution of disputes over new swimsuits designed to reduce drag and increase speed, football helmets intended to reduce injuries, "Flex-Foot" prostheses invented for para-athletes who then sought to compete against non-disabled runners, and the use of e-bikes (battery powered bicycles) on off-road trails. In each of these cases, governing bodies struggled to manage the pace of technology change and to set standards that would maintain fair competition, reduce risks to athletes, and strike a balance among competing interests for access and participation. These disputes were only resolved through the imposition of regulatory approaches, such as setting performance standards for equipment or banning certain gear entirely. As a result, inventors of sports technologies increasingly need to plan for additional testing and review when bringing their products to market. The governance of sports inventions, however, remains decentralized, with the consequence that regulatory decisions often lack procedural transparency and predictability that are important to inventors and manufacturers.

So why are some sports technologies accepted as beneficial and fair, while others are rejected as unsafe or unfair? Indeed, some inventions are embraced and quickly adopted across competitive and recreational sport; others are accepted into competition only after

lengthy negotiations involving athletes, team owners, corporate sponsors, leagues, and associations; and yet other technologies are banned entirely from competition, either immediately or after a period of testing and use. While few inventions are banned from recreational sports, some are shunned or limited through peer pressure. Case studies of disputed sports technologies reveal that interested parties—athletes, trainers, teams, leagues, associations, fans, and the media—seek regulation when inventions appear to overshadow athletes, pose health risks to participants, provide an unfair advantage to a specific group or class, or challenge a sport's heritage and traditions. Ultimately, the chapter finds that acceptance and use of sports inventions hinges on governing their fair use, increasingly through scientific testing procedures that are themselves subject to ongoing innovation and negotiation over standards, limits, and expertise.

Sports Governance and Technology Change

Both competitive and recreational sports are governed by official rules and regulations that are codified alongside more informal customs and traditions. Sports that developed since the nineteenth century are characterized by formal organizations stratified at local, regional, and national levels; rules and regulations written and enforced by national or international organizations; and official registrars for statistics and records.[3] For example, in the early twentieth century, officials established federations—such as the Fédération Internationale de Natation (FINA, founded in 1908 to oversee swimming) and the International Amateur Athletic Federation (IAAF, founded in 1912 to oversee track-and-field events)—to organize competitions, keep records, and establish rules for their sports. To update rules, these governing organizations rely on a sequence of evaluation, closed-door committee discussion, and voluntary consensus-building. These procedures worked well for decades, but then struggled to accommodate rapid changes to sports technologies in the 2000s.

The governance of sports encompasses rules, regulations, norms, and practices. Rules establish guidelines for conduct and operate at a general level, while regulations are issued by an enforcing organization and typically have situational specificity. Rules describe proper gameplay and the basis for competition, while regulations govern equipment or specify calls officials should make in a variety of highly specific situations. In practice, and especially outside legal settings, the terms "rules" and "regulations" are often used interchangeably. Throughout this chapter, rules refer to descriptions directing how to conduct fair and competitive play, while regulations govern specifics, including the use or prohibition of new technologies. They are enforced through officials' calls, fines, suspensions, and even bans from competition.

Sports officials and governing federations are also responsible for enforcing regulations that seek to prevent cheating and that determine which athletes are eligible to compete. At the amateur level, a junior ice hockey official might check the birth certificate of a particularly large and fast skater to ensure that they are competing in the correct age group. At the elite level, officials use advanced biochemical screening technologies to test athletes' bodies for performance-enhancing drugs and hand out suspensions for offenders. More controversially, officials administer gender verification tests including visual inspections of athletes' body hair and genitalia or biomedical tests for specific chromosomes and hormone levels to determine the eligibility of athletes who do not conform to gender norms. Officials and sports federations frequently court controversy—or have it thrust

upon them—when updating test protocols for eligibility decisions or adapting standards of fairness as cultural attitudes and social norms evolve.[4]

In sports as in other regulated professions, norms and practices develop over time alongside rules and regulations. Furthermore, many sports have a legacy of "unwritten rules," a set of well-known but uncodified traditions that resist change. These are pronounced in older sports like baseball and can cover everything from not overly admiring the home run you hit to teammates superstitiously not talking about a no-hitter in progress. Despite being called "unwritten," these norms are the subject of dozens of books and hundreds of articles.[5] There are also unwritten practices in recreational play, such as which mountain biker yields the right-of-way when two meet on a steep hill, or how runs can be scored in halfball, a version of baseball played on city streets.

A truism of regulation holds that it grows inexorably over time. For example, rules and regulations written by the U.S. government—the Code of Federal Regulations—expanded from 10,000 pages in 1950 to some 190,000 pages today. Organized sports appear to follow a similar trajectory. American football, for example, evolved from a set of 61 rules that fit on 2 pages in 1876 to its present 74-page rulebook accompanied by over 150 additional pages of detailed game scenarios. The NFL rulebook also includes a 34-page instant replay guide that describes what officials should do under specific situations.[6] That set of regulations was not necessary—or even conceivable—before the NFL introduced instant replay as an officiating tool in 1986.[7]

For all the growth in rules and regulations in different sports, there are few guidelines for rule-making or standards for enforcement by leagues and associations, at least when compared to government-based regulation. In the United States, for example, federal rules follow a sequence from problem identification to data gathering and analysis to proposed rules that are subject to public comment to agency responses and issuance of the final rule. Both the process and the final regulations are subject to legal challenges and judicial oversight.[8] By contrast, each major sport has its own governing body that establishes its own rule-making procedures and may or may not have approaches in place to respond to technology change. As a result, while sports technologies face fewer barriers to market entry than innovations in many other sectors (e.g., pharmaceuticals), decisions by sports' governing associations lack the transparency and procedural legitimacy of government regulation.

The remainder of this chapter presents case studies of regulatory controversies concerning innovative sports technologies that further develop the book's main themes. First, the Speedo LZR Racer and other full-body swimsuits raised questions about managing competition for swimmers wearing breakthrough high-tech suits. Second, innovations in football helmets intended to enhance health and safety contributed to intensified play, greater concussion risks, and a variety of rule-making interventions to protect players. Third, para-athletes who used the Flex-Foot lower-leg prosthesis sought to compete against non-disabled athletes. This raised questions about fairness, since many competitors, officials, and engineers believed that the prosthesis conferred an unfair mechanical advantage. Fourth, when riders of battery-powered e-bikes sought off-road trail access alongside traditional mountain bikers, they forced public land managers to balance among competing definitions of fun for everyone. It is increasingly important, the chapter concludes, that inventors, athletes, coaches, and even passionate fans understand and engage with still-emerging processes for governing new sports technologies.

Competitive Edge: Regulating Full-Body Swimsuits

The invention, use, and regulation of full-body swimsuits between the early 1990s and 2009 vividly illustrate the tensions that arise when innovations disrupt existing competitive dynamics and sports associations struggle to set rules for new technologies. Competitive swimming has been popular since the mid-nineteenth century and featured in every summer Olympics since 1896. For much of the sport's history, elite swimmers have pursued a competitive advantage by wearing ever smaller suits made from lightweight fabrics. But research in the 1990s led to a new generation of full-body, hydrodynamic suits that spurred accusations of technological doping when swimmers began breaking records more frequently than ever before.

In the nineteenth century, swimsuits were made of wool jersey and mercerized cotton; they featured modest designs covering much of the body. Men's suits typically included a tank top worn over snug, knee-length shorts, while women wore a mix of skirt-like bottoms and T-shirt-like tops. In 1927, the Australian immigrant and knitting-mill owner Alexander MacRae developed a suit he called the "racerback," which—somewhat scandalously for the time—exposed women's shoulders. It was banned from many beaches for exposing too much skin, but competitive swimmers soon used silk versions that absorbed less water than wool or cotton.[9]

Renamed Speedo, MacRae's company became a world leader in swimsuit design and manufacturing. The company continued to innovate by creating smaller racing suits made from new synthetic fabrics that were more form-fitting and more impervious to water. Suits made of nylon and elastane (better known by its trade name Lycra) came into use in the early 1970s. Notably, Mark Spitz wore a minimalist suit while winning seven gold medals at the 1972 Munich Olympics, and the East German women's swim team debuted synthetic suits at the 1973 World Aquatics Championships in Belgrade. These suits were quickly adopted across the sport.[10]

Beginning in the early 1990s, manufacturers of elite swimsuits invested in extensive research and product development to create a new generation of swimsuits.[11] Speedo, in particular, sponsored hydrodynamics research and worked with NASA engineers to develop a better understanding of how bodies move through water.[12] Swimsuit designers and swimmers themselves had long assumed that bare skin created the least amount of drag in water. As social attitudes relaxed, suits became ever smaller. However, Speedo's research revealed that highly engineered synthetic fabrics could be more hydrodynamic than human skin.

Accordingly, swimsuit designs shifted to cover more of a swimmer's body with high-tech water-channeling materials. In 1992, Speedo introduced the S2000, a Lycra and nylon blend, as the first "fast suit." Four years later, Speedo and Mizuno debuted the Aquablade, which featured alternating slick and rough water-repellent vertical stripes. Next, Adidas, which was carrying out its own hydrodynamic studies, introduced the JetConcept, a full-body suit with silicone riblets that channeled water along swimmers' backs.[13] In 2000, Speedo followed with the "Fastskin," made of spandex coated in Teflon. Speedo's ads suggested that the suit's design was inspired by an analysis of shark skin.[14] An upgraded version, the Fastskin FSII debuted at the 2004 Athens Olympics.

After a decade of generally incremental innovation in suit designs, Speedo sought to develop a major breakthrough. A focused R & D team, led by textiles expert Fiona Fairhurst, embarked on a multiyear product-development process that drew on the

FIGURE 6.1. Jess Sylvester, Olympic competitor from Great Britain, wearing the Speedo LZR Racer, surrounded by Speedo Aqualab designers and engineers, 2008. Credit: Courtesy of Gary Calton.

expertise of kinesiologists, elite swimmers, textile scientists, biomechanics researchers, and aerospace engineers versed in computational fluid dynamics.[15] The final product incorporated ultralightweight, water-resistant, quick-dry compressive fabric, with panels of rigid polyurethane at the torso, abdomen, and lower back (Figure 6.1). To reduce drag, the suits were manufactured without stitched seams. Swimmers found that they needed 20 minutes to put on the new suits, which compressed their bodies into a low-drag profile.

Celebrated at its 2008 debut as "the most technically advanced swimsuit in the world," the Speedo LZR Racer was embraced by swimmers and appeared poised to permanently change the sport.[16] The sports press was initially enthusiastic, celebrating the suit's reduced drag, ultrasonic welding, and "corset-like effect."[17] Some swimmers, however, raised concerns. For example, American Bill Pilczuk suggested that the new suits provided excessive buoyancy, thereby interfering with "a level playing field."[18] But the era's dominant swimmer, Speedo-sponsored Michael Phelps, had helped test the LZR Racer while it was under development and immediately adopted it. Phelps's coach Bob Bowman argued, "the swimmer makes the suit, not the other way around."[19]

Speedo's LZR Racer and other highly engineered swimsuits soon posed a challenge for swimming's governing federation. Founded in 1908 by eight European countries as the Fédération Internationale de Natation (FINA) and renamed World Aquatics in late 2022, FINA sets rules for swimming competitions, specifying everything from pool sizes to permissible swimwear. FINA's governance operated through published rules and the certification of events and results, with a "Technical Congress" meeting every four years to vote on any changes.

FINA operated on a largely voluntary basis through the early 1980s but began to professionalize when it opened and staffed new offices in Lausanne, Switzerland, in 1986.

The federation soon found itself responding to an increased pace of innovation in swimming. Responding to the backstroke swimmer David Berkoff's "blastoff," in which he swam face-up underwater for 35 meters before surfacing, FINA ruled in 1988 that swimmers' heads must break the surface within 10 meters of the starting blocks (extended to 15 meters in 1991). The same distance was set for other strokes in 1998. Other FINA rule changes specified that breaststroke swimmers could make only one dolphin kick when turning and eliminated restarts after a false start.

But even as FINA responded to innovative swimming techniques with new rules, it took a laissez-faire approach to swimsuits, approving each new Speedo, Adidas, or Arena product after a quick review. However, Speedo's LZR Racer generated considerably more attention than its predecessors when it debuted during the 2008 Beijing Olympics. Critics had been arguing since at least 2000 that the new generation of suits should be considered performance-enhancing equipment. They also noted that the suit was initially only available to Speedo-sponsored athletes.[20] In response, the International Olympic Committee required Speedo to make the suit available to all competitors. That put swimmers and entire national teams in the unenviable position of breaking sponsorship contracts and purchasing the $600 LZR Racer to keep up with rivals.[21]

These concerns appeared validated as swimmers wearing the LZR Racer in Beijing set world records 25 times at 34 events, resulting in 23 distinct world records. In addition, swimmers wearing the LZR suit broke Olympic records 65 times, setting 30 distinct Olympic records. In some races at the Beijing Olympics, the top five finishers all swam faster than the prior Olympic record. Speedo claimed that athletes wearing the LZR Racer won 92% of swimming medals in Beijing. Swimmers who used the LZR suit broke an additional 43 records in 40 events at the 2009 World Swimming Championships. In total, elite swimmers who wore the Speedo LZR Racer broke 130 swimming records in 2008 and 2009.

Debates over the new suits dominated sports journalism and the popular press, even as Speedo's competitors quickly developed their own new full-body suits. Phelps had a change of heart after losing to Paul Biedermann in the 200 m freestyle at the 2009 World Championships, warning he would leave the sport if FINA did not intervene.[22] Biedermann, who had not previously won an Olympic or World Championship medal, was wearing a new suit made by Arena and cut more than four seconds from his 200 m freestyle times from the prior year.

Meeting as the 2009 World Championships were underway, FINA's General Congress voted overwhelmingly in favor of new rules that held, "No swimmer shall be permitted to use or wear any device or swimsuit that may aid his speed, buoyancy or endurance during a competition."[23] Widely understood as a reaction to the outsized technological advantage conferred by the full-body suits, the new rules prohibited men's swimsuits above the waist or below the knee. Women's suits, meanwhile, were prohibited beyond the shoulders or below the knee. Since then, swimsuits have undergone a more rigorous review by FINA, whose staff measure them against specified parameters for permitted materials, thickness, buoyancy, and permeability.[24] After the ban, new record-setting dropped back to a much slower pace. At the 2012, 2016, and 2020 (held in 2021) Olympics, swimmers set between six and nine new world records, on par with historical averages.

Inventions that help athletes go faster, leap higher, or otherwise surpass the current competitive norm can lead to accusations of technological doping. As the LZR Racer case documents, sports associations like FINA/World Aquatics may not be prepared for an

accelerating pace of technology change. And while fans, athletes, and governing associations might prefer a linear sequence of technological development from invention to testing to review to use in competition, inventions that provide a competitive edge instead follow uneven pathways. Inventors work closely with athletes to test and tweak new gear. Those athletes, in turn, want to use the best equipment possible when competing. In the case of full-body swimsuits, advances came in rapid succession and FINA was slow to react even as the innovations impacted competition. The eventual establishment of precise rules and specifications for swimsuit size and materials means that World Aquatics now acts as a pre-clearance regulator that must approve gear before it can be used in competition.

Health and Safety: Regulating Football Helmets

Many sports have histories of intense and violent competition that result in injuries to players. In turn, concern for athlete safety has been a key motivator for inventors who have developed new protective gear, from medieval jousting armor to today's football padding and helmets. Derived from nineteenth-century English rugby, American football was notoriously dangerous in its early years and has remained problematically injury-prone ever since. Inventors have long been motivated to make the sport safer, especially through helmet innovations. While the history of helmets is largely one of incremental change, three broad eras characterize the past 130 years: leather headwear starting in the 1890s, polycarbonate helmets adopted in the 1950s, and new high-tech helmet materials invented since the mid-2010s. Yet as with the case of high-tech swimsuits, football associations and leagues only recently began to set standards and regulate protective gear.

In reviewing the history of football, one can identify a triangular relationship among new gear, changes to gameplay, and new rules and regulations (Figure 6.2). Ironically, inventions intended to minimize injury have historically led to more intense play, contributing to concussions and increased risk of what is now identified as chronic traumatic encephalopathy (CTE).[25] While many sports have a similar trilateral relationship among innovations in gear, gameplay, and rules, football offers an especially well-documented case.

American football emerged at U.S. college campuses in the 1870s as a variation on rugby. Injuries were common, since few players wore protective gear and kicks and punches were largely overlooked. Moreover, teams ran plays such as the notorious "flying wedge," in which the ball carrier charged at opponents while running behind a V-shaped formation of blockers.[26] Seeking to limit abrasions, broken noses, poked eyes, and knocked-out teeth, former player Arthur Cumnock patented a mask in 1891 for "protections of the nose and to some extent of the forehead and mouth."[27] Padded headgear made its debut in the mid-1890s when Spalding designed and sold a "head harness" and the retailer Thomas Larwood, Jr., patented a leather "head-protector."[28] Some physicians observed concussions and warned of their harm, but most held that traumatic neuroses—a broad category for behavioral change related to brain injury, first identified by Sigmund Freud—were not caused by football.[29]

As more players began using leather helmets and face masks, injury rates spiked due to intensified play. In 1903, the Intercollegiate Football Association banned helmets made of "hard or unyielding" materials. Then in 1905, a wave of 19 fatalities led several universities to suspend their football programs. President Theodore Roosevelt—whose son played football for Harvard—convened a group of college coaches at the White House and called for reforms. Encouraged by Roosevelt, the National Collegiate Athletic Association

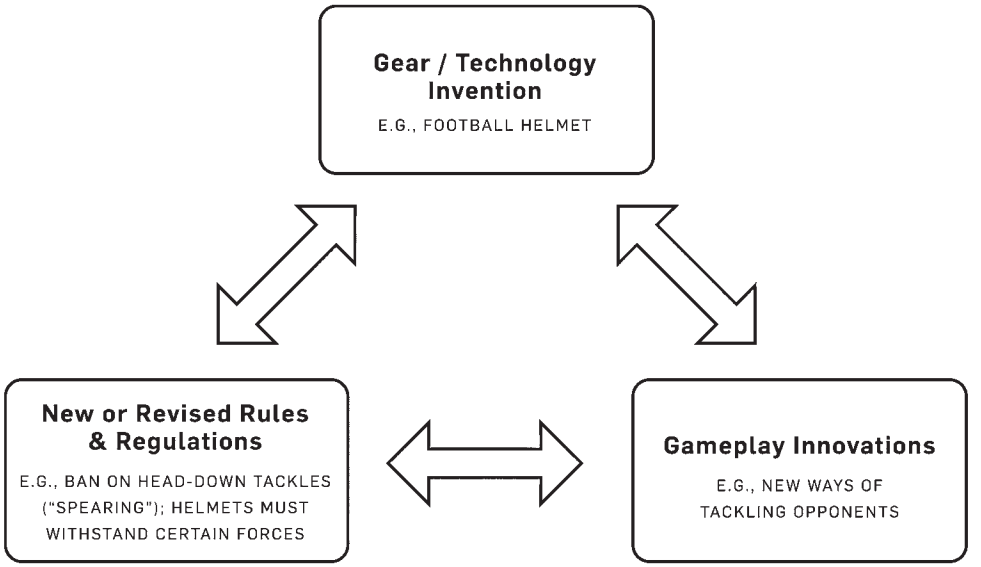

FIGURE 6.2. The invention cycle in American football. Credit: Arthur Daemmrich.

(NCAA) was formed in 1906 and developed rule changes to govern both technology and gameplay. The NCAA banned the flying wedge, mandated seven players on the offensive line, established a "neutral zone" between the offensive and defensive teams, and permitted the forward pass. Together, these changes reduced collisions among large groups of players and shifted play from brute force toward greater use of strategy.[30] Meanwhile, inventors developed dozens of new helmets from the mid-1900s through the 1930s. These helmets used firmer leather and incorporated ear protection, attachments for nose guards, and canvas webbing that cradled the skull during collisions. Yet helmets were heavy, hot, and uncomfortable and many players refused to wear them.

A second era of innovations in helmets, play-calling, and rule-making started in 1939, when the NCAA required collegiate football players to wear helmets. That same year, John Riddell and his son invented a plastic shell helmet. By 1943, the NFL also required players to wear helmets, though the onset of World War II limited the supply of synthetic plastics. After the war, Riddell reformulated its plastic blend after dozens of helmets shattered upon contact in 1948. Riddell's helmets gradually replaced leather helmets over the next few years and the firm gained a dominant market position (Figure 6.3).

Improved body padding also came into use mid-century, including shoulder pads made of foam with a hard plastic exterior. Gameplay again intensified, undermining the protective benefits of the helmet and pad technologies available at the time. Notably, the NCAA's Equipment and Injuries Committee documented 32 fatalities in organized football from head and neck injuries in 1967 and 36 more deaths in 1968. Seeking to encourage the development and use of better safety gear, a group of organizations including the NCAA, the American College Health Association, the National Federation of State High School Associations, and the Sports Fitness Industry Association formed the nonprofit National Operating Committee on Standards for Athletic Equipment (NOCSAE) in 1970. NOCSAE then established helmet test standards and started to certify new equipment. Subsequent helmet innovations included padding to absorb impact forces, adjustable air

FIGURE 6.3. Football helmets through the eras, from left: 1946 MacGregor leather helmet, 1979 Riddell plastic helmet worn by running back Franco Harris in the 1980 Super Bowl, and 2023 VICIS Zero2-Matrix ID Trench Helmet. Credits: Leather and plastic shell helmets photos by Hugh Talman and Jaclyn Nash, respectively, National Museum of American History, Smithsonian Institution; Vicis helmet photo courtesy of Vicis.

bladders to provide a customizable fit to the head, new facemask designs, stronger chin straps, and visors to protect the eyes. Rule changes at the NFL and NCAA incrementally stopped pile-on tackling, prohibited players from grabbing face masks, and limited what counted as legal contact when blocking. To many observers, football appeared to be getting safer. Head-injury fatalities declined through the 1980s, with no documented deaths in 1990 for the first time since reporting started in 1931.[31]

A third era of helmet innovations, changes to play, and rule revisions took hold in the 2000s as dementia, depression, and suicides among well-known former NFL players were causally linked to repeated concussions suffered while playing.[32] Dozens of inventors at universities and start-ups responded to the concussion crisis by accelerating the pace of innovation in helmet technologies. In 2022, for example, inventors earned 87 U.S. patents on "impact absorbing" helmets (48 of which explicitly referenced football), compared with 23 helmet patents in 2000 (6 mentioning football) and 6 patents in 1990.[33] Whereas leather helmets sought to protect against abrasions and plastic helmets from skull fractures, the new helmets aimed to soften contact and slow down movement of the skull in the hopes of preventing concussions. At the same time, a new wave of rule changes for professional, collegiate, and youth leagues sought to reduce risks through new regulations on tackling and implementation of a concussion protocol that included testing and mandated recovery time.[34]

Nevertheless, innovative helmet manufacturers have found the market challenging. For example, VICIS began as a University of Washington spin-off, raised $85 million from investors, and began supplying NFL teams with a breakthrough helmet in 2017. Yet the company found it difficult to turn a profit because of a limited market. Leading-edge helmets are priced at $900 or higher, making it challenging for anyone other than elite players or wealthy teams to access top-rated gear. Facing bankruptcy, VICIS was acquired in 2019 by Innovatus Capital Partners and consolidated under the Certor Sports brand alongside Schutt and Tucci, two other sports helmet manufacturers.

Amid growing recognition of concussion risks, participation in high school football declined by about 10% during the 2010s, falling below one million participants nationally in the 2021–2022 school year. The number of youths aged 6 to 12 playing tackle football has declined by 18%, offset in part through the growth of flag football.[35] In response, the NFL has committed funding for further research into brain health and concussion risk

even as it supports initiatives for safer youth play. Meanwhile, even leading-edge helmets carry the dire warning, "No helmet can prevent head, brain, or neck injuries, including paralysis or death, that a player might receive while participating in football."[36]

The football helmet's evolution over time offers a striking case study of the interplay of changing societal norms for safety, evolving knowledge about risk, helmet technology innovations, changing approaches to tackling and blocking, and new rules and regulations. Football is violent and remains one of the most dangerous sports in the United States, with injury and concussion rates among high school athletes double those of any other sport.[37] Since the early twentieth century, leagues have sought to use a combination of protective gear and rule changes to improve safety. However, improved helmets and other safety equipment gave players greater confidence and led to the unintended outcome of more violent tackling and riskier play.

Only in the 2000s did the NFL become actively engaged in funding scientific research and helmet technology development. Similar to World Aquatics oversight of swimsuits, the NFL now actively regulates technology change. The league provides players with a list of approved helmets ranked by outcomes on NOCSAE certification tests. For example, the 2023 list featured 19 helmets from four manufacturers that the league ruled acceptable; it also listed several helmets that were not recommended and even ones that were prohibited from NFL games.[38] Regulating helmets thus involves both setting and updating test standards and annual testing and ranking of the latest models. As noted, inventors are developing protective technologies at a rapid pace; from 2020 through 2023, they earned 217 U.S. patents for new football helmets with impact protection.[39] The league and testing organizations plan to regularly update their rules and encourage innovations in helmet design that align with emerging knowledge about concussions and CTE.

Fairness and Accuracy: Regulating Prosthetic Legs

Sports associations establish rules and regulations to standardize competition and to reduce the risk of injury or harm. These rules also aim to establish a level playing field for competitors. For example, since its founding in 1912, track-and-field's governing body, the International Amateur Athletic Federation (IAAF), which was renamed the International Association of Athletics Federations in 2001 and World Athletics in 2019, has established standard race distances from 100 meters to the marathon. The intent was to allow for comparison of results regardless of when or where the competition was held. Track officials also determine who can compete against whom. Traditionally, men do not run against women and para-athletes do not compete against non-disabled runners. But advances in prostheses over the past thirty years raised questions for track-and-field officials that were not easily answered through direct observation or even laboratory testing. Instead, international sports associations that were formed to promote sports events and advance the interests of a variety of athletes had to set rules for who could compete in the face of uncertainty regarding new prostheses and other technologies, a situation for which they were poorly prepared.[40]

The Paralympic movement has its origins in sports events organized in Great Britain to help rehabilitate wounded veterans after World War II. The first international Paralympics were held in Rome in 1960 and were more oriented to showcasing participation and encouraging athlete development than to competition. As the movement grew, organizers began grouping athletes with similar disabilities who used similar assistive

technologies, such as wheelchairs or prostheses. By 1965, the International Sport Organization for the Disabled (ISOD) had developed a classification system that grouped athletes by disability.[41] The first categorization system had 27 distinct classes but was trimmed to 12 classes for the 1976 Summer Paralympics and further reduced to 9 classes by the 1992 Summer Paralympics. For example, class A1 includes athletes with bilateral (both legs) above-the-knee lower-limb amputations, while class A2 covers athletes with unilateral (one leg) above-the-knee lower-limb amputations. Athletes from the two classes typically do not compete against each other, though competitive times have converged across several of the classes in recent years as prosthesis technology has improved.[42]

The International Paralympic Committee (IPC) was established in 1989 as a global governing body to organize and administer the Paralympic Games and the paralympic world championships. To grow the sport and involve more athletes, the IPC developed more inclusive classifications for athletes that extended beyond the spinal cord injuries or limb amputations covered by ISOD.[43] More recently, the IPC shifted to classifying athletes according to what they can or cannot achieve physically, rather than by the severity of their disabilities.

These disability classification systems remain in place for track-and-field and many other sports.[44] A panel of doctors and kinesiologists determines each athlete's classification by conducting a physical examination and a series of functional, sport-specific tests.[45] From the perspective of many athletes, the process is onerous, intrusive, and unpredictable. As the runner P. David Howe chronicled in a 2008 article, classification upon arrival at the Paralympics is conducted by a medical doctor, a physiotherapist, and a sports technical expert, typically all non-disabled. The outcome determines everything from status (elite or non-elite) to housing assignments.[46]

Maintaining a level playing field among competitors became a divisive issue for both the Paralympics and the Olympics in the early 2000s when competitive runners began using a new generation of advanced prosthetic limbs. In the early 1980s, the biomedical engineer Van Phillips invented the "Flex-Foot Cheetah" lower-leg prosthesis. Van Phillips's left leg had been severed below the knee in a water-skiing accident in the mid-1970s and he was frustrated by cumbersome prostheses that mimicked the appearance of a human leg and foot but failed to propel the body forward.[47] He reportedly conceived the Flex-Foot design by observing the lower legs of kangaroos and cheetahs and the mechanics of athletes using diving boards and vaulting poles. By the 1990s, Phillips had perfected his carbon-fiber, C-shaped Flex-Foot "blades," which store and release energy with each stride.

South African runner Oscar Pistorius, whose lower legs were amputated as an infant due to a congenital condition, began wearing the Flex-Foot prostheses in early 2004 at age 18. His running career took off almost immediately and he won bronze in the men's 100 m and gold in the 200 m at the 2004 Athens Paralympics. Observers credited Pistorius's performance partly to his leading-edge prostheses and started calling him the "Blade Runner." Over the next several years, even as he set records in Paralympic events—by 2008, he held Paralympic world records at 100, 200, and 400 meters—he also began to receive invitations to non-disabled international track meets. In March 2007, he finished just 0.8 seconds off the Olympic qualifying time at the South African Championship 400 m race, making him a contender for South Africa's 2008 Olympic team.[48]

Pistorius's potential participation in the Olympics generated a high-profile controversy. The IAAF, track's international governing body, scheduled a meeting nine days after Pistorius's 2007 run to consider numerous proposed rule changes. But the IAAF Council

adopted only one. Rule 144.2(e) now banned the "use of any technical device that incorporates springs, wheels or any other element that provides the user with an advantage over another athlete not using such a device."[49] IAAF's leadership, many suspected, feared that Pistorius and other para-athletes would use advanced prostheses to run faster than their non-disabled competitors.[50]

Pistorius appealed the IAAF rule to the Court of Arbitration for Sport (CAS) based in Lausanne, Switzerland. The case posed key questions: Did Pistorius's prostheses confer a mechanical advantage? Were they a remediation for his disability or an unfair enhancement?

The IAAF then commissioned scientific and medical experiments to compare Pistorius's mechanical forces—including the downward forces he exerted while running and the energy return from his Flex-Foot prosthesis—against non-disabled "control" sprinters. The tests also measured the sprinters' aerobic exertion, including oxygen use and lactic acid buildup, since some researchers hypothesized that Pistorius burned less energy because his blade prostheses were lighter than human legs.

Following the tests, Peter Brüggemann at the University of Cologne reported that Pistorius used 25% less energy than runners with biological legs. He found that the prostheses enabled Pistorius to run with 30% less mechanical work to lift his body. The IAAF then ruled in January 2008 that the prostheses were ineligible in competitions with non-disabled runners. Pistorius responded by funding his own team of scientists who met in Houston to conduct an alternate set of tests. This second set of tests also considered Pistorius's running style, which involved slower starts because the Flex-Foot provided less initial power than biological legs. Pistorius's experts concluded that his prostheses did not confer a net advantage. With this new evidence, Pistorius appealed the IAAF ruling. Balancing the data, the CAS ruled in May 2008 that Pistorius and his Flex-Foot prostheses were allowed to compete against non-disabled runners in IAAF-sponsored competitions. This precedential ruling also meant that for disabled athletes to be eligible to compete against non-disabled runners, the IAAF would thereafter require data from lab testing of each individual disabled runner wearing their specific prostheses.

Pistorius did not meet the 2008 Beijing Olympic qualifying time, but he did make the South African team for the 2012 London Olympics. He ran the 400 m in 45.44 seconds but did not medal. Pistorius next competed in the London Paralympic Games and won a silver medal in the 200 m race, behind gold medalist Alan Oliveira of Brazil. In an ironic twist, Pistorius accused Oliveira of "running tall" by using overly long prostheses. The IPC refuted Pistorius, and documented that Oliveira met the standards for allowable prostheses relative to his body size. Pistorius's career ended ignominiously when he was convicted in 2015 of murdering his girlfriend Reeva Steenkamp.[51]

Dr. Alena Grabowski and other scientists who worked on the Pistorius case have carried out additional research into biomechanics and continue to help Paralympic athletes provide evidence to the IAAF for reviews (Figure 6.4).[52] IAAF Rule 144.2(e) remains in place, so athletes must prove that their prostheses do not confer a net advantage over biological legs on an event-by-event and technology-by-technology basis.[53] For example, rather than use data from the Pistorius studies, the right-leg amputee long jumper Markus Rehm had to commission new research to document that his carbon-fiber prosthetic leg did not give him an unfair advantage.[54]

Ensuring fairness in competitive sports has never been about making sure that every athlete is the same in every way. But technologies for disabled athletes raise challenging

FIGURE 6.4. Biomechanical engineer and University of Colorado professor Alena Grabowski observes tests on a lower-leg prothesis. Credit: Glenn Asakawa / University of Colorado.

questions about how to maintain a level playing field. Non-disabled sprinters are rightly concerned about running against para-athletes if advanced prostheses provide an unfair advantage. On the other hand, para-athletes understandably seek to compete against non-disabled competitors, since those races typically provide greater visibility and opportunities for larger financial rewards. Taking regulatory governance a step farther than either World Aquatics or the NFL, track-and-field athletes with disabilities must develop and present evidence to the IAAF/World Athletics about their prostheses prior to competing in non-disabled competitions. This approach places the onus of responsibility on athletes who wish to compete, rather than on sports equipment manufacturers or sports associations.

Fun for Everyone: Regulating e-Bike trail Access

Inventions that increase participation in sports and make them more fun to play can also raise tensions that lead to calls for regulatory interventions. In recent decades, new sports technologies have made it possible for more people of all ages, abilities, and conditioning to participate in recreational sports. But disputes over land use and norms for play can be difficult to resolve in the absence of leagues or associations that govern competitive sports. For example, e-bikes—bicycles that use a battery-powered electric motor to assist human pedaling—have raised complex issues regarding access and use of off-road trails frequented by traditional mountain bikers. On land overseen by the government, a regulatory process unfolded that sought to balance competing interests. Notably, participants in

the debate struggled to define whether e-bikes were motorized, and thus equivalent to motorcycles; or human-powered, and thus equivalent to bicycles.

Mountain biking developed as a new sport in the mid-1970s when several groups of bikers in Marin County, California, organized group rides and informal races up and down local mountains. To handle the rugged downhill trails, they started modifying older cruiser bikes, which had thicker frames than light 10-speed bikes. Joe Breeze became known as a key founder of the sport after he won an infamous local downhill race (the "Repack," named for riders having to repack the grease and sometimes replace bearings in the hubs of their coaster brakes during the competition) on his self-built "Breezer 1" in 1978. Within a few years, Breeze, Gary Fisher, and others were building and selling mountain bikes; by 1986, they outsold road bikes in the United States.[55]

As mountain biking gained in popularity, local negotiations unfolded over trail use. While these played out differently around the United States and internationally as the sport grew, government officials gradually differentiated trail uses. Some trails were specifically designated for hiking, horseback riding, or mountain biking, while other trails were marked for common use by all participants. Within the mountain bike community, riders developed peer-enforced norms. On shared trails, for example, mountain bikers would stop for horses and talk to the riders, and yield to hikers going in the opposite direction. By the early 1990s, the U.S. Forest Service—an agency within the Department of Agriculture that manages 193 million acres of land—had marked its trails and unpaved roads to either allow or ban various combinations of hikers, mountain bikers, motorcycle riders, and equestrians.

Starting in the early 2010s, a new kind of bicycle appeared on unpaved roads and smaller trails. Thanks to innovations in materials, battery technology, and compact electric motors, e-bikes evolved from purely street vehicles into mountain bikes.[56] By 2016, online forums were exploding with commentary and back-and-forth accusations about the presence of e-bikes on off-road trails. Some mountain bikers argued that much like motorcycles, the heavier e-bikes would physically damage mountain paths and undermine the trail access that they had negotiated in the 1980s and 1990s. Others expressed concern that the presence of battery-powered e-bikes on off-road trails would threaten the very soul of mountain biking. Mantis Bicycles founder and land-access activist Richard Cunningham observed in 2016, "Sure, it's hard, but the slow going is when mountain bikers talk to themselves, to each other, and look into the faces of other users. It's the part of mountain biking that makes us honest."[57] Advocacy groups including the Sustainable Trails Coalition officially opposed e-bikes on wilderness trails anywhere in the United States (Figure 6.5).[58] But e-bike enthusiasts countered that their battery-powered bikes offered access to nature for the disabled, elderly, and less physically fit, while also reducing carbon emissions if used in place of cars. Articles in *The Colorado Sun* and other newspapers offered compelling stories of children with cancer and people recovering from hip surgery who gained access to the outdoors by riding e-bikes.[59]

By the late 2010s, numerous western states, including California, Colorado, and Utah, ruled that trails open to motorcycles were open to e-bikes, but banned them from other trails. Although e-bikes have pedals and can be human powered, they were increasingly regulated as motor vehicles and grouped with motorcycles. Then in late August 2019, the Department of the Interior (DOI), which oversees the National Park Service (NPS) and the Bureau of Land Management (BLM), which together manage some 250 million acres of land across the country, issued a directive allowing the use of e-bikes on all

FIGURE 6.5. A trail in Wasco County, Oregon, that allows bicycles, horses, and hikers, but prohibits e-bikes. Credit: Gabriel Amadeus Tiller.

DOI-managed routes that accommodate human-powered cycling.[60] In effect, the DOI proposed to classify e-bikes as bicycles, not motorized vehicles.

A rule-making process followed, generating more than 17,000 public comments in a two-month period. The final rule, issued a full year later in December 2020, reiterated that e-bikes were not motor vehicles and tasked NPS superintendents around the country to conduct local reviews, engage communities, and evaluate potential environmental harms when determining trail access. Yet NPS also stated that e-bikes have electric motors and therefore are "not nonmotorized."[61] The Forest Service, however, ruled slightly differently in March 2022—that e-bikes would remain classified as motor vehicles. The agency then limited which trails e-bikers could access.[62] The outcome of the contested rule-making process was ambiguous. Federal agencies responsible for public lands—the Forest Service, Park Service, and Bureau of Land Management—did not agree on how to classify e-bikes. In the face of epistemic uncertainty over human versus machine power, the agencies left decisions to local officials.

Among trail users, the issue also remained imperfectly settled. With limited staff relative to the expanse of land they oversee, public land managers rarely stand at trailheads to enforce access rules. Instead, hikers and bikers police themselves by calling out violators. But new inventions will only make that task more difficult in coming years, as future generations of e-bikes become indistinguishable from human-powered ones to the naked eye.

Many sports inventions, like e-bikes, are designed to encourage fun and increase access. However, traditionalists sometimes complain if they feel that the enabling technology makes a sport "too easy" or somehow undermines its "spirit," as WADA put it in 2006. Incumbents often seek to limit or even ban the new technology. In contrast to the regulation of equipment, clothing, and rules of play established by sports leagues and

international associations, governance of off-road e-biking has involved a mix of government agencies setting rules through a formal regulatory process and very local negotiations on a trail-by-trail basis. Although the process generated confusion, it also empowered local communities to provide greater input on trail use.

Conclusion

As inventors develop, test, and market new sports technologies, and as athletes and coaches create new training methods and techniques for competition, key stakeholders do not passively accept new innovations. Instead, athletes, coaches, sportswriters, fans, and league officials often take conflicting stances regarding the regulation of technology. Governing bodies then intervene, albeit typically only after a controversy has already developed.

The case studies in this chapter and throughout the book reveal that accusations of "technological doping" have a long history, even though the specific phrase is of recent vintage. Calls for regulation arise when new technology creates an abrupt break with sports traditions, when its availability and use is limited to specific teams or only wealthier competitors, when its use would necessitate major redesign of game fields or arenas, or when it interferes with the space needs of other sports. In turn, rules and regulations can inspire additional waves of technology and gameplay innovation, sometimes leading to yet more rules and regulations on how games are played and what equipment is used. Throughout history, commentators have warned that new technology will remove the human element from play and competition. That risk seems overstated, even as the accelerating pace of technology change suggests that sports leagues and associations will come under further pressure in coming years to develop additional regulations.

Sports technology governance by leagues and international associations has a mixed record of balancing the interests of athletes, inventors, teams, and fans. Furthermore, the regulatory processes developed by leagues and associations tend to lack transparency and clear procedural frameworks, while offering little predictability to inventors or athletes on timeframes for decisions. In this chapter's case studies, FINA and the NFL did not initially have mechanisms to review and incorporate leading-edge scientific and medical knowledge into their regulatory processes. Only when the integrity of swimming and football competitions was in jeopardy did the governing bodies design new regulatory approaches. Similarly, the IAAF was not prepared for athletes using prostheses, and has pushed requirements for proof of fairness onto the disabled athletes themselves.

Sports inventions will inevitably come at unpredictable times and will have varying impacts on competition and play. New technologies for competitive advantage, health and safety, fairness and accuracy, and increased fun and participation will continue to cross and blur traditional boundaries, such as the line between human-powered and motorized bicycles or distinctions between disabled and non-disabled athletes. For sports leagues and associations newly thrust into more visible regulatory roles than in the past, the path forward is to implement greater transparency and clearer, fairer procedures for testing and decision-making about new inventions.

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Chapter 7

The Past, Present, and Future of Sports Innovation

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Inventors and the technologies they develop have become integral to modern sports. As the previous chapters illustrate, every sport is witnessing numerous innovations in high-tech apparel, prostheses, safety gear, and officiating equipment, as well as advances in training techniques, competitive strategy, recovery methods, and medical care. These inventions make the difference between victory or defeat, safety or injury, and participation or exclusion. And while athletes will always remain central to recreational and competitive sports, we have shown that inventors and their innovations are just as crucial and inseparable.

Who invents for sports and why? As the book's case studies demonstrate, sports inventors bring a wide variety of backgrounds and experiences to their work. Being an athlete is not required. For example, Arthur Ehrat—a farmer—invented the breakaway basketball rim. Yet more often than not, we find that sports inventors are amateur or professional athletes who become frustrated when they lose a contest, sustain an injury, endure a bad call, or find themselves unable to participate fully in their chosen sport. These athlete-inventors then channel their frustration, adopt an inventive identity, and engage in user innovation to develop creative solutions.[1] Accordingly, we argue that inventors are driven to develop sports technologies because of four primary motivations: to improve performance and gain a competitive advantage; to prevent injuries and enhance safety; to aid officiating; and to make sports more fun and inclusive for everyone.

Earning money is also an important, if secondary, motivation for inventors operating in the $416 billion global sports industry.[2] Indeed, Jake Burton Carpenter, Nathan Ramasarma, Doug DeAngelis, and many other inventors established start-up firms to commercialize their inventions. Nevertheless, inventors' own accounts support our argument that the core motivations—competitive edge, health and safety, fairness and accuracy, and fun for everyone—supersede the profit motive for sports inventors.

As the book's case studies document, individuals of every race, gender, age, nationality, education level, and (dis)ability have invented for sports. The diverse backgrounds, circumstances, and life experiences of these inventors often provide direct inspiration for their breakthroughs. For example, Mike Schultz's snowmobile accident and lower-leg amputation motivated him to invent the Moto Knee and Versa Foot prostheses so he could continue to compete in extreme sports. As a young Black girl growing up in the Bronx, Tahira Reid channeled her childhood love for double Dutch to invent a machine to turn the ropes when her friends were absent. Joggers Lisa Lindahl and Hinda Schreiber Miller

personally experienced the discomfort women felt when running with inadequate, unsupportive undergarments. With the help of costume designer Polly Palmer Smith, they invented the Jogbra, an invention "by women, for women."

Even as the spectrum of human differences and sporting cultures acts as a powerful inspiration for inventors, identity can be a double-edged sword. The benefits of diversity that inspire new ideas are often accompanied by discrimination and pernicious structural barriers that hamper innovation. Women and minorities are underrepresented at every stage of the invention and innovation pipeline. They are much less likely to earn an advanced STEM degree, find technical jobs in corporate R & D labs, or secure patents than their White, male counterparts. Women and minorities also find it more difficult to attract venture capital, launch start-up companies, or license their ideas to interested firms.[3] Nevertheless, the stories of sports inventors developed here provide powerful evidence that diversity is an important driver of successful innovation. Across a variety of sectors, teams with greater gender and racial diversity produce better designs, generate higher patent commercialization rates, and attract more customers, leading to greater sales revenues and higher market shares.[4]

Sports inventors also draw inspiration and technical know-how from contexts outside sports.[5] For example, Howard Head's familiarity with lightweight aerospace materials inspired the invention of his namesake metal skis and tennis rackets. Paul Hawkins's experience calculating trajectories while working on a missile defense system aided his invention of the Hawk-Eye ball-tracking system. Conversely, sports inventors often broaden their impact by developing new applications for their technologies in domains beyond sports. Shawn Springs, for example, has incorporated Windpact's patented foam pads in a variety of helmets worn not only by football players, but also by construction workers and military personnel.

What happens when an inventor introduces a new sports technology or game-changing technique? What is its impact *within a sport* and sometimes *across all sports*? Sports inventions, almost by definition, tend to disrupt established norms and traditions, often stirring passionate debates among athletes, fans, and the media. Some technologies, like Jake Burton Carpenter's snowboard, are readily adopted and quickly redefine the sport's new normal. In other cases, such as Speedo's high-tech LZR Racer swimsuit, key stakeholders reject the technology because of concerns over competitive integrity, safety, costs, or perceived threats to the sport's traditions. And sometimes sports communities rewrite rules as new knowledge and changing social attitudes shift around a longstanding technology. For example, college and professional football have witnessed several waves of concern over concussion injuries followed by periods of relative calm, even as the risks remained largely unabated. Collectively, the case studies demonstrate that sports governance is a parallel process of both technological and social change, in which stakeholders co-construct new rules, training regimens, officiating practices, and fan experiences in concert with the emergence of new technologies and techniques.[6] In this dynamic and iterative process, athletes, fans, and sports federations negotiate whether and how a particular innovation should be adopted, regulated, or rejected.

The Accelerating Pace of Sports Innovation

Meanwhile, sports innovation continues apace. We observe this anecdotally simply by reading sports headlines. In October 2019, Eliud Kipchoge—with help from Nike's

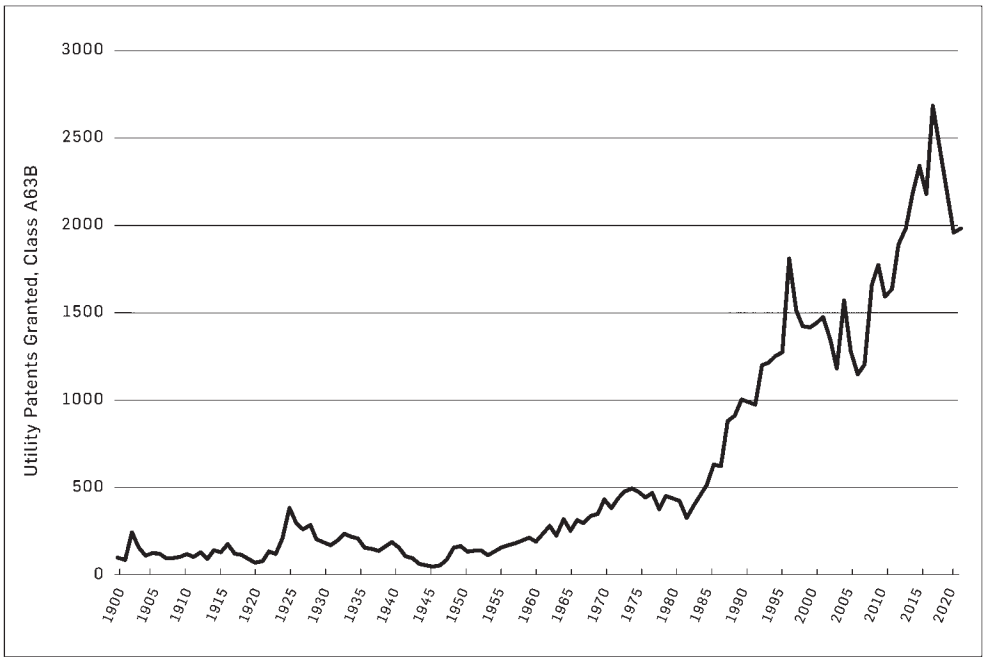

FIGURE 7.1. Sports equipment utility patents (class A63B) issued from 1900 through 2023. Source: U.S. Patent and Trademark Office, Public Patent Search.

high-tech "VaporFly super-shoe" and a team of pacers—became the first human to run a marathon in under two hours.[7] During the COVID-19 pandemic, the 2021 Australian Open tennis tournament replaced human line umpires with the Hawk-Eye ball-tracking system.[8] The 2022 World Cup soccer tournament in Qatar featured multiple innovations, including the Adidas "Al Rihla" soccer ball, whose built-in microchips and sensors communicated with the Hawk-Eye system to help officials make offside calls and goal determinations. To reduce the tournament's environmental impact, Qatar constructed "Stadium 974" using shipping containers; it will eventually be dismantled, shipped, and reassembled elsewhere, potentially for the 2030 World Cup.[9]

These examples are emblematic of an acceleration in the pace of technological change in the sports industry. Looking, for example, at U.S. utility patents issued in the technology class most relevant to competitive sports over the past 120 years (Figure 7.1), we see a rapid increase in the number of patents issued beginning in the mid-1980s, with a particularly noticeable spike starting around 2005.[10] The sharp decline in patents issued in 2021 followed by the beginnings of a rebound in 2022 and 2023 is likely due to corporate and university research lab closures during the COVID-19 pandemic and slow subsequent recovery. Counting patents is an imperfect proxy for measuring the rate of innovation; not all inventions are patented, and only a few patents result in impactful, widely used innovations.[11] Yet the empirical data affirm the hunch drawn from the headlines: we are living in an era of rapid and accelerating technological change in the sports industry.

How do we explain this acceleration? First, more people are participating in recreational sports, thanks to changing social attitudes and enabling technologies such as the Jogbra and the Quickie wheelchair. In addition, more fans are watching competitive sports broadcast on television and streamed via the internet.[12] This increased participation and

viewership over recent decades has brought more potential athlete-inventors into the picture. Some of the growth in sports technology reflects opportunities identified by a growing population of athletes and fans.

Second, sports technology has grown into a major industry that relies on continuous innovation as a fundamental business strategy. As of 2023, the global sports equipment ($143 billion), apparel ($213 billion), and footwear ($60 billion) markets were worth an estimated $416 billion combined, with a leading firm, Nike, reporting annual revenues of $46.7 billion.[13] Nike, Speedo, Specialized, and other major sports equipment firms maintain robust research and development (R & D) teams to develop the next leading-edge running shoe, swimsuit, or bicycle frame. These innovations then become the basis of marketing campaigns that assure both elite and amateur athletes that they are using the most advanced gear available. In short, major sports equipment and apparel firms have found that innovation pays and have redoubled their R & D efforts.

New entrepreneurial start-ups are also driving the accelerating pace of sports innovation, aided by an influx of venture capital (VC) investments. According to the market research firm SportsTechX, venture capitalists invested some $19.2 billion in sports-technology start-ups between 2015 and 2020. Traditional venture capital firms including Sequoia Capital, L Catterton, and Founders Fund have made significant investments in sports, fitness, and media start-ups including Strava, Peloton, Tonal, and The Athletic.[14] In parallel, new boutique VC firms such as Next Ventures, ADvantage Sports Fund, and Elysian Park Ventures have raised funds specifically designed to make investments in early-stage sports technology start-ups. For example, Sapphire Ventures has raised nearly $300 million across two sports technology funds—Sapphire Sport I ($117 million, 2019) and Sapphire Sport II ($181 million, 2023)—leading to investments in sports technology start-ups including Aglet (premium sneaker market), PlayVS (esports), and Overtime (sports live streaming).[15]

Sapphire has raised these funds from major institutional investors such as Adidas, SAP, and Major League Baseball, but its limited partners also include wealthy former athletes.[16] A growing number of current and former professional athletes—including Shaquille O'Neal, Magic Johnson, Kevin Durant, Joe Montana, Serena Williams, Stephen Curry, Tony Gonzalez, LeBron James, and the late Kobe Bryant—have injected a new energy into the sports technology sector by becoming angel investors and limited partners. Having amassed considerable fortunes as professional athletes and product endorsers, these athlete-investors now devote a portion of their portfolios to investing in high-risk, high-reward entrepreneurial start-ups. Unlike prior generations of retired athletes who started restaurants and invested in other low-return investments, the new generation of "A-list angels" sees technology—and especially sports technology—as a high-growth sector. O'Neal, for example, famously earned significant returns from pre-IPO investments in Google, Ring, and Uber, but also has made seed investments in Vitaminwater's fitness drink and MaxPro's home fitness equipment.[17] As more professional athletes invest in early-stage sports start-ups, they are diversifying the notoriously White, male VC sector.[18] For example, superstar tennis player Serena Williams and her co-investor Alison Rapaport Stillman have built a diverse team at Serena Ventures, which has intentionally focused its investments on start-ups founded by women and people of color.[19]

A third driver of recent technology change in sports came from the global COVID-19 pandemic. The crisis catalyzed experimentation and new ways of operating in nearly every sector of society, including sports.[20] Initially, the pandemic forced the cancellation of high

school sports and the NCAA's "March Madness" college basketball tournament, plus the postponement of the Tokyo 2020 Summer Olympics and the UEFA Euro soccer championship. In an early adaptation, the 2020 Boston Marathon encouraged registered participants to run the course individually and submit self-reported finish times.[21] As the pandemic stretched on, the National Basketball Association (NBA), National Hockey League (NHL), and Major League Soccer (MLS) eventually resumed their seasons by establishing quarantine "bubbles." Athletes, officials, and a limited numbers of camera operators and other staff lived in tightly controlled facilities. They submitted to frequent COVID-19 testing and were required to wear face coverings except when playing, eating, or sitting alone in their hotel rooms. To bring some sense of fan involvement into the competitions, leagues installed video screens that displayed crowd-sourced images of spectators along with pre-recorded cheers and clapping.[22]

While league bubbles and virtual fans were relatively short-lived, other pandemic-era innovations have remained in place. For example, some play-by-play analysts continue to call games remotely, and colleges with smaller athletic budgets offer virtual recruiting "visits" to prospective high school athletes. The pandemic also accelerated the adoption of new technologies that were in early testing stages in 2019. As noted, the Australian Open and other tennis tournaments relied more heavily on Hawk-Eye and other forms of technological officiating to reduce the number of officials in the arena. At stadiums, mobile tickets have largely replaced paper tickets and contactless payment has superseded the use of cash to purchase concessions or merchandise. And after years of hesitation, teams, leagues, and jurisdictions across the United States have largely embraced sports betting to generate additional revenue, often using geofencing technologies to manage where bets are placed.[23]

Finally, as we have shown throughout this volume, inventors have accelerated the pace of sports innovation by leveraging other cutting-edge technologies. General-purpose technologies, including artificial intelligence (AI), virtual reality (VR), the Internet of Things, and additive manufacturing have become cheaper and more accessible over the past decade. Numerous inventors, including Nathan Ramasarma and Arielle Rausin, have applied these technologies to design and fabricate their inventions at small scale.[24] Moreover, many of these twenty-first-century technologies require relatively little capital or material infrastructure to get started. For example, with minimal investment, an aspiring inventor can purchase a computer and virtual reality headset, download an open-source software developer's kit, and then begin coding a sports VR app. Furthermore, subscription platforms including Amazon, Google Play, and the Apple App Store enable inventors and firms to easily advertise, sell, and distribute both physical and digital products at scale. Overall, these lower barriers to entry have attracted many inventors to experiment with VR, AI, and other emerging technologies, thereby accelerating the pace of sports innovation.

Where is sports technology headed next? We believe that the next generation of sports inventors will be motivated to develop innovations in three key areas: novel ways of collecting and analyzing sports data; all-ability and unified sports; and technological adaptations to climate change.

Big Data and Artificial Intelligence

The widespread collection, analysis, and application of large sports data sets is a major area of opportunity for inventors. From the aerodynamic drag measurements Specialized

collects to improve bicycle designs to the ball-trajectory information Hawk-Eye compiles to aid officiating, inventors are already leveraging data analysis in their innovations. In the future, we anticipate the extension of these techniques from professional and collegiate levels into youth and recreational sports. We also anticipate a broader application of these technologies across the sports landscape. By harnessing insights from data and AI, inventors can help individuals and teams gain a competitive advantage, improve players' health and safety, help officials make better calls, and even improve the fan experience.

A prerequisite for data analysis is data collection. Information systems have evolved with the times, from hand-recorded scorecards and ledgers to mainframe computers to small handheld devices. Nevertheless, information systems have traditionally relied on a scorekeeper to observe and manually record data, for example, hits, runs, and errors in a baseball game.[25] In recent years, inventors and start-ups have introduced a variety of technologies to automate the collection of game-related data. During the 1990s, for example, computer scientist Allen Plaskett introduced the Snickometer, a microphone and oscilloscope installed near the wickets that could distinguish whether a cricket ball hit the bat or the batter's leg.[26] In 2014, the NFL partnered with Zebra and Wilson to embed RFID (radio-frequency identification) transmitters in players' shoulder pads and in the football itself to chart plays with greater precision.[27] Coaches and broadcasters also have deployed aerial drones mounted with cameras to provide bird's-eye images of practices and games.[28] More recently, firms such as formsense and Apple have introduced wearable sensors embedded in clothing and smartwatches that can track an athlete's movements, heart rate, and temperature.

Collectively, these automated, panoptic data collection systems create an avalanche of data amenable to intense statistical analysis.[29] But raw data is unhelpful; the real value of data-driven solutions lies in the software algorithms that interpret and make sense of the data. The Snickometer algorithm, for example, uses pattern-matching routines. Specifically, it compares the waveforms of sounds recorded by the microphone against a reference library of waveforms for bat and batter touches to determine what the system just "heard." Similarly, Hawk-Eye's algorithms compare the real-time position data of the ball against the reference positions of the painted lines on a tennis court to determine if the ball landed in or out. The speed and data-processing capacity of these systems far exceeds that of any human being, and thus they represent a simple form of AI that can aid human decision-makers.

What kinds of big data solutions might we expect to see in the near future? Inventors, we believe, will continue to extend today's high-tech data solutions to give athletes, coaches, and general managers a competitive advantage. Sports analytics, for example, entails gathering and interpreting observable statistical data about athletes and gameplay to aid decision-making. Team executives use statistical analysis to evaluate players for drafting, trades, and contract negotiations, while coaches leverage data to understand competitors' tendencies, develop in-game strategies, and identify areas for player improvement.[30] Many sports fans first learned about analytics from Michael Lewis's 2003 bestselling book, *Moneyball,* or its 2011 film adaptation starring Brad Pitt as Oakland Athletics general manager Billy Beane. The book and film describe how Beane deployed unorthodox mathematical techniques to find high-performing but undervalued baseball players, which helped his low-payroll baseball team compete successfully against wealthier clubs.[31]

This statistical approach has since been adopted across all the major professional sports, though baseball has remained at the frontier of the analytics revolution. Hawk-Eye

has provided ball- and player-tracking services to Major League Baseball since 2020, and the wealth of data it collects is enabling general managers and coaches to probe ever deeper into player performance. While a simple box score calculation can provide a hitter's on-base percentage, Hawk-Eye can provide a hitter's bat speed, swing plane angle, and the exit velocity of a batted ball. Where a radar gun once provided a pitcher's velocity, Hawk-Eye now measures a pitcher's "spin rate" (revolutions per minute) and the number of inches a breaking pitch moves. Hawk-Eye's geospatial data can also help quantify an outfielder's "range," or the amount of ground he can cover to make a defensive play.[32]

Some observers believe the next frontier in player evaluation involves using brain wave data from wearables or facial expression tracking to discern emotions and other "intangibles" such as work ethic or coachability.[33] Mining historical data can also help with in-game strategy and learning opponents' tendencies. For example, coaches can run simulations to see how specific lineups might fare against their opponents or learn how a specific opposing batter fares against curveballs versus sliders. We can also imagine a future where a coach programs the opposing pitcher's observed characteristics—arm angle, velocity, spin rate, pitch sequence—into a high-tech pitching machine to give his batters the best possible preparation for the upcoming game.[34]

Advances in data collection and AI have also propelled innovations in officiating. As noted, Hawk-Eye's ball-tracking technology has already started to replace line officials in tennis and this trend toward fully automated officiating is likely to continue. Indeed, Hawk-Eye's automated ball-strike system (ABS) debuted in 11 Triple-A baseball stadiums in 2022 and expanded to all 30 stadiums in 2023; it may eventually replace home plate umpires.[35] Similarly, with RFID chips transmitting the precise location of the football, the recently revived United States Football League (USFL) eliminated the officiating "chain gang" that manually measures the 10 yards required for a first down.[36] Officials who remain on the field will increasingly rely on AI systems to help with notoriously tough judgment calls. FIFA, for example, has already begun to test a Hawk-Eye–powered system to adjudicate onside/offside calls in soccer; it is easy to imagine the NBA using the same technology to automate calls for goaltending, charging, or blocking.[37]

Big data systems can also be used to detect cheating. For example, Hawk-Eye can detect if a baseball pitcher has an abnormally high velocity, spin rate, or movement in his pitches over historic norms, suggesting that he may have "doctored" the baseball with a sticky substance to improve his grip.[38] Similarly, we envision sports leagues undertaking longitudinal studies of referees' foul calls and penalties; the discovery of any anomalous patterns might suggest that an official had thrown a game for the benefit of sports gamblers.

Data-driven insights will also continue to advance inventions for athletes' health and safety. As noted, wearable sensors can continuously record multiple biometric data points, including an athlete's precise three-dimensional position, speed, acceleration, body temperature, respiration, and heart rate. Carefully monitoring this data can help athletes train with the proper technique, minimize injuries, and expedite recovery. For example, several NHL teams have implemented Catapult sensors to monitor their players' exertions from skating and the force of collisions; these data help coaches and trainers determine which players need more rest. Similarly, the motusBASEBALL arm sleeve tracks a pitcher's arm angle and the stress placed on the ulnar collateral elbow ligament to help them improve their mechanics and prevent injuries.

While athletes and trainers typically evaluate biometric data after workouts and then make adjustments, real-time interventions are the next frontier. For example, Wearable X

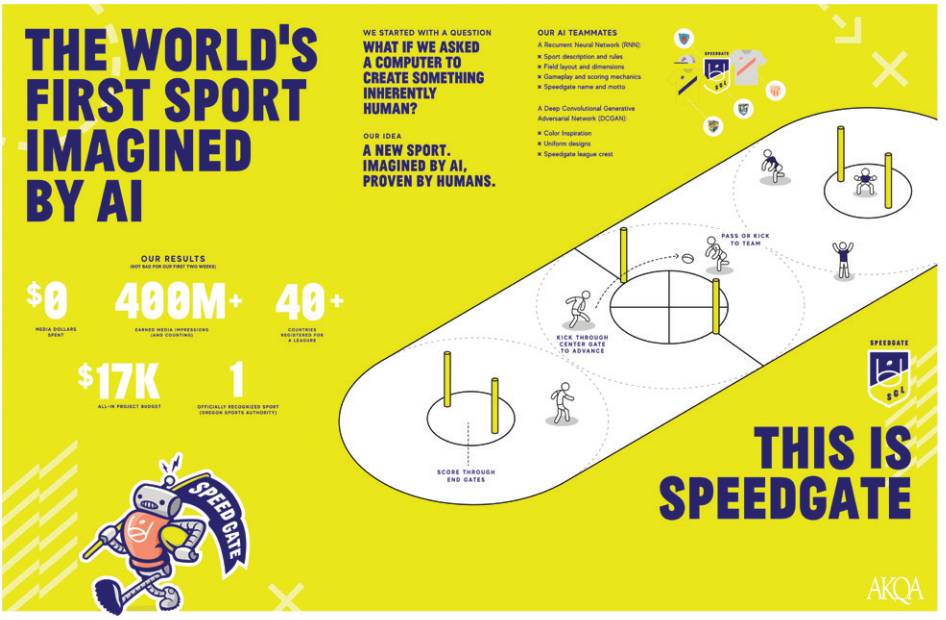

FIGURE 7.2. Speedgate, a new sport invented with artificial intelligence. Credit: AKQA.

already sells smart yoga pants that gently vibrate when a yogi is out of position; conversely, the pants say "Ohm" when the body angles are correct.[39] It is easy to imagine a future football helmet warning "Heads up, see what you hit!" to help a linebacker maintain a safer tackling technique. We can also envision an ingestible sensor that sends a warning to trainers if it detects irregular heart rhythms, elevated body temperature, dehydration, or another dangerous health condition an athlete might experience.

Data-driven innovations will continue to make sports more fun and inclusive, especially for recreational athletes and fans. The Strava app, for example, uses a smartphone's GPS feature to enable runners and cyclists to share their routes among friends. Similarly, Nike Training Club and other wellness apps help weekend warriors meet fitness goals by tracking their workouts.[40] Developers are also using AI to invent entirely new sports. In 2019, the design firm AKQA input the rules from 400 sports into a large language model (LLM), then queried the AI system to generate concepts for a new sport that would be accessible, enjoyable, easy to learn, and give participants a good workout. The system generated dozens of potential ideas; the AKQA designers elected to develop a concept called "Speedgate" (Figure 7.2), which blends elements from rugby, ultimate frisbee, soccer, and croquet.[41] Although Speedgate has not yet achieved wide adoption, AKQA's approach points to a future where even a small group of kids with limited gear can co-invent a new sport with an AI app.

Data solutions are also transforming how fans watch the games, as teams and leagues share more data with broadcasters. In 2015, the NFL leveraged its RFID tracking technology to offer broadcasters and fans "next-generation stats," such as a receiver's separation from his defender, or the number of yards a defender traveled to make a tackle.[42] As tracking technologies become more commonplace and less expensive, they will be applied even in lower-revenue sports. We can imagine, for example, that Hawk-Eye will someday

provide viewers with a diver's splash height and angle of entry into the water, or the spin rate and height of a figure skater's triple lutz.

As wearable sensors become more common, we expect that broadcasters will share even more personal details about athletes, such as a golfer's heart rate just before a crucial putt.[43] Eventually, with help from virtual reality headsets, broadcasters should be able offer fans a "player's eye view" of the action.[44] And just as they are beginning to automate officiating, AI systems may eventually replace sportswriters and broadcasters. Drawing on large language models, AI systems will soon be able to follow the game action as recorded by automated tracking systems, "write" a real-time game log, and have a Siri-like robot voice provide real-time play-by-play and analysis. An early version of that future has already arrived. Parents who use the GameChanger app can hear an automated real-time play-by-play of their child's Little League game and receive a post-game summary that reads remarkably like a beat writer's recap.[45]

Like most sports innovations, big data solutions have engendered both praise and criticism. Fans—especially gamblers—will continue to embrace the "next-generation stats" and enhanced insights they now enjoy on television broadcasts. Coaches and general managers will also keep investing in analytics to gain a competitive advantage. Athletes will continue to enjoy the health and safety benefits of biometric performance tracking but will grow more concerned about Orwellian "dataveillance" and the loss of privacy from data breaches. Referees and umpires undoubtedly worry that automated officiating will someday render them obsolete, while fans will likely have mixed reactions. Many tennis spectators clap rhythmically in anticipation while waiting for Hawk-Eye to display an automated line call, while other fans believe that missed calls and the occasional controversy are exactly what make sports interesting and entertaining.[46] For better or worse, the compilation of big data sets and implementation of AI across a wide range of applications will continue to drive future innovations in sports.

Unified Sports

As seen throughout this book, sports inventions include new physical or digital technologies as well as new approaches to training and competition. Innovations in techniques, strategies, and the rules of play can inspire the coproduction of new sports technologies and vice versa. Along these lines, we believe that "unified" or "all-ability" sports will provide ample opportunities for future inventors while making both competition and recreation more inclusive and fun for everyone.

Unified Sports teams pair athletes with and without physical and cognitive disabilities, necessitating innovations to rules, training, ways of playing, officiating, and scoring. The Unified Sports movement emerged historically from the Paralympic movement, which organized sports competitions for people with physical disabilities, and Special Olympics, which is geared toward athletes with intellectual disabilities.[47] Unified Sports features a diverse mix of assistive technologies, including prostheses, wheelchairs, and devices for auditory and visual feedback. Given its growing popularity, there are significant invention opportunities on the near-term horizon.

Sports competitions for athletes with physical disabilities have developed considerably since "adaptive sports" began as part of rehabilitation efforts for injured World War II veterans.[48] Organized competitions grew steadily during the second half of the twentieth century, spurred by the creation of the Paralympic Games in 1960. While the Paralympic

Games themselves focus on competition, the movement also strongly emphasizes self-realization and the expression of personal determination for athletes who participate, whether or not they earn medals. In a positive feedback loop, the growing popularity of the Paralympics has inspired advances in high-tech prostheses, wheelchairs, and other adaptive technologies, which in turn make competitions more intense and interesting for television audiences and sponsors.[49] Adaptive sports have also been part of the X Games since 2008, when the Adaptive Super X Race and the Adaptive Skate Park Jam were added. At the X Games, adaptive athletes compete on the same terrain and conditions as non-disabled athletes, further helping to create a market for precision equipment.[50]

People with physical disabilities have a long history of building supportive communities, yet people with cognitive disabilities were historically excluded from community life, and especially from sports. Change came starting in the early 1960s when Eunice Kennedy Shriver started a summer day camp at her home in suburban Washington, D.C., for children with intellectual disabilities.[51] The first International Special Olympics games were held in 1968, attracting 1,000 athletes from across the United States and Canada. By 2023, the Special Olympics World Games in Berlin featured over 7,000 athletes from 190 countries competing in 27 sports.[52] Both the Paralympics and Special Olympics have developed detailed classification schemes to ensure fair and balanced competitions among athletes with similar abilities. Officials revise these classifications regularly when they adopt new sports, when athletes introduce new adaptive technologies, and when doctors share updated medical knowledge and practices.

Starting in the 1980s, Special Olympics began to modify existing sports to have people with and without intellectual disabilities on the same teams playing and competing together (Figure 7.3). In July 1988, Unified Sports was launched as an official initiative of Special Olympics, starting with bowling, volleyball, and softball.[53] Since then, the movement has grown considerably, with programs across the United States, Canada, Europe, and the Middle East. Many grassroots programs are run by schools, regional leagues, and community groups, including religious organizations.[54] Some larger unified competitions are organized by national sports associations in conjunction with major international sports competitions. For example, as part of its programming for the 2022 FIFA World Cup, Qatar's "Generation Amazing" initiative invited teens from the Middle East, Africa, and Asia to participate in "all-in" soccer matches that integrated non-disabled athletes with competitors with cognitive and physical disabilities.[55]

Core principles define Unified Sports, starting with involving everyone in a meaningful way. Guidebooks outline methods for team practice, ranging from individual skill-building to advice on how to train together. Special Olympics and other associations have developed modified rules for each sport, which detail who can play and specify limitations based on medical diagnoses. The Unified soccer rulebook, for example, differentiates between official and friendly events, defines "athletes" (people with disabilities) and "partners" (non-disabled), and sets standards for the field of play, ball size, clothing, penalties, substitutions, fouls, and other aspects of the game.[56] Teams are certified and put into divisions by official "Unified Observers," who report to Special Olympics on the results of a standardized set of skill tests overseen by a Divisioning Committee.[57]

Numerous research studies document the benefits of participating in Unified Sports, including greater social integration and reduced stigma for disabled athletes, increased empathy and respect for people with disabilities among non-disabled athletes, and greater participation in employment and other economic activity by people with

FIGURE 7.3. Peter Gade (left) and Chai Xinyu (right) compete in Unified badminton at the 2019 Abu Dhabi Special Olympics World Games. Credit: Meng Tao, Xinhua / Alamy Live News.

intellectual disabilities.[58] Building on these benefits, future innovations will include approaches to integrate play and competition for athletes with varying abilities. We envision inventions that integrate auditory, visual, and haptic technologies that an athlete can customize to their disability. Some of these adaptive technologies can be built into athletes' clothing and footwear, while others will be built into equipment, such as the soccer ball and goals.

A second direction for the future of all-ability sports involves intentional technological enhancement, such as the use of robotic or virtual stand-ins for human athletes. As opposed to athletes who use adaptive equipment or prostheses, participants and competitors in the "Cybathlon"—an initiative launched by ETH Zurich (Technical University Zurich)—develop technologies to use their bodies and minds to control robots remotely.[59] For example, paraplegic athletes race in robotic exoskeletons, bikes, and motorized wheelchairs, or use technologies that read brain waves so they can compete in virtual contests. While the Paralympics bans motorized equipment, the Cybathlon embraces them. The event's organizer, Professor Robert Riener, notes: "Instead of celebrating the human body moving under its own power, the cyborg games will celebrate the strength and ingenuity of human-machine collaborations."[60]

Future inventions will further enhance the ability of athletes regardless of their physical or intellectual disabilities. In the future, cyborg athletes will use bionic implants in competition, such as high-tech robotic limbs controlled remotely by signals from the brain or the tension of muscle fibers.[61] One can envision artificial limbs with synthetic muscles that increase a person's strength hundreds of times, allowing those who train successfully to run faster, jump higher, and otherwise exceed their natural body's limitations. Sports leagues and international competitions will need to develop rules for integration or

separation of future cyborg athletes even as fan interest ultimately shapes the resources available to grow the sport.

Looking ahead, inventors will improve prostheses and other assistive devices by deploying strong, lightweight, and flexible materials; harnessing new knowledge about the biomechanical functioning of the human body; and developing technologies to scan and accurately interpret brain activity. These technological innovations, combined with changes in attitudes and participation, have the potential to make competitive and recreational sports more fun and inclusive for everyone.

Climate Change

In 2018, organizers of the U.S. Open tennis tournament were forced to implement a new policy that allows players to take a "heat break" during their matches. The rule came into effect after several athletes withdrew from the tournament due to extreme heat conditions in New York City. Two years later at the Australian Open, poor air quality from wildfires again forced players to withdraw.[62] Dangerous weather conditions related to climate change are having a profound effect on sporting events around the world. The sports industry thus has the opportunity and responsibility to act as a catalyst for change in reducing carbon emissions. Going forward, sports inventors will play a critical role in developing technologies to enable training, play, and competition in hotter environments.

The adverse effects of global warming and extreme weather events will continue to introduce new risks and influence when, where, and how we participate in sports.[63] Hotter weather creates longer seasons for outdoor activity. With increased exposure to UV rays, athletes, spectators, and officials are at greater risk for skin cancers. In the future, they will also be more susceptible to mosquito- and tick-borne diseases as climate change contributes to their spread.

Certain sports are endangered because of climate change and are developing innovative responses to survive. Snow availability is a growing problem for alpine sports, with mountain slopes experiencing shorter winters and less snow. Consequently, the number of ski areas operating in the United States has declined from 529 in 1992 to 473 in 2021.[64] Ski resorts have invested in snow-making equipment and have even started to move the sport indoors; as of 2024, there were over 110 active indoor ski areas worldwide.[65] The future may also include summer tournaments on "roller skis" and the elimination of fluorinated waxes, which make skis go faster but leave chemical residue in the environment.[66] Similarly, golf courses are intensive users of irrigated water and synthetic fertilizers, and have drawn criticism from environmental advocacy groups. In response, turf managers have been experimenting with new grass strains and water-management techniques to make the game more sustainable in arid climates.[67]

While sports both contributes to and increasingly is shaped by climate change, it also can play a key role in raising awareness, influencing behaviors, and shrinking carbon emissions.[68] As early as 1992, the International Olympic Committee (IOC) recognized the effect of climate change on sports. In 1994, the IOC added "respect for the environment" to its list of Olympic values and articulated the organization's commitment "to encourage and support a responsible concern for environmental issues, to promote sustainable development in sport, and to require that the Olympic Games are held accordingly." For example, the Olympic Village for the 2000 Sydney Olympiad became the world's largest solar-powered suburb, showcasing green housing technologies on a large

FIGURE 7.4. Dr. Saud Abdulaziz Abdul Ghani demonstrating the cooling system at the Al Janoub stadium in Qatar. Credit: Brian Benton.

scale. Two years later, the Salt Lake City winter games demonstrated an energy recycling system, which used the curling hall's air-conditioning unit to heat water for the venue's showers and bathrooms. The 2020 Tokyo Games (held in 2021) drew most of its energy from renewable sources, including solar arrays and wood biomass power. The Tokyo Games also purchased carbon offsets for both direct and indirect emissions produced throughout the event.[69]

Sustainable stadium design remains important for reducing sports' carbon footprint and raising awareness about climate change among media and fans. Climate Pledge Arena in Seattle became the first carbon-neutral sports venue in the world when it opened in October 2021. To achieve this, the venue runs completely on renewable energy, uses recycled water, eliminates single-use plastics, sources food locally, donates leftover concessions to food banks, and encourages spectators to use public transportation instead of their personal vehicles.[70] Similarly, the desert nation of Qatar invested in innovative stadium designs—and scheduled the tournament for late autumn—after winning its bid to host the 2022 FIFA World Cup. Qatar University engineering professor Dr. Saud Abdulaziz Abdul Ghani (also known as "Dr. Cool") helped invent and implement environmentally friendly cooling systems in seven of the eight World Cup stadiums (Figure 7.4). Dr. Saud's designs used chilled underground water tanks for heat exchange and featured air intake and outflow vents designed to cool just the pitch and seating areas, not the entire air volume of the stadium. This innovative cooling technology is approximately 40% more sustainable than prior air-conditioning systems.[71]

In the future, sports venues, leagues, and teams will come under greater pressure to demonstrate that they can operate without additional negative climate impacts. This likely means fewer major sporting events during the summer or during the warmest midday hours, and greater emphasis on reduced or even negative carbon footprints, especially for major international competitions. Going forward, there will be fewer outdoor, open-air stadiums. Instead, teams will invest in fully enclosed, solar-powered, carbon-neutral

venues that can insulate athletes and fans from high temperatures, torrential rains, and deteriorating air quality.

Under some scenarios, we can also anticipate a reduction in travel for future sporting events. If future carbon taxes make travel more expensive, athletes and teams may elect to compete in just a few high-profile international competitions such as the World Cup and the Olympics. College and professional sports leagues might eventually limit long-haul air travel and instead prioritize scheduling intra-regional competitions they can reach via electric buses or trains. Certain sports, such as swimming, might dispense with in-person meets altogether, and instead have competitors swim locally and post their times, much like the pandemic-era Boston Marathon. Fans will also be more reluctant to travel for sporting events and gravitate toward augmented, immersive broadcasts to follow their favorite teams. With virtual reality headsets, fans will feel like they are sitting in a far-flung stadium—or standing on the field with the athletes—while instead sitting at home on the couch.[72]

The effects of climate change are already beginning to impact sporting events around the world. Going forward, inventors will play a critical role in making adaptations to reduce sports' carbon footprint and ensure the comfort and safety of athletes and fans. As sports inventors develop climate solutions, they will also play a key role in raising awareness about the need for change. As author David Goldblatt suggests in *Playing Against the Clock*, "sports has the power to reach a large number of people who aren't listening to scientists or politicians. It can be a powerful tool in climate politics."[73]

Finish Line

As demonstrated throughout this book, invention and technology are—and will remain—integral to modern sports. Indeed, historical and contemporary trends suggest that more individuals of every race, gender, age, nationality, education level, and physical and intellectual ability will continue to drive innovation in sports.

Elite athletes and weekend warriors will continue to be frustrated when they lose a contest, suffer an injury, endure a bad call, or find themselves unable to fully participate in their chosen sport. They will become motivated to invent their own solutions—or enlist the help of sports scientists and engineers—to develop game-changing technologies that boost performance, enhance safety, improve officiating, and make sports more fun and inclusive. They will continue to draw inspiration from high-tech sectors like aerospace engineering, biomechanics, and genetics, and from the societal challenges posed by pandemics, climate change, and other future threats. They will continue to deploy proven inventive strategies, such as learning from failure or deploying general-purpose technologies in new ways. And when sports inventors introduce their new technologies, athletes, fans, leagues, and the media will continue to debate whether these innovations should be adopted, regulated, or rejected altogether.

There are more opportunities to invent for sports than ever before and the pace of innovation appears to be accelerating. As our historical and contemporary case studies have illustrated, everyone has the capacity to be an inventor. Anyone, from any background, can focus their natural creativity to develop solutions and invent helpful improvements for sports, for the home, for school, or for the workplace. Everyone is inventive and that includes you, the reader. We hope that these stories inspire you to change *your* game, whatever that may be.

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Index

An *n* after a page number indicates a note, and a number after the *n* indicates the note number.

ABA (American Basketball Association), 45
ABC television, 67
Abdul-Jabbar, Kareem, 45
ABS (automated ball-strike system), 127
Abu Dhabi Special Olympics World Games, 131
Access Smithsonian, viii
accuracy. *see* fairness and accuracy
Accutrack photo-finish camera, 64–65, 75n16
actor-network theory, 4
adaptive sports: critical sports studies, 3; for injured World War II veterans, 88, 110, 129; motocross, ix, 19; skateboarding, ix; snocross, 19; snowboard-cross, 19; snowboarding, 19–21; X Games, 19, 130. *see also* disabilities, athletes with; prostheses; wheelchair athletes
Adidas: full-body swimsuit, 104; soccer balls, 28, 123; sports technology investments, 124
Advancing Informal STEM Learning (AISL) program, viii
ADvantage Sports Fund (VC firm), 124
AEG Magnetophon K4 magnetic recording machines, 67
aerodynamics. *see* wind tunnels
African Americans: *Change Your Game / Cambia tu juego* (exhibition), viii, ix; discrimination against, 40, 79; mentorship, 40–41; motivation for playing sports, 97; racial power dynamic in sports, 3–4; thermal properties of curly hair, 95; and venture capital (VC), 124, 136n18. *see also* double Dutch jump-roping; minorities; *specific people*
Aglet (premium sneaker firm), 124
AI. *see* big data and artificial intelligence
AISL (Advancing Informal STEM Learning) program, viii
AKQA (design firm), 128
Albrecht, Randy, 44–45
Alcindor, Lew, 45
all-ability sports. *see* Unified Sports
Allen, Paul, 38
alpine skiing. *see* skiing
AMB i.t. (firm), 63
American Bandstand (television show), 67
American Basketball Association (ABA), 45
American College Health Association, 108
Americans with Disabilities Act (1990), 79
American Watch Company, 61
America's Cup (sailing), 1
AMF, Inc., 85
Ampex VR-1000 video recorder, 67, 68, 69–70
amputees. *see* prostheses; *specific people*
AMSOIL Championship Snocross race (2008), 18
Anacostia Community Museum, viii
apparel. *see* clothing
Apple (firm), 126
apps, 128, 129
Aqualab. *see* Speedo
Arena swimsuits, 106
Army–Navy football game (1963), 69–70
artificial intelligence. *see* big data and artificial intelligence
Ashe, Arthur, 8, 86
Aspen Institute's 2021 Project Play report, 97
Association of Tennis Professionals (ATP), 86
Athens Olympics (2004), 104
Athens Paralympics (2004), 111
The Athletic (media start-up), 124
Atlanta Braves, 70
ATP (Association of Tennis Professionals), 86
Australian Open (tennis tournament), 72, 123, 125, 132
automated ball-strike system (ABS), 127
automobile safety: car safety seats, 38–39; crash test dummies, 42, 43; as inspiration, 38–39; sled tests, 43, 55n14; warning systems, 51; Windpact technology, 40
auto racing: digital photo finish, 66; injuries and deaths, 41–42, 44; injury prevention, 42–44, 53; remote-control car races, 63

Backhill (snowboard), 22–23
Bad Call (Collins, et al.), 72–73
Bahne and Co., 92
ball-tracking system. *see* Hawk-Eye Ball-Tracking System
Barber, William, 64
Barfoot (snowboard manufacturer), 24
basal skull fractures, 42–44
baseball: aluminum bats, 8; automated ball-strike system (ABS), 127; ball tracking and trajectory prediction, 73, 126–127; big data and artificial intelligence, 3, 126–127; chest protectors, 40; face

baseball (*continued*)
mask for umpires, 40; instant replay, 70; motus-BASEBALL arm sleeve, 127; norms and practices ("unwritten rules"), 103; telecasts, 68
basketball: Break-A-Way basketball rim, vii, 8, 9, 44–47, 56n27, 121; COVID-19 pandemic, 125; Jogbra for, 82; WNBA time periods, 59
BCE (Bing Crosby Enterprises), 67
Beane, Billy, 3, 126
Beatty, Jim, 45, 47
Beijing Olympics (2008), 106, 112
Bell, Lesli, 82
Bell, Sandy, 70
Bellisse (Lightning2 Company), 82
Berkoff, David, 106
Berlin Marathon, 63
Berthoud Pass (ski resort), Colorado, 24
bicycling. *see* biking; mountain biking
Biedermann, Paul, 106
big data and artificial intelligence, 125–129; accelerating pace of innovation, 125; cheating detection, 127; future innovations, 127–128; Hawk-Eye Ball-Tracking System, 7, 9, 10, 70–73, 122, 125–129; introduction, 11; *Moneyball* (Lewis), 3, 126; new sports, 128
Bijker, Wiebe, 4
biking: aerodynamics, 28; components used in ultralight wheelchairs, 88; digital photo finish, 66; introduction, 4; Jogbra for, 82; regulating e-bike trail access, 9, 10, 101, 103, 113–116; science and technology studies (STS) scholarship, 4; sensors to analyze tricks, 52; Strava app, 128; Win Tunnel (low-speed wind tunnel), 7, 8, 17, 28–32. *see also* mountain biking
Bing Crosby Enterprises (BCE), 67
BioDapt (firm), 19–21
biomechanics, 19. *see also* kinesiology
Biomedical Design Inc., 43
biometric data: future innovations, 127–128, 129; privacy concerns, 54, 129; real-time interventions, 127–128; smart apparel, 9, 51–54, 126–129
bionic implants, 131
Bleakney, Adam, 9, 25, 26
BLM (Bureau of Land Management), 114, 115
BMX competitions, 17
bobsled, 30, 59
Bolt, Usain, 59–60, 74n2
Booth, Douglas, 4
Borg, Björn, 86
Bostian, Charles W., 51
Boston Marathon, 26, 27, 63, 125
Bowman, Bob, 105
boxing, 73
Boys & Girls Clubs of Greater Washington, viii
brain injuries, 38, 39
bras, athletic. *see* Jogbra
Brayton, Scott, 44
Break-A-Way basketball rim, 44–47, 121; design process, 45, 46, 47; introduction, 8, 9; in museum collections, vii; patent, 45, 56n27
Breeze, Joe, 6, 114
"Breezer 1" (mountain bike), 6, 114
broadcasting. *see* television
Brüggemann, Peter, 112
Brunswick Corporation, 22
Bryant, Kobe, 124
Bubbs, Marc, 3
Burdick, Andrew, 94
Bureau of Land Management (BLM), 114, 115
Burton Snowboards, 22–24, 32

Cade, Mary, 49
Cade, Robert: background, 48, 53; character traits, 48, 50; design process, 48–50; Gatorade, 8, 9, 47–50, 53, 54; and informed consent, 57n42
Cadillac Wheels Company, 91, 92–93
California, e-bike trail access, 114
Cambia tu juego / Change Your Game (exhibition), viii, ix–x
camping innovations, 4
cancer and cancer treatment, 82, 132
Carpenter, Daniel, 5
Carpenter, Jake Burton: background, 21–22; Burton Snowboards, 22–24, 32, 121; character traits, 21–22; death, 24; design process, 22–23, 32; as "father of snowboarding," 24; motivation, 22; promotion of snowboarding, 24; snowboard invention, 8, 9, 17, 21–25, 122
car safety seats, 38–39
CART (Championship Auto Racing Teams), 44
CAS (Court of Arbitration for Sport), 112
Catapult sensors, 127
CBS television, 67–70
Certor Sports brand, 109
Chai Xinyu, 131
ChampionChip (firm), 9, 59–63
Championship Auto Racing Teams (CART), 44
Change Your Game / Cambia tu juego (exhibition), viii, ix–x
Chastain, Brandi, 84
Chesebrough-Pond's, 86
chronic traumatic encephalopathy (CTE), 38, 107, 110
Ciholas, Mike, 65
civil rights movement, 79
climate change, 11, 132–134
Climate Pledge Arena, Seattle, 133
clocks. *see* timekeeping
clothing: aerodynamic, 30, 31; Compressure Comfort Bra, 82; global market value, 124; Jogbra, ix, 9, 10, 79, 80–84, 96, 121–122; smart apparel, 5, 8–9, 50–54, 126–129; 3D-printed wheelchair racing gloves, 8, 9, 17, 25–28
cognitive disability, defined, 138n47
collaboration: competitive edge, 8, 24, 27; fun for everyone, 88, 96; future human-machine collaborations, 131; health and safety, 9, 37, 42–43, 54; introduction, 8, 9; museum programs, vii, ix
Collins, Harry, 73
Colorado, e-bike trail access, 114
Colorado Time Systems, 62, 75n9
competitive edge, 17–36; big data solutions, 126; extreme sports prostheses (Schultz), ix, 8, 9, 17–21,

121; inspiration and motivation, 7, 18, 19, 21, 22, 25–29, 32–33, 121; introduction, 8, 9, 17; regulating full-body swimsuits, 8, 9, 10, 122; safety conflicts, 54; snowboard (Carpenter), 8, 9, 17, 21–25, 122; 3D-printed wheelchair racing gloves (Rausin), 8, 9, 17, 25–28; Win Tunnel (Yu and Cote), 7, 8, 9, 17, 28–32
Compressure Comfort Bra, 82
concussions, 38, 39, 103, 107, 109, 110, 122
Connors, Jimmy, 86
ConocoPhillips, viii
construction workers, injury prevention, 54, 122
Cote, Mark: background, 7, 28; design process, 29, 32; low-speed "Win Tunnel," 7, 8, 9, 17, 28–32; motivation, 28–29
Couric, Katie, 94
Court of Arbitration for Sport (CAS), 112
COVID-19 pandemic, ix, 11, 123, 124–125
Crash Cloud Technology, 7, 8, 9, 37–41, 54
crash test dummies, 42, 43
Creative Urethanes (firm), 90, 91
cricket, 71–72, 77n48, 126
critical sports studies, 3–4
Crosby, Bing, 67
CTE (chronic traumatic encephalopathy), 38, 107, 110
Cumnock, Arthur, 107
Cunningham, Richard, 114
Curry, Stephen, 124
Cybathlon, 131
cyborg athletes, 131–132
cycling. *see* biking; mountain biking

Dallas Cowboys, 38
D'Alusio, Chris, 29
Darby, Newman, vii
data collection. *see* big data and artificial intelligence
Dawkins, Darryl "Dr. Dunk," 45–46
Daytona 500 (2001), 44
DeAngelis, Doug: background, 64; commercialization of inventions, 121; design process, 65; digital photo-finish camera, 7, 9–10, 63–67; IsoLynx player-tracking system, 65–66, 137; motivation, 64–65, 73
DeForest Concepts, 82
dehydration and rehydration, 48–50, 54
Department of the Interior (DOI), 114–115
de Quesada, Alex, 48, 50
Devitt, John, 60–61
digital photo finish, 63–67; design process, 65; FinishLynx, 7, 9–10, 65, 66–67; human interpretation, 73; introduction, 7, 9–10, 63–64; motivation, 64–65, 73; precursors, 63–65; uses, 73
disabilities, athletes with: Americans with Disabilities Act (1990), 79; classification system, 111; definitions, 138n47; motivation for innovation, 7; museum exhibitions, vii, viii, ix; Unified Sports, 11, 111–113, 129–132. *see also* adaptive sports; para-athletes; prostheses; Special Olympics; wheelchair athletes
Dodd, Bobby, 50
DOI (Department of the Interior), 114–115
Dolphin 44 (snowboard), 24
doping. *see* performance-enhancing drugs
double Dutch jump-roping: about, 93; automated jump-roping machine, 9, 10, 79, 93–96, 121; history, 93
Douglas, Dewayne, 48
Douglas Edwards and the News (television show), 67
Downing, James R., 42–44, 53
Drago, George, 69
drones, 126
DSMRI (dynamic systems model of role identity), viii
Duffy, Ken, 38
Dunn, Shannon, 24
Durant, Kevin, 124
dynamic systems model of role identity (DSMRI), viii

Earnhardt, Dale "The Intimidator," 44
e-bikes, 31; defined, 113; trail access regulations, 9, 10, 101, 103, 113–116
Echauri, Alfredo, 29
Ehrat, Arthur: background, 44–45, 53, 121; Break-A-Way basketball rim, vii, 8, 9, 44–47, 56n27, 121; design process, 45, 46, 47; museum exhibitions, vii; patent, 45, 56n27
Ehrat, Sharon, 46–47
electrolytes, 48, 49, 50
Elysian Park Ventures (VC firm), 124
Emmy Awards, 67, 70, 82
Enberg, Dick, 70
environmental impact, 123
Epilepsy Foundation, 82
Epstein, David, 3, 74n2
Equal Rights Amendment (ERA), 79, 80
Erving, Julius "Dr. J," 45
Evans, Robert, 73
Evert, Chris, 86
Everyone Plays! (exhibition), vii
extreme sports prostheses, ix, 8, 9, 17–21, 121

F1 (Formula 1) racing, 30, 44
Fairhurst, Fiona, 104–105
fairness and accuracy, 59–78; cheating prevention, 74n1, 102, 127; digital photo finish (DeAngelis), 7, 9–10, 63–67; eligibility determination, 74n1, 102; fully automatic timekeeping (Parkinson), 7, 9, 59–63, 75n9; Hawk-Eye Ball-Tracking System (Hawkins), 7, 9, 10, 70–73, 122, 125–129; inspiration and motivation, 7, 59, 71, 73–74, 121; instant replay (Verna), 2, 9, 10, 67–70, 103, 117n7; introduction, 9–10, 59. *see also* governance of sports technology
fans: auto racing, 44; basketball slam dunk, 45, 46; big data and AI, 126, 128, 129; and climate change, 133, 134; COVID-19 virtual fans, 125; fairness as concern, 9, 10, 59, 73, 74, 101, 102; fan engagement technology, 2; increased viewership, 123–124; in-game innovations, 2; instant replay, 2, 70, 117n7; as market force, 2; soccer goal-line technology, 72; sports governance technology, 103, 107, 116, 117n7,

fans (*continued*)
122; as stakeholders, 4, 7, 10, 73; technology debates, 2, 4, 7–8, 10, 11, 70, 74, 101, 102, 116, 122, 134; tennis ball-tracking system, 72, 129; virtual reality, 129, 134
Farmers Elevator Co., Lowder, Illinois, 45
Fédération Internationale de Natation. *see* FINA/World Aquatics
Federer, Roger, 72
FIFA: Adidas soccer balls, 28, 123; automated onside/offside calls, 72; ban on removal of jerseys in celebration, 84; "Generation Amazing" initiative, 130; Hawk-Eye goal-line technology, 72; Hawk-Eye onside/offside calls, 127; sustainable stadiums, 123, 133; Unified Sports, 130; World Cup, 28, 123, 130, 133
FINA/World Aquatics, 62, 102, 105–107, 116
FinishLynx digital photo-finish camera and timing system, 7, 9–10, 64–67
Finn, Jonathan, 64
first responders, protective gear for, 40
Fisher, Gary, 6, 114
Flex-Foot running prostheses, vii, 101, 103, 111, 112
football: Army–Navy game (1963), 69–70; body padding, 108, 126; dehydration, 48–50, 54; future innovations, 128; Gatorade, 8, 49–50, 53, 54; head-injury fatalities, 108, 109; helmet evolution, 107–110; helmet regulations, 9, 10, 101, 103, 107–110; helmet safety improvements, 7, 8, 37, 38–39, 53–54, 103, 107–109; history, 107–109; injuries, 38, 107, 108, 109, 110, 122; instant replay, 69–70, 103, 117n7; invention cycle, 108; IsoLynx player-tracking system, 65–66, 76n25; RFID transmitters, 126, 127; rules and regulations, 103, 107–108, 109; telecasts, 69–70; youth football, 109–110
Forest Service (U.S.), trail access, 114, 115
formsense (firm), 9, 52–53, 54, 126
Formula 1 (F1) racing, 1, 30, 44
Fotheringham, Aaron "Wheelz," ix
Fouché, Rayvon, 4
Founders Fund (VC firm), 124
Fox Shox (shocks manufacturer), 18, 19
Free, James, 48, 50
French Open (tennis tournament), 72
Freud, Sigmund, 107
Fritzlen, Martha ("Marty") Becker, 86
fun for everyone, 79–100; automated double Dutch jump-roping machine (Reid), 9, 10, 79, 93–96, 121; improved skis and tennis rackets (Head), vii, 8, 9, 10, 79, 84–87, 96, 122; inspiration and motivation, 7, 79, 86, 93, 96–97, 121; introduction, 10, 79–80; Jogbra (Lindahl, Miller, and Smith), ix, 9, 10, 79, 80–84, 96, 121–122; Quickie wheelchair (Hamilton), vii, ix, 9, 10, 79, 88–90, 96; regulating e-bike trail access, 9, 10, 101, 103, 113–116; urethane skateboard wheels (Nasworthy), 7, 9, 10, 79, 90–93, 96
future of sports innovation. *see* past, present, and future of sports innovation

Gade, Peter, 131
Game Changer (Fouché), 4
GameChanger app, 129
game clocks. *see* timekeeping
Garner, Joanna, viii
Gary Fisher (mountain bike vendor), 6
Gatorade, 47–50; design process, 48–50, 53; introduction, 8, 9; name, 50; for pediatric dehydration, 50, 54; taste, 49
gender inequities. *see* Title IX legislation
gender stereotypes, 54
gender verification, 3, 102
General Motors Research Laboratories, 42
Georgia Tech, 50
girls. *see* women and girls
Girl Scouts of the Nation's Capital, viii
Glenn L. Martin Company, 84
global warming. *see* climate change
gloves, 3D-printed wheelchair racing gloves, 8, 9, 17, 25–28
glucose, 49
Goldblatt, David, 134
golf, 8, 82, 132
Gonzalez, Tony, 124
Gordon, Jeff, 44
governance of sports technology, 101–120; cheating prevention, 74n1, 102, 127; competitive edge (full-body swimsuits), 8, 9, 10, 101, 103, 122; eligibility determination, 74n1, 102; fairness and accuracy (prosthetic legs), vii, 9, 10, 101, 103, 110–113; fun for everyone (e-bike trail access), 9, 10, 101, 103, 113–116; governmental oversight, 5; health and safety (football helmets), 9, 10, 101, 103, 107–110; introduction, 5–6, 9, 10, 101–102; norms and practices ("unwritten rules"), 103; recreational *vs.* professional use, 8; regulations, defined, 102; rules, defined, 102; self-regulation, 5; and technology change, 102–103, 105–107
Grabowski, Alena, 112, 113
"Grand Stands: Stadium Technology and the New Fan Experience" (educational video), ix, xn6
Gravenstein, Nikolaus, 50
Graves, Ray, 49, 50
Greater Lowell Road Runners, 64
Guttmann, Allen, 60
gymnastics, Jogbra for, 82

halfball, 103
Hamilton, Marilyn: background, 88, 96; character traits, 88; museum exhibitions and programs, vii, ix; Quickie wheelchair, vii, ix, 9, 10, 79, 88–90, 96
hang gliding, 88, 96
HANS (head and neck support) device, 41–44; design process, 42–43, 53; introduction, 8, 9; motivation, 44
HANS Performance Products, 43, 44
Hargadon, Andrew, 7
Harris, Franco, 109
Harward, Bill, 90–91
Hawk, Tony, vii
Hawk-Eye Ball-Tracking System, 70–73; algorithms, 126; data collection and analysis, 125–129; future

innovations, 128–129; introduction, 7, 9, 10; Major League Baseball, 126–127; technical know-how, 122; tennis, 123, 129
Hawk-Eye Innovations Ltd., 72, 77n48
Hawk-Eye's automated ball-strike system (ABS), 127
Hawkins, Paul: background, 71, 122; Hawk-Eye Ball-Tracking System, 7, 9, 10, 70–73, 122, 125–129; introduction, 7; motivation, 71, 73; patent, 72
head and neck support device. *see* HANS device
Head, Howard: background, 84, 122; financing, 85; Head Ski Company, 85; improved skis, vii, 9, 10, 79, 84–85, 86, 96, 122; improved tennis rackets, vii, 8, 9, 10, 79, 84–87, 96, 122; motivation, 86, 96; museum exhibitions, vii; patents, 85
head injuries: auto racing, 41–44; basal skull fractures, 42; chronic traumatic encephalopathy (CTE), 38, 107, 110; concussions, 38, 39, 103, 107, 109, 110, 122; football, 37–38; prevention (*see* Crash Cloud Technology; HANS device)
Head Ski Company, 85, 122
health and safety, 37–57; Break-A-Way basketball rim (Ehrat), vii, 8, 9, 44–47, 56n27, 121; Crash Cloud Technology football helmets (Springs), 7, 8, 9, 37–41, 53–54; football helmet regulations, 9, 10, 101, 103, 107–110; Gatorade (Cade), 8, 9, 47–50, 53, 54; HANS device (Hubbard), 8, 9, 41–44, 53; inspiration and motivation, 7, 9, 37, 38–39, 44, 50–51, 53–54, 121; introduction, 8–9, 37; invention cycle, 108; obstacles to adoption of safety technologies, 54, 57n60; smart apparel (Ramasarma), 8–9, 50–53; wheelchair racing gloves, 25
Heitfield, Richard, 90–91
Heitfield, Vernon, 90–91
Helman, Don, 79, 88–90, 96
Hemment, John, 64
Higgins, Christopher, 73
hiking innovations, 4
history of sport, as field, 3–4
hockey, 40, 127
Honeywell Bull (engineering firm), 64, 65
Hopper-Dean Family Fund, viii
horse racing, 63–64, 66
Houston Astros, 70
How Breakthroughs Happen (Hargadon), 7
Howe, P. David, 111
Hubbard, Robert: background, 42, 53; design process, 42–43; HANS device, 8, 9, 41–44, 53; motivation, 44
Huffy (manufacturer), 46
hydrodynamics, 104–105

IAAF/World Athletics, 102, 110, 111–112, 116
ice hockey, 40, 102, 127
identity, 3–4. *see also* disabilities, athletes with; inventive identity; transgender athletes; women
I Love Lucy (television show), 69
Indy Racing League (IRL), 44
informed consent, 57n42
Ingenium Manufacturing, 27–28
Innoskate (Lemelson Center initiative), vii–viii, ix
innovation, defined, 6
Innovative Lives (Lemelson Center speaker series), ix, xin7
Innovatus Capital Partners, 109
Innsbruck Olympics (1964), 85
inspiration. *see* motivation for inventors
instant replay, 67–70; adoption of, 73; and fans, 2, 70, 117n7; first use, 69–70, 117n7; human interpretation, 73; introduction, 9, 10; motivation, 73; NFL rulebook, 103; as officiating tool, 103, 117n7; precursors, 67
Instant Replay (Verna), 69–70
intellectual disabilities, 138n47. *see also* Special Olympics
Intercollegiate Football Association, 107
International Amateur Athletic Federation. *see* IAAF/World Athletics
International Association of Athletics Federations. *see* IAAF/World Athletics
International Cricket Council, 77n48
International Olympic Committee (IOC), 106, 132–133. *see also* Olympics
International Paralympic Committee (IPC), 111
International Special Olympics. *see* Special Olympics
International Sport Organization for the Disabled (ISOD), 111
International Tennis Federation, 72
Internet of Things, 125
invention, defined, 6
"Invention, Sport, Technology, and Society" (Lemelson Center workshop), viii
Invention at Play (exhibition), vii
inventive identity, viii, x, 121, 141
Inventive Minds (Lemelson Center gallery), vii
IOC (International Olympic Committee), 106, 132–133. *see also* Olympics
IPC (International Paralympic Committee), 111
IRL (Indy Racing League), 44
ISOD (International Sport Organization for the Disabled), 111
IsoLynx player-tracking system, 65–66, 76n25

Jacquemart, Patrick, 42
James, LeBron, 124
Jasanoff, Sheila, 5
JBI, Inc., 82
JEM Corporation, 22, 23, 24
Jennings, Tom, 65
The Jim Henson Company, 82
jockstraps, as Jogbra inspiration, 80, 81
Jogbra, 80–84; commercializing, 81–82, 83; design process, 80–81; *Innovative Lives* (Lemelson Center speaker series), ix; introduction, 9, 10, 79; market research and testing, 80; motivation, 80, 96, 121–122; naming, 81; patents, 82
John Deere equipment, 45, 47
Johnson, Magic, 124
jump-roping. *see* double Dutch jump-roping

Kaminski, Frank, 85
Kaplan, Avi, viii

Kelly, Maurice, 38
Kennedy, John F., 69
Kentucky Derby, 66, 70
Kera Collective (evaluation firm), viii
kidney function, 47–48
Kim, Gyu Dae, 27
kinesiology, 4. *see also* biomechanics
Kipchoge, Eliud, 122–123
Kirby, Gustavus T., 64
knee pain, 51
Knight, Bobby, 45
knowledge transfer, 9, 44–45, 53–54, 96, 122

lacrosse helmets, 40
Larson, Lance, 60–61
Larwood, Thomas, Jr., 107
Lawson, Lajean, 82
L Catterton (VC firm), 124
lead users, 6, 65
Lemelson Center for the Study of Invention and Innovation, vii–x, 94
Lemelson Foundation, viii
Lewis, Michael, 3, 126
Lifetime (manufacturer), 46
Lightning2 Company (Bellisse), 82
Lindahl, Lisa: background, 80; Epilepsy Foundation, 82; *Innovative Lives* (Lemelson Center speaker series), ix; Jogbra, ix, 9, 10, 79, 80–84, 96, 121–122; Lightning2 Company (Bellisse), 82; motivation, 80
London Olympics (2012), 106, 112
London Paralympics (2012), 112
Loreto, Oscar, Jr., ix
lymphedema, 82
Lynx System Developers, Inc., 65–66, 76n25, 137
LZR Racer swimsuit, 8, 10, 103, 105, 106, 122; *see also* Speedo

MacGregor leather helmet, 109
MacRae, Alexander, 104
Mahoney, Kenneth, 46
Major League Baseball (MLB), 70, 124, 126–127
Major League Soccer (MLS), 125
Mancina, Dan, ix
Mantis Bicycles, 114
marathons: Boston Marathon, 26, 27, 63, 125; COVID-19 pandemic, 125; records, 122–123; timekeeping, 63; wheelchair athletes, 26, 27
Marin County, California, 114
Marx, Ernest, 64
Massachusetts Institute of Technology (MIT), 28, 51, 65, 66
MaxPro (home fitness equipment), 124
McDonald, Tommy, 69
McGee, Patti, vii
McLaren (Formula 1 racecar manufacturer), 30
McMahon, Ed, 67–68
MCR (Motion Control Requirement), 82
mentorship, 38, 41
Meyer, Petrus ("Peter"), 85
Michigan State University, 42
Mid-Ohio Sports Course, 42
Miles, Phoebe Cade, 48, 50
military helmets, 40, 53, 122
military veterans: Middle East deployments, 19; World War II, 88, 110, 129
Miller, Hinda Schreiber: background, 80; DeForest Concepts, 82; *Innovative Lives* (Lemelson Center speaker series), ix; Jogbra, ix, 9, 10, 79, 80–84, 96, 121–122; political career, 82
minorities. *see also* African Americans: discrimination against, 40, 79; mentorship, 40–41; underrepresentation, 122; venture capital investments in, 124
MIT (Massachusetts Institute of Technology), 28, 51, 65, 66
Mizuno Aquablade swimsuit, 104
MLB (Major League Baseball), 70, 124, 126–127
MLS (Major League Soccer), 125
Modell, Jerome H., 50
Molson 1000 (1980), 42
Moneyball (Lewis), 3, 126
Montana, Joe, 124
Moore, Greg, 44
Motion Control Requirement (MCR), 82
Motion Designs (firm), 88–90, 98n22
motivation for inventors: competitive edge, 7, 18, 19, 21, 22, 25–29, 32–33, 121; fairness and accuracy, 7, 59, 71, 73–74, 121; financial gain, 7; frustration, 7; fun for everyone, 7, 79, 86, 93, 96–97, 121; health and safety, 7, 9, 37, 38–39, 44, 51, 53–54, 121; introduction, 6–8; monetary profit, 121; own athletic benefit, 6–7, 121
motocross, 17
motocross, adaptive, ix, 19
Moto Knee (prosthesis), ix, 8, 9, 19, 20, 121
motusBASEBALL arm sleeve, 127
mountain biking: history, 6, 114; norms and practices ("unwritten rules"), 103; regulating e-bike trail access, 9, 10, 101, 103, 113–116; suspension components used in prostheses, 18–19, 20
Move United Junior Nationals, 28
Mullen, Rodney, vii
Mullin, Jack, 67
Munich Olympics (1972), 104
Muybridge, Eadweard, 63–64
MyLaps Sports Timing (firm), 63

Nadal, Rafael, 72
NASA, Speedo collaboration, 104
NASCAR (National Association for Stock Car Auto Racing), 44, 66
Nasworthy, George Franklin, Sr., 90
Nasworthy, George Franklin ("Frank"), Jr.: background, 90–91; Cadillac Wheels Company, 91, 92–93; Hewlett-Packard career, 92; patents, 92; urethane skateboard wheels, 7, 9, 10, 79, 90–93, 96
National Association for Stock Car Auto Racing (NASCAR), 44, 66
National Association of Radio and Television Broadcasters, 67

National Basketball Association (NBA), 46, 70, 125, 127
National Collegiate Athletic Association (NCAA): COVID-19 pandemic, 125; football injuries, 108; football regulations, 108, 109; formation, 107–108; "March Madness" college basketball tournament, 125; slam dunk regulations, 45
National Collegiate Inventors and Innovators Alliance, 94
National Federation of State High School Associations, 108
National Football League (NFL), 37–38; concussion crisis, 38, 109; Crash Cloud Technology helmets, 7, 8, 9, 37–41, 53–54; funding for brain health and concussion risk research, 109–110; helmet regulations, 108, 110; instant replay, 69, 70, 103, 117n7; IsoLynx player-tracking system, 65–66, 76n25; "Next Generation Stats," 66; regulatory process, 116; RFID transmitters, 126, 128
National Hockey League (NHL), 70, 125, 127
National Museum of American History (NMAH): Arthur Ehrat Papers, 46; *Change Your Game / Cambia tu juego* (exhibition), viii, ix–x; Howard Head Papers, 85, 87; Innoskate (Lemelson Center initiative), vii–viii; Lemelson Center for the Study of Invention and Innovation, vii–x
National Operating Committee on Standards for Athletic Equipment (NOCSAE), 108, 110
National Park Service (NPS), 114, 115
National Science Foundation (NSF), viii
Navratilova, Martina, 86
NBA (National Basketball Association), 46, 70, 125, 127
NCAA. *see* National Collegiate Athletic Association
Nelson, Lindsey, 69, 70
nephrology, 47–48
neurological conditions, early detection, 52
Nevco Sports, 66
Next Ventures (VC firm), 124
NFL. *see* National Football League
NHL (National Hockey League), 70, 125, 127
Nike: Lemelson Center grant, viii; Nike Training Club, 128; R & D, 124; sports bra, 84; VaporFly super-shoe, 122–123
NMAH. *see* National Museum of American History
NOCSAE (National Operating Committee on Standards for Athletic Equipment), 108, 110
nonbinary athletes, vii
NPS (National Park Service), 114, 115
NSF (National Science Foundation), viii

Oakland Athletics, 3, 126
officiating technologies, 2, 7, 10, 59, 70, 72, 73. *see also* governance of sports technology
Ohio State University, 38
Okamoto, James ("Jim"), 79, 88–90, 96
older adults, protective gear for, 40, 54
Oliveira, Alan, 112
Olympics: Athens (2004), 104; Beijing (2008), 106, 111–112; and climate change, 132–133; Innsbruck (1964), 85; London (2012), 106, 112; Munich (1972), 104; officials, 60–61; Rio de Janeiro (2016), 106; Rome (1960), 60–61, 68, 69; Salt Lake City (2022), 133; snowboarding, 24; speed skating, 31; Stockholm (1912), 60; swimming, 104, 106; Sydney (2000), 132–133; timekeeping, 60–61; Tokyo (2020), 106, 125, 133; track and field, 111–112
Omega (firm), 62
O'Neal, Shaquille, 124
Orange Bowl, 50
Oswald, Lee Harvey, 69
Overtime (sports live streaming firm), 124
Owens, Jesse, 60, 74n2
Ozyurtcu, Tolga, 4

para-athletes: critical sports studies, 3; extreme sports prostheses, ix, 8, 9, 17–21, 121; Flex-Foot running prostheses, vii, 101, 103, 111, 112. *see also* disabilities, athletes with; Unified Sports; wheelchair athletes
Paralympic movement: blade prostheses, 7, 10, 111; emphasis on self-realization, 130; first, 110; museum exhibitions and programs, vii, ix; origins, 110–111, 129–130; positive feedback loop, 130. *see also* Unified Sports
Paralympics: ban on motorized equipment, 131; classification system, 111, 130; snowboarding, 19–21; track and field, 27, 28, 110–111, 112; training sites, 25
Parkinson, Bill: automatic timekeeping, 7, 9, 59–63; design challenges, 61–62; motivation, 73; patent, 62, 75n9
Parkinson, Martha, 62
Parkinson's disease, 52, 54
past, present, and future of sports innovation, 121–140; accelerating pace of sports innovation, 122–125; big data and artificial intelligence, 125–129; climate change, 132–134; introduction, 10–11, 121–122; Unified Sports, 129–132
patents: Accutrack photo-finish camera, 64, 75n16; automated double Dutch jump-roping device, 94; automated swim-timing system, 62, 75n9; "ball tracking and trajectory prediction" system, 72; Break-A-Way basketball rim, 45, 56n27; Crash Cloud, 41; football helmets, 109, 110; HANS device, 43; Head's skis, 85; increase in, 123; Jogbra, 82; Nasworthy's, 92; United States Patent and Trademark Office, viii
Patrick J. McGovern Foundation, viii
Peak (Bubbs), 3
pediatric dehydration, 50, 54
PEDs (performance-enhancing drugs), 2, 3, 101, 102
Peek, Gene, 48, 50
Peloton (firm), 124
performance-enhancing drugs (PEDs), 2, 3, 101, 102
performance-enhancing swimsuits, 106
perspiration, composition, 48
Petty, Adam, 44
pharmaceutical innovations, 2, 3
Phelps, Michael, 105, 106

Philadelphia Eagles, 69
Phillips, Van, vii, 111
photo finish, digital. *see* digital photo finish
Pilczuk, Bill, 105
Pinch, Trevor, 4
Pistorius, Oscar, 111–112
Pitt, Brad, 126
Plaskett, Allen, 126
"The Playful Mind" (Lemelson Center symposium), 94
Playing Against the Clock (Goldblatt), 134
Playtex Apparel, 82
PlayVS (esports firm), 124
Polaroid film, 64–65
polyurethanes: roller-skate wheels, 90–91; skateboard wheels, 7, 9, 10, 79, 90–93, 96; use of term, 98n27
Poniatoff, Alexander M., 67
Poppen, Sherman, vii, 22
present state of sports innovation. *see* past, present, and future of sports innovation
Prince Manufacturing Company, 86, 87
prostheses: blade prostheses, 7, 10, 111; critical sports studies, 3; extreme sports, ix, 8, 9, 17–21, 121; Flex-Foot running prostheses, vii, 101, 103, 111, 112; introduction, 7; motivation for innovation, 7; Moto Knee, ix, 8, 9, 19, 20, 121; regulating prosthetic legs, 9, 10, 103, 110–113; Versa Foot, ix, 8, 9, 19–20, 121

Qatar, FIFA World Cup, 123, 130, 133
Qatar Foundation, ix
Qualcomm, 51
Quickie wheelchair, vii, ix, 9, 10, 79, 88–90, 96

racial discrimination, 40, 79
racial power dynamic in sports, 3–4
radio, high-fidelity recording, 67
radio-frequency identification (RFID): automatic timekeeping systems, 62–63, 73; football, 126, 127, 128; IsoLynx player-tracking system, 65–66, 76n25
Ramasarma, Nathan: background, 51, 53; design process, 52, 125; formsense (firm), 9, 52–53, 54, 121; motivation, 50–51; smart apparel, 8–9, 50–53; Tune Smartwear, 53; wearable sensors, 51–53, 54
Range (Epstein), 3
Rausin, Arielle: background, 25; design process, 25–26, 32, 125; Ingenium Manufacturing, 27–28; motivation, 25–28, 32; 3D-printed wheelchair racing gloves, 8, 9, 17, 25–28
recombinant innovation, 7
regulations. *see* governance of sports technology; rules and regulations
regulatory governance, 5–6. *see also* governance of sports technology
Rehm, Markus, 112
Reid, Tahira: academic career, 95–96; automated double Dutch jump-roping machine, 9, 10, 79, 93–96, 121; background, 93, 121; motivation, 93, 96; patents, 94; "The Playful Mind" (Lemelson Center symposium), 94
remote-control car races, 63
Rensselaer Polytechnic Institute (RPI), 93–94, 95
RFID. *see* radio-frequency identification
Riddell, John, 108
Riddell football helmets, 108, 109
Riener, Robert, 131
Rio de Janeiro Olympics (2016), 106
RK&A (evaluation firm), viii
road races. *see* running
Robinson, Neil, 84
robots, 131
Rogers, Jane: curation of sports-themed exhibitions, vii
Roke Manor Research Ltd., 71–72
Roker, Al, 94
roller-skating, 90–91
Rome Olympics (1960), 60–61, 68, 69
Rome Paralympics (1960), 110
Roosevelt, Theodore, 107–108
Roto (design firm), viii
RPI (Rensselaer Polytechnic Institute), 93–94, 95
Ruby, Jack, 69
Rudolph, Wilma, 68
rules and regulations: defined, 102; enforcement, 102; establishing level playing field, 110, 112–113, 116; invention cycle, 108; NCAA slam dunk, 45; new technology, 116; proliferation of, 103. *see also* governance of sports technology
running: Flex-Foot running prostheses, vii, 101, 103, 111, 112; Jogbra, ix, 9, 10, 79, 80–84, 96, 121–122; knee pain, 51, 53; Nike VaporFly super-shoe, 122–123; road races, 62–63, 64, 73; Strava app, 128; technologies, 1; timekeeping innovations, 62–63, 73. *see also* marathons; track and field

safety. *see* health and safety
Safety 1st (car safety seat manufacturer), 38–39
sailboards, vii
sailboat racing, 28
Salt Lake City Olympics (2022), 133
SAP (investor), 124
Sapphire Ventures (VC firm), 124
Sara Lee Corporation, 82
Saubert, Jean, 85
Saud Abdulaziz Abdul Ghani, 133
Saunders, Patrick, 70
Schultz, Mike: adaptive motocross racing, 19; adaptive snowboarding, 19–21; background, 17–18, 121; BioDapt (firm), 19–21; design process, 18, 32; extreme sports prostheses, ix, 8, 9, 17–21, 121; fabrication process, 18–19; injury and amputation, 18; "Monster Mike" nickname, 18; motivation, 18, 19, 21, 32; motocross racing, 17; museum programs, ix; Paralympics, 20–21; snowmobile racing, 18, 19
Schutt (sports helmet manufacturer), 109
science, technology, engineering, and mathematics (STEM), viii, 8
science and technology studies (STS), 3, 4–5
SCOT (social construction of technology), 4
Seattle Seahawks, 38
Seau, Junior, 38

Seiko (firm), 62
Senna, Ayrton, 44
Sequoia Capital (VC firm), 124
Serena Ventures (VC firm), 124
Sesame Street (television show), 82
Shapiro, Paul, 56n27
Sherry, David, 72
shipping containers, in stadium construction, 123, 133
Shires, Dana, 48, 50
Shō Foundation, viii
Shriver, Eunice Kennedy, 130, 139n51
Shriver, Pam, 8, 72, 86
Shriver, Timothy, 139n51
Sigel, Kirk, 65
SIMS (snowboard manufacturer), 24
Sims, Tom, 24
Sinyard, Mike, 28, 29
skateboarding: adaptive skateboarders, ix; improvements, 91–92; Innoskate (Lemelson Center initiative), vii–viii; origins, 90; sensors to analyze tricks, 52; urethane skateboard wheels, 7, 9, 10, 79, 90–93, 96
Skateboarding Hall of Fame and Museum, 92–93
skeleton, 30
Ski Cooper, Colorado, 24
skiing: and climate change, 132; Head's improved skis, vii, 9, 10, 79, 84–85, 86, 96, 122; introduction, 4; Olympics (1964), 85; Paralympic alpine skiing, ix; timekeeping as essential, 59; Win Tunnel use, 30
slam dunk, 44–46
sled tests, 43, 55n14
Slocum, Alexander, 29
SLS, Inc., 82
smart apparel, 50–53; design process, 52; inspiration, 52; introduction, 8–9; privacy concerns, 54; regulatory governance, 5; tracking movement and biometrics, 126, 127; wearable sensors, 51–53, 54, 126, 127, 129; yoga pants, 128
Smart Legging, 52
smartphone apps, 128
Smith, Dean, 45
Smith, Polly Palmer: background, 80; *Innovative Lives* (Lemelson Center speaker series), ix; The Jim Henson Company, 82; Jogbra, ix, 9, 10, 79, 80–84, 96, 122
Snickometer, 126
snowboarding: adaptive snowboarding, 19–21; evolution of sport, 24–25; invention, 8, 9, 17, 21–25, 122; Paralympics, 19–20; precursor (Snurfer), vii, 8, 17, 22–24, 25; snowboard composition, 22, 24; snowboard design, 22–23, 24
Snowboarding (exhibition), vii
snowmobile racing, 18, 19
Snow Valley (ski resort), Vermont, 24
Snurfer (snowboard precursor), vii, 8, 17, 22–24, 25
soccer: Adidas soccer balls, 28, 123; automated onside/offside calls, 72; ball tracking and trajectory prediction, 73, 123; and climate change, 133; COVID-19 pandemic, 125; Hawk-Eye goal-line technology, 72; Hawk-Eye onside/offside calls, 127; Jogbra, 82, 84; referees, 4; sustainable stadiums, 123, 133; Unified Sports, 130; Women's World Cup, 84
social construction of technology (SCOT), 4
Society of Automotive Engineers, 42
Soda Springs Ski Bowl, California, 24
softball, 40
Sony (firm), 72
The Sources of Innovation (von Hippel), 6
Spalding (firm), 107
Spark!Lab, ix
Specialized Bicycle Components, Inc.: aerodynamic gear, 30; data collection and analysis, 125–126; founders, 28; mountain bikes, 6; R & D, 124; Venge Integrated Aero Solution (ViAS), 30, 31; "Win Tunnel," 7, 9, 28–32
Special Olympics, vii, 129–131, 138n47, 139n51
Specialty Instruments Corporation, 64
Speedgate (sport), 128
Speedo: Aquablade swimsuit, 104; Aqualab design team, 105; company history, 104; Fastskin, 104; LZR Racer swimsuit, 8, 10, 103, 105, 106, 122; R & D, 104–105, 124; S2000, 104
speed skating, 31
Spitz, Mark, 104
Sporting Invention (exhibition), vii
sports bras. *see* Jogbra
Sports Fitness Industry Association, 108
The Sports Gene (Epstein), 3
sports historians, 3–4
SportsTechX, 124
SportTechie, ix
Springs, Ron, 38
Springs, Shawn: background, 37–38, 40, 53; Crash Cloud Technology, 7, 8, 9, 37–41, 53–54; introduction, 7, 8; motivation, 38–39; non-football helmets, 40, 53, 54, 122; obstacles, 40; Windpact founding, 7, 37, 38
stadium design: and climate change, 133–134; constructed from shipping containers, 123; "Grand Stands: Stadium Technology and the New Fan Experience" (educational video), ix, xn6; Tinkercad web app, ix, xin6; "12 Game-Changing Stadiums in History" (educational video), ix, xn6
stakeholders, 4–5, 7, 10, 73
Staubach, Roger, 69
Staubley, Richard, 56n27
Steenkamp, Reeva, 112
STEM (science, technology, engineering, and mathematics), viii, 8
Stichweh, Rollie, 69–70
Stillman, Alison Rapaport, 124
St. Louis University, 44
Stockholm Olympics (1912), 60
Stokely-Van Camp, 50
stopwatches, 60, 61, 73
Stowe, Vermont, 84
Strasburger, Mary, 48
Strasser, Christoph, 30
Stratton (ski resort), Vermont, 24
Strava app, 124, 128
"strip photography" technique, 64

stroke patients, 54
Strong, Evan, 19–20
STS (science and technology studies), 3, 4–5
suicide, 38, 109
Suicide Six (ski resort), Vermont, 24
Summer Olympics. *see* Olympics
Sunrise Medical, 90
Super Bowl (1980), 109
surfing, 4, 90–91
Sustainable Trails Coalition, 114
Swersey, Burt, 93, 94, 95
swimming: and climate change, 134; governing organization, 62, 102, 105–107; Olympics, 104, 106; regulating full-body swimsuits, 8, 9, 10, 101, 103, 106–107, 122; rules concerning swimming techniques, 106; swimsuit history, 104; timekeeping, 60–62, 73
Sydney Olympics (2000), 132–133
Sylvester, Jess, 105

technological doping, 8, 101, 116
technology brokers, 7
technoscientific actors, 4–5
Teixeira, Chuck, 29
television: AI systems, 129; baseball telecasts, 68; instant replay, 2, 9, 10, 67–70, 117n7; magnetic recording solutions, 67; "tape delayed" television broadcast, 67
tennis: automated tennis-ball machine, 86; ball tracking and trajectory prediction, 72, 73, 123, 125, 129; and climate change, 132; COVID-19 pandemic, 123, 125; Head's tennis rackets, vii, 8, 9, 10, 79, 84–87, 96, 122; Jogbra for, 82; U.S. Open Women's Wheelchair Championship, 88
3D-printing, wheelchair racing gloves, 8, 9, 17, 25–28
timekeeping: boxing, 73; fully automatic, 7, 9, 59–63, 73; human error, 59, 60–61, 73; introduction, 7; Olympics, 60–61; running races, 62–63, 73; stopwatches, 60, 61, 73; swimming, 60–62, 73
Tinkercad web app, ix, xin6
Title IX legislation, vii, 3, 79, 80, 83
Today (NBC morning show), 94
Tokyo Olympics (2020), 106, 125, 133
Tonal (firm), 124
"Toss Back" device, 46
Tour de France, 66
track and field: automated false-start detection, 65; Beijing Olympics (2008), 111–112; disability classification systems, 111; governing organization, 102, 110; laser distance measurement tool, 65; Paralympics, 110–111, 112; photo-finish cameras, 64–65; Rome Olympics (1960), 68; rules and regulations, 110, 111–112; Stockholm Olympics (1912), 60; technological innovations, 74n2; television replay, 68; timekeeping, 59–60; wheelchair athletes, 25–27. *see also* running
training regimens, 1, 3
trajectory prediction, 71–72
transgender athletes, vii. *see also* gender verification
Transportation Safety Institute, 42
Tucci (sports helmet manufacturer), 109
Tune Smartwear, 53
"12 Game-Changing Stadiums in History" (educational video), ix, xn6
Tyner, Frederick C., 56n27

UEFA Euro soccer championship, 125
Under Armour (sports apparel firm), 31
Unified Sports, 11, 103, 111–113, 129–132
United States Football League (USFL), 127
United States Patent and Trademark Office, viii
Univega (mountain bike vendor), 6
University of Florida, 8, 48–50, 53
University of Illinois, 25, 26, 27, 42
University of Maine, 64
University of Washington, 109
urethane: skateboard wheels, 7, 9, 10, 79, 90–93, 96; use of term, 98n27
U.S. Air Force, 42
U.S. Army Signal Corps, 67
user innovation, 6, 65, 121
USFL (United States Football League), 127
U.S. Forest Service, trail access, 114, 115
U.S. Indoor Track and Field, 65, 66
U.S. National Inventors Hall of Fame, 82, 86
U.S. National Ski and Snowboard Hall of Fame, 86
U.S. Open (tennis tournament), 72, 86, 132
U.S. Open Snowboarding Championship, 24
U.S. Open Women's Wheelchair Championship (tennis), 88
Utah, e-bike trail access, 114

Vaitones, Steve, 64
Van Brocklin, Norm, 69
Vanvolsem, Maarten, 64
Venge Integrated Aero Solution (ViAS), 30, 31
venture capital (VC) investments, 124, 136n18
VentureWell, 94
Verna, Tony: background, 67–68; career overview, 70; death, 70; instant replay, 9, 10, 67–70; *Instant Replay* (memoir), 69–70; motivation, 73
Versa Foot (prosthesis), ix, 8, 9, 19–20, 121
veterans, 19
ViAS (Venge Integrated Aero Solution), 30, 31
VICIS football helmets, 109
videotapes, 67
Virginia Squires, 45
virtual reality (VR), 125, 134
Vitaminwater (fitness drink), 124
volleyball, Jogbra for, 82
von Hippel, Eric, 6, 65

WADA (World Anti-Doping Agency), 101, 115
Waldych, Joe, 85
Walsh, Patrick, 63
Waltrip, Darrell, 44
Wasco County, Oregon, 115
water tunnels, 28

Wearable X, 127–128
We Belong Here (exhibition), vii
Webster, "Iron Mike," 38
Wells, John, 69
wheelchair athletes: critical sports studies, 3; motivation for innovation, 7; Quickie wheelchair, vii, ix, 9, 10, 79, 88–90, 96; 3D-printed gloves, 8, 9, 17, 25–28; track and field, 25–27; wheelchair motocross, ix; World War II veterans, 88
Whitehead, Cindy, vii
Will, Sarah, ix
Williams, Serena, 124
Wilson (firm), 126
Wimbledon (tennis tournament), 72, 86
Windpact (firm): Crash Cloud Technology, 7, 8, 9, 37–41, 53–54; founding, 7, 37, 38; product range, 40, 53, 54, 122
windsurfing, vii
wind tunnels: Win Tunnel, 7, 8, 17, 28–32; Wright Brothers Wind Tunnel, 28
Winter Olympics. *see* Olympics
Win Tunnel (low-speed wind tunnel), 28–32; design, 29; features, 29; introduction, 7, 8, 17; motivation, 28–29
WNBA time periods, 59
women and girls: *Change Your Game / Cambia tu juego* (exhibition), viii, ix; historic exclusion and underrepresentation, 3, 122; inclusion in sports, 79; Jogbra, ix, 9, 10, 79, 80–84, 96, 121–122; lacrosse helmets, 40; mentorship, 41; motivation for playing sports, 97; snowboards, 24; softball catcher's mask, 40; swimsuits, 104; Title IX legislation, vii, 3, 79, 80, 83; and venture capital (VC), 124, 136n18
Women's World Cup (soccer), 84
Wooden, John, 45
World Anti-Doping Agency (WADA), 101, 115
World Aquatics/FINA, 62, 102, 104, 105–107, 116
World Athletics/IAAF, 102, 110, 111–112, 116
World Cup (soccer), 28, 123, 130, 133
World Swimming Championships (2009), 106
World Tennis Championship, 86
World War II: adaptive sports for injured veterans, 88, 110, 129; B-26 bombers and PBM-3 flying boats, 84; limited supply of plastics, 108; technological spoils, 67
Wright Brothers Wind Tunnel, 28

X Games, adaptive sports, 19, 130

YMCA of Metropolitan Washington, viii
yoga, 128
Yu, Chris: background, 7, 28; design process, 32; low-speed "Win Tunnel," 7, 8, 9, 17, 28–32; motivation, 28–29; Win Tunnel design, 29

Zebra (firm), 76n25, 126, 137n27
Zevenheuvelenloop (Seven Hills Run), 62–63

What if the true champions of sports aren't just the athletes, but the inventors behind the game-changing technologies they use? *Inventing for Sports* invites you to explore the hidden world of innovation that fuels modern athletics. From cutting-edge swimsuits and aerodynamic bikes to precision officiating tools and adaptive equipment, this book uncovers how invention shapes competition, safety, fairness, and participation. With compelling stories of inventors driven by frustration, inspiration, and ingenuity, the authors reveal how technology transforms every sport and challenge readers to see the extraordinary potential for invention in their own lives. Whether you are a fan, an athlete, a technology enthusiast, or a reader interested in history, this book will change how you view sports.

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